

Biograph

MI

Replacements of Parts

Gantry

CT Gantry

For use with the following systems:
Biograph 16, Biograph 40/64 TruePoint

© Siemens, 2006

All documents may only be used for rendering services on Siemens Healthcare Products. Any document in electronic form may be printed once. Copy and distribution of electronic documents and hardcopies is prohibited. Offenders will be liable for damages. All other rights are reserved.

Print No.: MI12-023.841.01.02.02
Replaces: MI12-023.841.01.01.02
Part No.: D0007344-VFT-000-02

English
Doc. Gen. Date: 02.11

Copyright

“© Siemens, 2006” refers to the copyright of a Siemens entity such as Siemens Aktiengesellschaft - Germany, Siemens Shenzhen Magnetic Resonance Ltd. - China, Siemens Shanghai Medical Equipment Ltd. - China, Siemens Medical Solutions USA Inc. - USA, Siemens Healthcare Diagnostics Inc. - USA and/or Siemens Healthcare Diagnostics Products GmbH - Germany.

Document Version

Siemens reserves the right to change its products and services at any time.

In addition, manuals are subject to change without notice. The hardcopy documents correspond to the version at the time of system delivery and/or printout. Versions to hardcopy documentation are not automatically distributed. Please contact your local Siemens office to order current version or refer to our website <http://www.healthcare.siemens.com>.

Disclaimer

Siemens provides this documentation “as is” without the assumption of any liability under any theory of law.

The installation and service of equipment described herein requires superior understanding of our equipment and may only be performed by qualified personnel who are specially trained for such installation and/or service.

1	Revision History	12
2	General	13
	Safety information	13
	Information about the possibilities for switching off the gantry power	14
	Tools	15
3	XRS parts, Sensation 10/16/40/64/Cardiac/Cardiac64	16
	HV Tank	16
	HV transformer E604	16
	Generator Electronic Box	25
	Generator electronic box E603	25
	Akron Tube	36
	Overview	36
	Prerequisites	36
	Preliminary Steps	37
	Removing	38
	Installing	42
	Final Steps	43
	Straton Tube	46
	Overview	46
	Prerequisites	46
	Preliminary steps	47
	Removing	50
	Installing	57
	Final steps	61
	Tube collimator E555	65
	Prerequisites	65
	Preliminary steps	65
	Removing	66
	Installing	67
	Final steps	67
	PHI_Z_MON D554	69
	Prerequisites	69
	Preliminary steps	69
	Removing	70
	Installing	70
	Final steps	71
	D400 in generator electronics box	72
	Prerequisites	72
	Preliminary steps	72
	Removing	72
	Installation	73

Final steps	74
STRATON FIL assembly D470	75
Prerequisites	75
Preliminary steps	75
Removing	75
Installing	76
Final steps	77
XRS Power Supply E606	78
Prerequisites	78
Preliminary steps	79
Removing	80
Installing	85
Final steps	87
Fuses F4, F5 generator electronics	88
Prerequisites	88
Preliminary steps	88
Removing	88
Installing	89
Final steps	89
Fan of generator electronic box	90
Prerequisites	90
Preliminary steps	90
Removing	90
Installing	91
Final steps	91
4 Gantry left stand	92
Overview	92
LUST Rotation Converter	93
Prerequisites	93
Preparation	93
Removing	94
Installing	94
Final steps	94
LUST Tilt Converter	96
Tilt Sensor	97
Prerequisites	97
Preparation	97
Removing	98
Installing	98
Final steps	98
MCU-RTC Board	100
Prerequisites	100
Preparation	100
Removing	101

Installing	101
Final steps	102
CPI Unit	103
Prerequisites	103
Preparation	103
Removing	103
Installing	103
Final steps	104
Balancing unit	105
Prerequisites	105
Preparation	105
Removing	106
Installing	106
Final steps	106
Balancing Sensors	107
Prerequisites	107
Preparation	107
Removing	108
Installing	108
Final steps	108

5 **Gantry front side** 110

Overview	110
Heat Exchanger	111
Prerequisites	111
Preparation	111
Removing	112
Installing	112
Final steps	112
Fan for the heat exchanger	114
Prerequisites	114
Preparation	114
Removing	115
Installing	116
Final steps	116
Gantry front door	118
Prerequisites	118
Preparation	118
Removing	119
Installing	120
Final steps	121
Gantry Front Panels	122
Prerequisites	122
Preparation	122
Removing	123

Installing	123
Final steps.	123
Static light markers Sensation 10/16/Cardiac/40/64/Cardiac 64	124
Prerequisites	124
Preparation	124
Removing	124
Installing	125
Final steps.	125
Adjustments	126
Display Panel	129
Prerequisites	129
Preparation	129
Removing	130
Installing	130
Final steps.	130
6 DMS parts Sensation 40/64/Cardiac 64/Open	131
Overview.	131
Modules	136
Prerequisites	136
Preliminary steps.	136
Removing	137
Installing	140
Final steps.	140
Complete DMS	142
Prerequisites	142
Preliminary Steps	142
Removing	144
Installing	147
Final Steps	148
DAS Controller	149
Prerequisites	149
Preliminary steps.	149
Removing	149
Installing	151
Final steps.	151
DMS backplanes.	152
Prerequisites	152
Preliminary steps.	152
Removing	152
Installing	153
Final steps.	154
Reference monitor board	155
Prerequisites	155
Preliminary steps.	155

Removing	155
Installing	156
Final steps	156
Complete UHR assembly	157
Prerequisites	157
Preliminary steps	157
Removing	158
Installing	158
Final steps	158
Complete UHR-plus assembly	160
Prerequisites	160
Preliminary steps	160
Removing	161
Installation	161
Final steps	161
UHR motor	164
Prerequisites	164
Preliminary steps	164
Removing	165
Installing	165
Final steps	165
UHR-plus motor.	166
Prerequisites	166
Preliminary steps	166
Removing	167
Installation	167
Final steps	168
UHR limit switch	169
Prerequisites	169
Preliminary steps	169
Removing	170
Installing	170
Final steps	170
UHR-plus limit switch.	171
Prerequisites	171
Preliminary steps	171
Removing	172
Installation	172
Final steps	172
UHR control board.	174
Prerequisites	174
Preliminary steps	174
Removing	175
Installing	175
Final steps	175
UHR-plus control board.	176

Prerequisites	176
Preliminary steps.	176
Removing	177
Installation.	177
Final steps.	178
DMS power supply	179
Prerequisites	179
Preliminary steps.	179
Removing	180
Installing	181
Final steps.	182
Fans	183
Prerequisites	183
Preliminary steps.	183
Removing	184
Installing	184
Final steps.	184
Temperature switch.	185
Prerequisites	185
Preliminary steps.	185
Removing	185
Installing	186
Final steps.	186
Appendix.	187
Channel list	187
Replacement of a center module	188
Overview of all used screws	189
Adjustment of UHR limit switch	190
Adjustment of UHR guide pins	190
Z-position check of modules	191
Firmware download for DAS controller	194

7 C-Box components 195

Overview.	195
C-Box	196
Prerequisites	196
Preliminary steps	196
Removing	197
Installing	198
Final steps.	198
FOC D514.	200
Prerequisites	200
Preliminary steps	200
Removing	200
Installing	201
Final steps.	201

ADU D513	203
Prerequisites	203
Preliminary steps	203
Removing.	203
Installing.	204
Final steps	204
DPU D540	205
Prerequisites	205
Preliminary steps	205
Removing.	205
Installing.	206
Final steps	206
8 Gantry right stand	207
Overview	207
Dehumidifier	208
Prerequisites	208
Preparation	208
Removing.	209
Installing.	209
Final steps	209
Tilt Spindle.	210
Prerequisites	210
Preparation	210
Removing.	212
Installing.	213
Final steps	215
GPC Board	217
Prerequisites	217
Preparation	217
Removing.	218
Installing.	218
Final steps	218
3-way valve	220
Prerequisites	220
Preparation	220
Removing.	221
Installing.	221
Final steps	221
9 Additional replaceable parts	222
Overview	222
Rotating light markers	223
Prerequisites	223
Preliminary Steps	223

Removing	223
Installing	225
Final steps	225
LPS Board (D522)	228
Prerequisites	228
Preliminary Steps	228
Removing	229
Installing	230
Final steps	230
Plexi Strip	233
Prerequisites	233
Preliminary Steps	233
Installing	233
Final steps	234
10 Gantry rear side	235
Overview	235
Receiver ring	236
Prerequisites	236
Preparation	236
Removing	237
Installing	241
Final steps	244
Slip ring	245
Prerequisites	245
Preparation	245
Removing	246
Installing	247
Final steps	248
Transmitter and receiver board	250
Prerequisites	250
Preparation	250
Removing	251
Installing	252
Final steps	254
D344 NOISE_CAN board	255
Prerequisites	255
Preparation	255
Removing	256
Installing	256
Final steps	256
D346 NOISE_FILT board	257
Prerequisites	257
Preparation	257
Removing	258

Installing	258
Final steps	258
Brake assembly	259
Prerequisites	259
Preparation	259
Removing	260
Installing	260
Final steps	260
11 Index	262

Rev.	Date	Technical Writer/ Prod- uct Engineer	ECN	Action
01	05/06	J. Deyoe	C0009242	Initial release.
02	02/11	J. Deyoe S. MacAdams	554797	Added chapters: - Akron Tube - CT Tube R&R Procedure - Data Measurement System R&R - High-voltage Tank - Generator Electronics Box

Safety information

Only qualified and system-trained service staff is allowed to perform Biograph system installation, service , maintenance and quality assurance. Ensure that the most recent version of the service documentation is available.

To avoid any risk of injury to persons and/or damage to the system, please read and observe the Biograph Safety Procedure and Biograph Safety Report documents.

Information about the possibilities for switching off the gantry power

- Switching off the gantry power by switching the system to **COMP/ON** at the control box.
 - **Be aware that power is still supplied to rotating parts of the gantry (e.g DMS) after switching to COMP/ON! Switch off this power with circuit breaker F9 in the PDC.**
 - **Be aware that power is still supplied to the PET part of the gantry.**
- Switching off the gantry power by using the **SYSTEM/OFF** switch at the control box.
 - **Be aware that power is still supplied to the rotating part of the gantry (e.g DMS) after activating the System OFF button. Switch off this power with circuit breaker F9 in the PDC**
 - **Be aware that power is still supplied to the PET part of the gantry.**
- Switching off gantry power by using the **SYSTEM/OFF** switch at the control box. Additionally, use the on-site **ON/OFF** switch to completely switch off system power and also switch off power to the PET gantry.
 - **In this case, system power is fully switched off .**

NOTE

Read and observe the safety information in the Biograph Safety Procedure and Biograph Safety Report documents before performing service work in the gantry. If the gantry power is switched off for any service work, always secure against accidental switch on.

Tools

Normally, a standard tool kit is required for performing service as described in this document. If special tools are needed, they are described in the relevant replacement description under "Tools" .

A 25 Nm to 130 Nm torque wrench is required with the following attachments:

Universal wrench (adapter) for 1/2" --> 1/4" or extension 355 mm 1/4"

NOTE

Torque wrenches must be checked for accuracy on a regular basis.

HV Tank

HV transformer E604

Prerequisites

Tools and auxiliary equipment

- Standard service kit
- The following tools are required. They are included with the part:
 - One strap with tensioner
 - One pad for the tabletop
 - One set of new bolts
 - Lifting brackets to help with the replacement of the HV tank
- Tools required for HV tank replacement:
 - Lifting device kit: part number 10244855
Contents: PHS Vertical Motor Power Interface, Lifting device, instructions

Time and manpower

- 1 person
- 1 hour

Safety



Avoid accident and injury or damage of parts.

Risk of accident and injury!

⇒ **Read and observe the safety information contained in the Biograph Safety Documents.**

Preliminary steps

1. Move patient table out.
2. Remove pallet from pedestal
3. Switch **OFF** the system at the control panel.
4. Open the gantry front door and disable rotation with switch S 301.
5. **Switch off F9 in the PDC.** Check that the LED's on DPU and DMS are off.
6. **Open the fuse holder F100 in the PDC.** Be aware that these fuses are still supplied with mains power.



7. Remove the plastic center panel from the gantry front side. Note the position of the large washers.
8. Move the HV tank to the **12 o'clock** position.

Removing



1. Before working on the XRS
 - ⇒ **Check that the DCC voltage is discharged** measure between D440_X1, X2 of the XRS.
2. Disconnect the following cables:



Fig. 1: Connection in HV tank

- HV transformer
 - Two HV cables to the tube
 - D432_X5, X6.
 - D432_X1, X2, X3, X4. (X2,X3 in Open)
 - E604_X3
 - Ground cable to the tube (top of tank)
 - Ground cable to the gantry grounding plate (bottom of tank)
 - Generator electronics
 - E603_X1
 - Cable E603_X2 at the generator electronics box (AKRON tube only)
 - Cable E603_X21 at the generator electronics box (STRATON tube only)
3. Cut tie wraps as needed to allow access to the tank bolts and tank removal

NOTE

Mark the positions of the tie wraps so they can be put back in the correct locations.

4. If a cable holding bracket is installed on the HV tank (Fig. 9 / p. 23) Note its orientation and remove it from the installed HV tank.
5. Disengage the support bracket

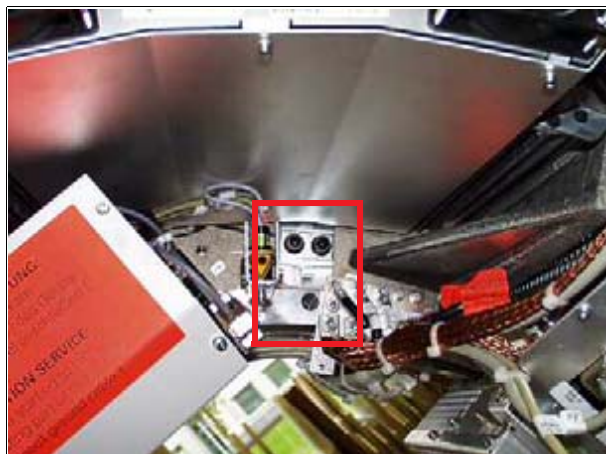


Fig. 2: Illustration of support bracket for Biograph 64/40/16

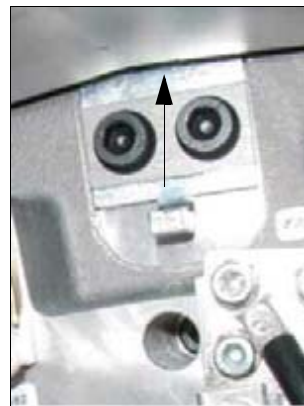


Fig. 3: Illustration of support bracket for Biograph 64/40/16

- Remove as many cable ties as necessary.
- Loosen the 2 hexagonal bolts (3 bolts on 40/64) for the support bracket and push the tab away from the center of rotation (arrow) to disengage the bracket from the tank. Tighten the bolts enough to keep them in position.



6. Move the HV tank to the **6 o'clock** position.
7. **Secure the rotating gantry with the safety bolt.**

NOTE

Consult the instructions that come with the PHS Vertical Motor Service Interface if needed.

8. Connect the PHS Vertical Motor Service Interface to the bed.

 CAUTION

Use care when raising or lowering the PHS with the Vertical Motor Service Interface. The limit switches are monitored when the system is off.

Limit switches may be damaged if PHS is raised all the way up.

- ⇒ Only Raise PHS high enough to lift and extract the HV Tank

9. Mount the lifting devices

- Install the lifting device on the PHS.
- Remove the cover from the transport box and place it next to the PHS. This will hold the old HV tank.
- Attach the lifting brackets provided with the HV tank. The cut-outs go to the outside to allow access to the tank mounting bolts (item 1). See fig. 4
- Move the table support and then the tabletop into the gantry tunnel.
- Secure the HV tank to the tabletop with the strap. Run the strap through the strap cut-outs (item 2) to ensure it is at the correct lifting position. See fig. 5

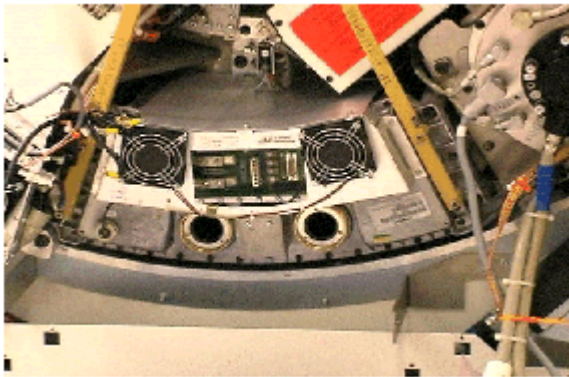


Fig. 4: Attaching Bracket

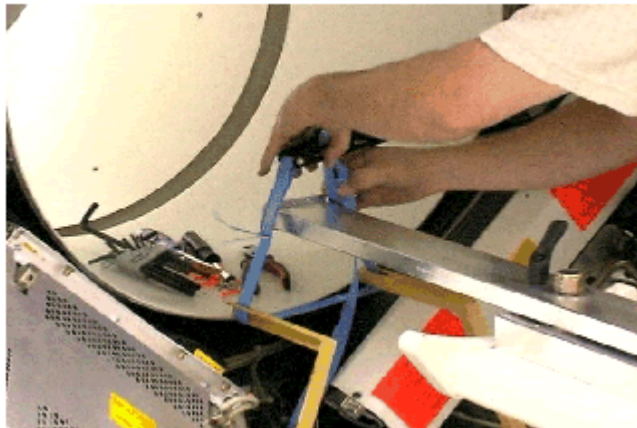


Fig. 5: Strapping Lifting Device

10. Remove the 4 hexagonal bolts securing the HV tank and move the tabletop and HV tank out of the gantry. Rotate the lifting device and lower the couch until the HV tank is positioned on the holder.



Fig. 6: Swing HV Tank



Fig. 7: Lowering HV Tank

11. Remove the strap from the HV tank lifting device.
12. Use the screws to secure the HV tank to the shipping holder, then remove the old HV tank from between the gantry and the patient table.

Installing

1. Remove the lifting brackets from the old HV tank and attach them to the new HV tank.
2. Place the new HV tank and next to the patient table.
3. Remove the shipping brackets



4. Secure the HV tank to the lifting device with the strap, rotate into place, and move the tabletop together with the HV tank into the gantry until it is positioned correctly.
5. Install the 4 hexagonal bolts and tighten them using the torque wrench. **Torque setting: 108 Nm**
6. Remove the lifting brackets and strap from the HV tank.
7. Unlock the safety bolt and position the HV tank at 12o'clock to facilitate support bracket adjustment

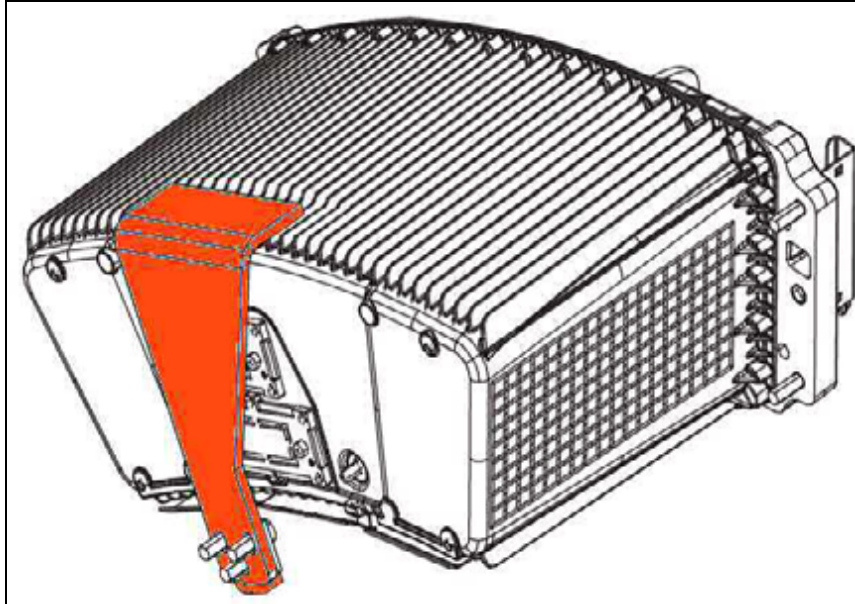


Fig. 8: Back of tank showing support bracket (Biograph 64/40/16)

- Loosen the hexagonal bolts holding the support bracket.
 - Pull the support bracket toward the center of rotation until it supports the outside of the tank. Use a large flat blade screwdriver between the tab and the plate to ensure it is set, but do not use excessive force.
 - Tighten the hexagonal bolts using the torque wrench. **Torque setting: 90 Nm**
8. Move gantry to 3o'clock position

9. Mount the HV cable holding bracket (if used)

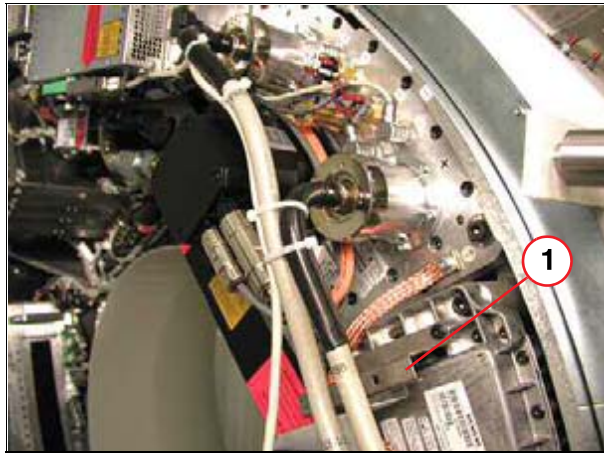


Fig. 9: HV cable holding bracket

Pos. 1 Bracket

- Reinstall HV cable holding bracket (item 1) to the tank if used **Torque setting: 24Nm**.



10. Connect all cables according to the original configuration. Pay special attention to E603_X21. Ensure that it is not rotated by 180 degrees and then connected.

Install D432_X1, X2, X3 and X4 (Sensation Open D432_X2, X3) -> Connector block -> cable -> washer -> nut with a torque of 4.8 Nm. Don't mix up cable X1 -> X4. It will short circuit the UDC voltage.

Connect the ground connections with a **torque of 10Nm**

Final steps

Mechanical

1. After finishing all mechanical work, switch rotation switch S301 to "1".
2. Replace all previously removed covers, and reattach the pallet to the PHS.
3. Switch on circuit breaker **F9** and close the fuse holder **F100** in the PDC.
4. Switch on the system.

Software

1. Continue with tune-up steps as described in guided tour **Service-> Tuneup-> Installations**

Refer to Tune-Up Instructions MI12-023.817.01.xx.xx.

24 XRS parts, Sensation 10/16/40/64/Cardiac/Cardiac64

Testing

1. Perform Quality tests.

Generator Electronic Box

Generator electronic box E603

Prerequisites

Tools and auxiliary equipment

- Standard service kit.
- Ratchet handle, extension, universal joint to open the upper left bolt of the electronic box.
- Torque wrench 1-5 Nm
- **Cable W562 (71-09-254) may be required** . (see text)
- The following tools are included with the part:
 - One strap with tensioner
 - One pad for the tabletop
 - One set of new screws
 - Lifting device to help with the replacement of the generator electronic box.
- Tools required for Gen Box replacement:
 - Lifting device kit: part number 10244855
Contents: PHS Vertical Motor Power Interface, Lifting device, instructions

Time and manpower

- 1 person
- 2 hours (mechanical)

Safety



Avoid accident and injury or damage of parts.

Risk of accident and injury!

⇒ **Read and observe the safety information contained in the Biograph Safety Documents.**

Preliminary steps

1. Move the patient table away from the CT gantry.
2. Remove the PHS pallet from the pedestal.
3. Switch **OFF** the system at the control box.

4. Open the gantry front door and disable rotation with switch S 301.
5. **Switch off F9 in the PDC.** Check that the LED's of the DPU and DMS go off.
6. **Open the fuse holder F100 in the PDC.** Be aware that these fuses are still supplied with mains power.
7. Remove the plastic center panel from the gantry front side. Note the location of the large washers.
8. Move the generator electronics box into the **3 o'clock** position.



Removing



1. Remove the cover from the UDC cable and measure between D440_X1, X2.
⇒ **Measure UDC voltage to make sure that there is no voltage present!**
2. Disconnect the following cables at the generator electronics box
 - E603_X1
 - Ground cable E603_X3
 - E603 D400_X3, X5
 - E603_X21(**STRATON ONLY**)
3. Remove E603_X2 (**AKRON tube only**).



Fig. 10: Releasing X2

- Disconnect E603_X2 (AKRON tube only) by inserting a flat blade screwdriver into the lock and pressing on it to keep the plastic lock away from the plug while pulling the plug.

4. Disconnect the following cables at the HV tank



Fig. 11: Connection in HV tank

- Disconnect at the HV tank
 - D432_X5, X6
 - E604_X3
 - D432_X1, X2, X3, X4.
5. Remove as many cable ties as necessary then move the generator electronic box into the **9 o'clock** position.

28 XRS parts, Sensation 10/16/40/64/Cardiac/Cardiac64

6. Disconnect the 48V DPU supply cable and /or the inhibit cable.

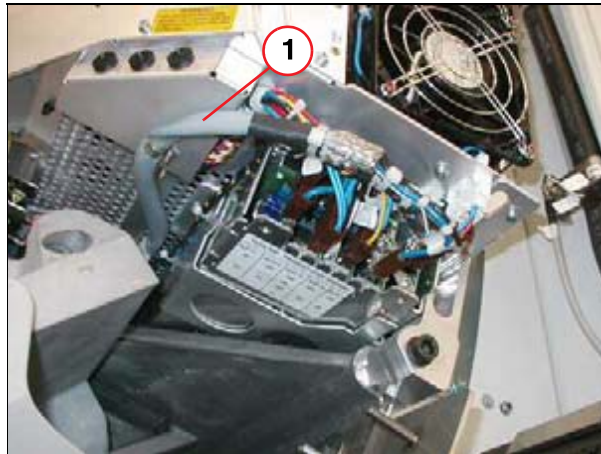


Fig. 12: Variant 1 supply, with old style power connectors
Pos. 1 Inhibit cable

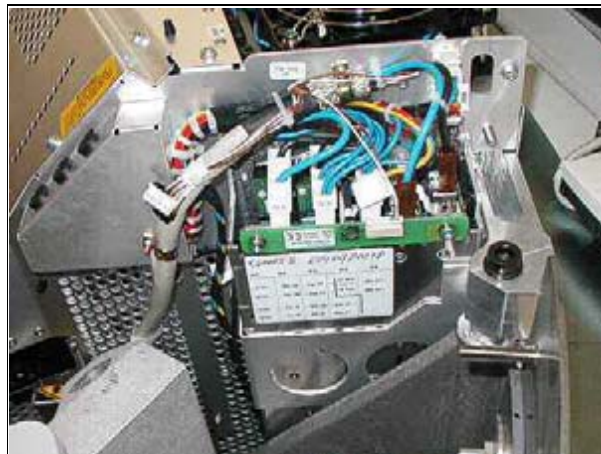


Fig. 13: Variant 2 power supply with new style power connectors

- Disconnect the inhibit cable from the XRS power supply and remove ties and clamp to release the cable (item 1). There are 3 variants of this power supply.
 - ⇒ **Variant 1**, (Early Sensation 16/ Cardiac) 4 plugs connected directly to the power supply. The black plugs have a lock in the center of the plug (this makes the procedure difficult), the white ones have one at each end (which is easier). Currently this cable is difficult to remove without damaging the black plugs, however the replacement E-box should have a variant 2 power supply installed which only uses the white INHIP +15v plug. Cable W562 (71-09-254) can be ordered as well.
 - ⇒ **Variant 2**. (Later Sensation 16/Cardiac) A single plug connected to an adapter board mounted on the supply. Remove the single inhibit plug (INHIP +15v)
 - ⇒ **Variant 3**. (Sensation 40/64/Cardiac 64) The inhibit plugs are not used for this model.
- Disconnect the wires E606 48V+ and E606 48V- , **note the positions as they are different on variant 3** (Sensation 40/64/Cardiac 64)



7. Move the generator electronic box into the **6 o'clock** position.
8. **Secure the rotating gantry with the safety bolt.**
9. Remove D440_X1, X2 (10 mm nut, UDC cable).
10. Install the PHS Vertical Motor Service Interface.

NOTE

The instructions for installing and using the PHS Vertical Motor Service interface are included with the device.

 CAUTION

Use care when raising or lowering the PHS with the Vertical Motor Service Interface.

The PHS Vertical limit switches may be damaged if the PHS is raised to the maximum upper position.

⇒ **ONLY RAISE** the PHS high enough to lift and extract the Electronics Box!

11. Install the Biograph lifting device on the PHS.

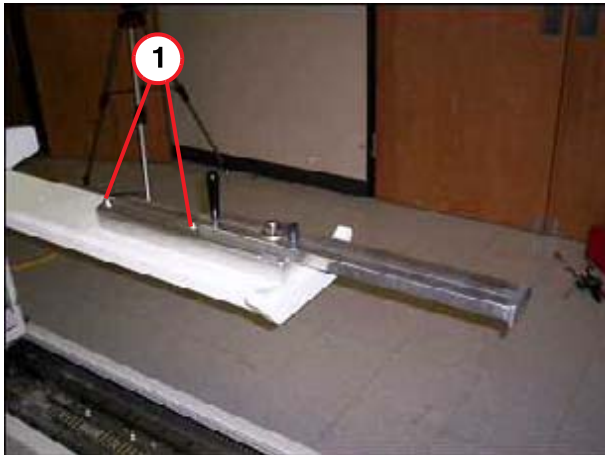


Fig. 14: Biograph Lifting device attached to PHS vertical support

Pos. 1 Bolts

12. Mounting the lifting devices

- Remove the cover from the transport box. Remove the metal bracket from it and place it next to the PHS. This will hold the old electronic box.

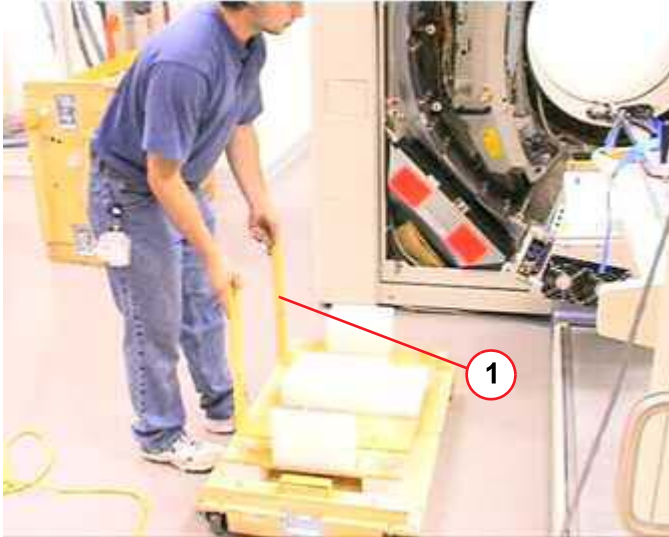


Fig. 15: Metal Bracket by PHS

Pos. 1 Metal Bracket

- Attach the lifting device provided to the electronic box with 4x M6 bolts.

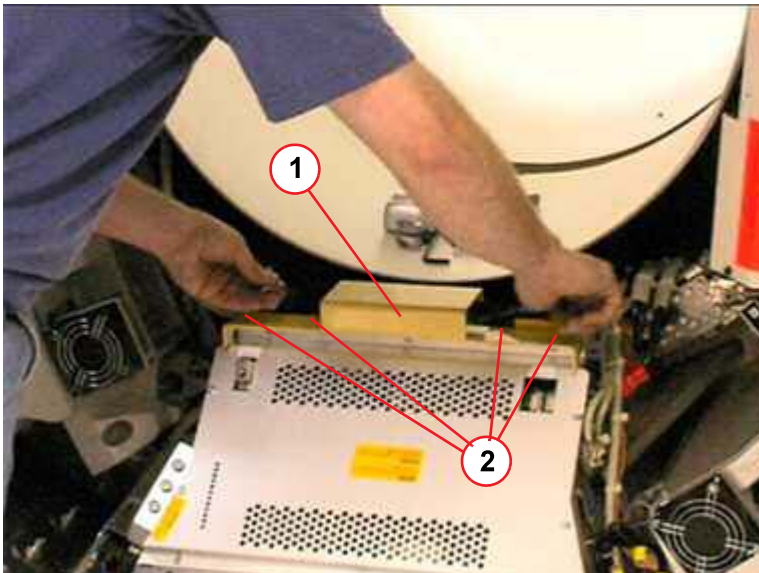


Fig. 16: Attaching Lift Device to Electronic Box with 4 bolts

Pos. 1 Lift Device

Pos. 2 4 Mounting Bolts

- Move the tabletop into the gantry tunnel.
- Secure the electronic box to the tabletop with the strap.

13. Remove the 4 M10 hexagonal bolts from the electronic box. Use a flexible coupling on the top left bolt. Depending on the size of the tools used, you may have to remove the light marker to obtain proper access to this bolt.
14. Move the tabletop together with the generator electronic box out of the gantry and rotate the lifting device until the generator electronic box is positioned on the transport holder.



Fig. 17: Rotate Biograph Lifting Device with Old Electronic box

15. Mount the metal bracket and secure it to its holder.
 - Carefully insert the metal bracket and bolt it to the transport holder.
 - Secure the generator electronic box to the holder using the 4 screws removed from the gantry.
16. Remove the holder with the old generator electronic box from the gantry area.

Installing

1. Remove the lifting device, attach it to the new generator electronic box, and place the new generator electronic box next to the PHS.
2. Remove the transport bolts from the new generator electronic box and remove the metal bracket.
3. Use the strap to attach the lifting device to the generator electronic box. Rotate the lifting device, and move it into the gantry until it is in the correct position.
 - ⇒ **Be careful not to damage the left light marker**
4. Ensure that the generator electronic box is fully inserted and all locating pins are engaged. Then secure the generator electronic box with the new set of screws included with the part.



5. Remove the lifting device from the generator electronic box.

6. Tighten all four M10 bolts with the torque wrench. Torque setting: **108 Nm**.

7. Reinstall the light marker if removed.



8. Connect D440_X1,X2 (UDC cable). **Secure with 10mm nut (Connector block->cable -> washer -> nut) with a torque of 4.8 Nm.** Reinstall the cover for the UDC cable.

9. Unlock the gantry and move to 9 o'clock position

10. Connect and secure W562 Inhibit cables. If the new supply is variant 2 connect the **INHIP +15V** cable to the adapter board and tie wrap the other 3 back to the cable harness. Refer to [\(MI12-023.841.04 / Installing\)](#) for more details.

⇒ The inhibit wire is not used on Sensation 40/64/Cardiac64.

11. Connect wires 48v+ and E606 48v- at the power supply

12. Move gantry to 3 o'clock position.

13. Connect all cables on the electronics box of the generator **Secure the ground strap (E603_X3) with a torque of 4.8 Nm.** Pay special attention to E603_X21 as it can be connected backward.



14. Connect all cables on the HV tank.

Install D432_X1, X2, X3 and X4-> Connector block -> cable -> washer -> nut with a torque of 4.8 Nm. Don't mix up cable X1 -> X4. It will short circuit the UDC voltage.

15. Secure all cables.

Final steps

Mechanical

1. After finishing all mechanical work, switch rotation switch S301 to "1".
2. Replace all previously removed covers and reattach pallet to PHS.
3. Switch on circuit breaker **F9** and close the fuse holder **F100** in the PDC.
4. Close the gantry door.
5. Switch on the system.

Software

1. Reload the actual XRS Firmware:
 - Go to the Service SW function **Control**.
 - Go to **Flashload**.
 - Select the most recent **XRS** firmware in the XRS pull-down menu (make sure it is the most recent version; if not, copy the FW from the FW CD to :\\Somaris\\config\\Control-SystemFlash).
 - Press < **Go** > to download the FW.

2. Continue with tune-up steps as described in guided tour **Service-> Tuneup-> Installations**

Refer to Tune-Up Instructions ([MI12-023.817.01 / Installations](#)) (MI12-023.817.01)

Testing

1. Perform Quality tests.
2. Back up the system settings

Akron Tube

Overview

The figure below shows the location of the XRS parts in the rotating gantry for AKRON system.

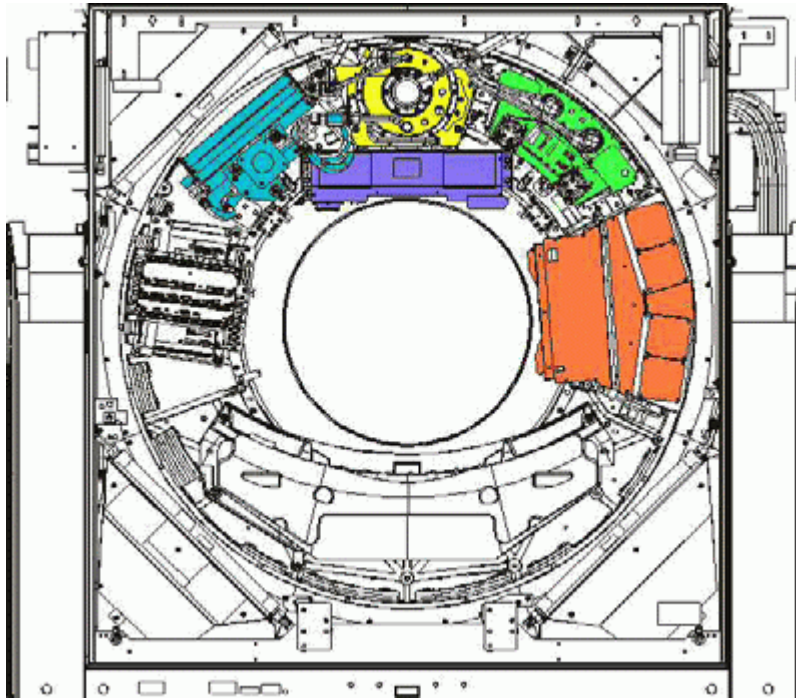


Fig. 18: Gantry, front

Prerequisites

Tools and auxiliary equipment

- Standard tool kit
- Adjustment tool (part. no. 27 92 658 K1081) for adjusting the membrane position of tube cooling system
- Tape measure
- The following tools are included with the part:

NOTE

These tools are included with the Akron tube however and additional kit will have to be ordered that includes the Biograph lifting device and the Vertical Motor Service Interface.

- One strap with tensioner
- One pad for the tabletop
- One set of new screws
- Tools required for Akron tube replacement:
 - Lifting device kit: part number 10244855 Contents: PHS Vertical Motor Power Interface, Lifting device, instructions

Time and manpower

- 1 Person
- 1 hour (Mechanical)

Safety



WARNING

Avoid accident and injury or damage of parts.

Risk of accident and injury!

⇒ **Read and observe the safety information contained in the Biograph Safety Documents.**

Preliminary Steps

1. Move patient table out
2. Remove pallet from pedestal
3. Switch off the system at the control box.
4. Open the gantry front door and disable rotation with switch S 301.
5. Switch off circuit breaker F9 in the PDC and check to ensure that the LED's on the DPU go off.
6. Remove the plastic center panel from the gantry front side. Note the location of the large washers.
7. Turn the gantry by hand until the tube is in 12 o'clock position.

Removing

NOTE

Before each tube replacement, you must check the expansion membrane of the tube cooling system. If the expansion membrane is contracted to its mechanical end then both components, the tube and the tube cooling unit, have to be replaced. This is the only way to guarantee that the oil-filled cooling circuit is filled properly with oil again (no air bubbles). Otherwise, the tube will be damaged and will have to be replaced prematurely.

1. Disconnect cables

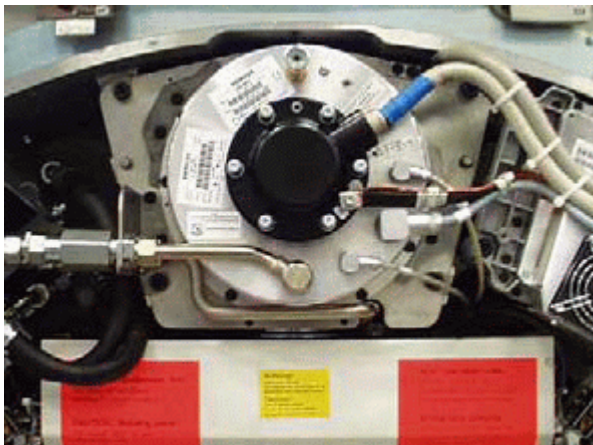


Fig. 19: Tube Connections

- Disconnect the following cable and connectors from the X-ray tube and tank
 - Cables (X4, X5, X6)
 - Ground strap (X2)
 - High voltage connectors
 - Move the tube to the 6 o'clock position.
2. Move the tube to the 6 o'clock position.
 3. Secure the rotation with the safety bolt

4. Open the oil couplings.

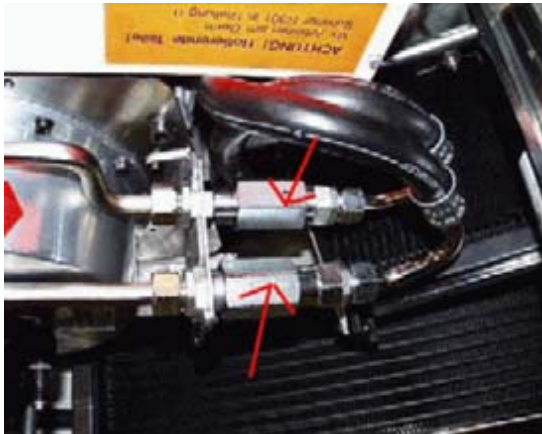


Fig. 20: Oil Couplings

- Open the cooling hoses between the cooling component (E590) and the X-ray tube. To loosen, the couplings have to be turned towards each other, i.e. in opposite directions.
 - Dry the couplings to prevent oil from dripping into the gantry
5. Remove the cover from the transport device of the new X-ray tube and place it next to the PHS. This cover will be used as the transport frame for the X-ray tube being replaced.
 6. Connect the PHS Vertical Motor Service Interface to the bed.

NOTE

Consult the instructions that come with the PHS Vertical Motor Service Interface if needed.

CAUTION

Use care when raising or lowering the PHS with the Vertical Motor Service Interface. The limit switches are not monitored when the system is off.

Limit switches may be damaged if PHS is raised all the way up.

⇒ **Only raise PHS high enough to lift and extract Akron Tube.**

7. Install the lifting device.

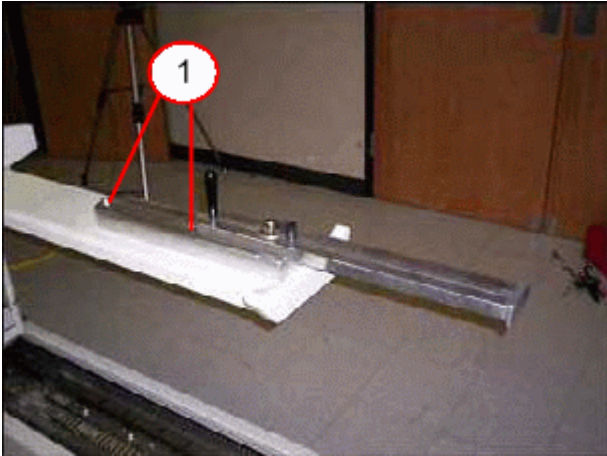


Fig. 21: Biograph Lifting device attached to PHS vertical support

8. Install the mounting bracket for lifting out the tube.

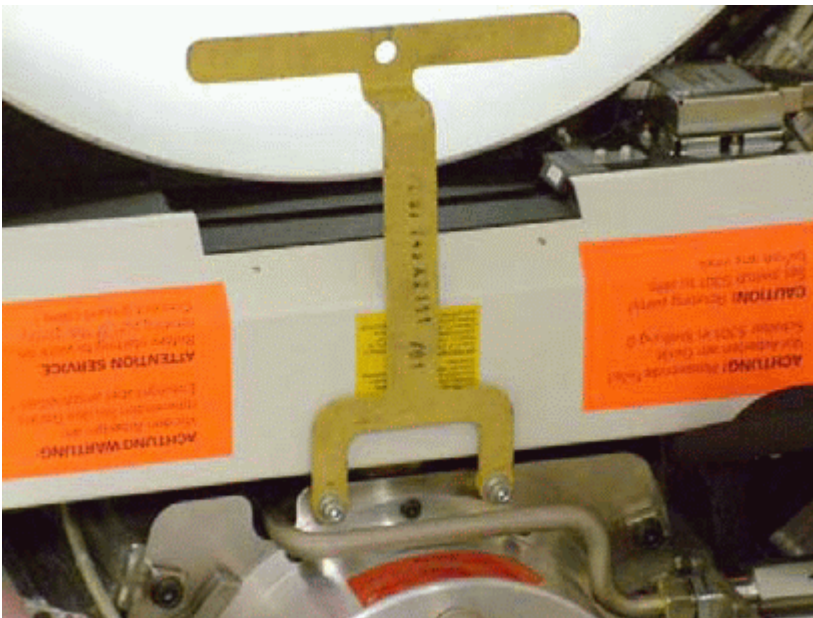


Fig. 22: Mounting bracket attachment for lifting the tube

9. Use an 8mm hexagonal key remove the four M10 hexagonal mounting bolts for the tube.

10. Rotate the lifting device and move the bed forward into position to attach the straps.

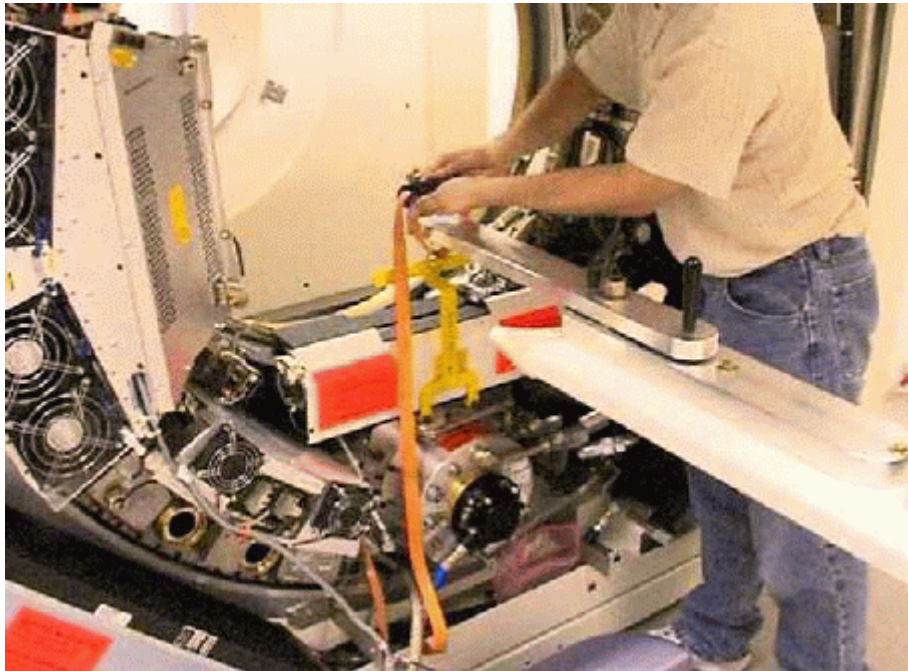


Fig. 23: Attaching the strap for lifting the tube.

11. Once the tube is securely strapped for transport, move the bed out of the gantry.

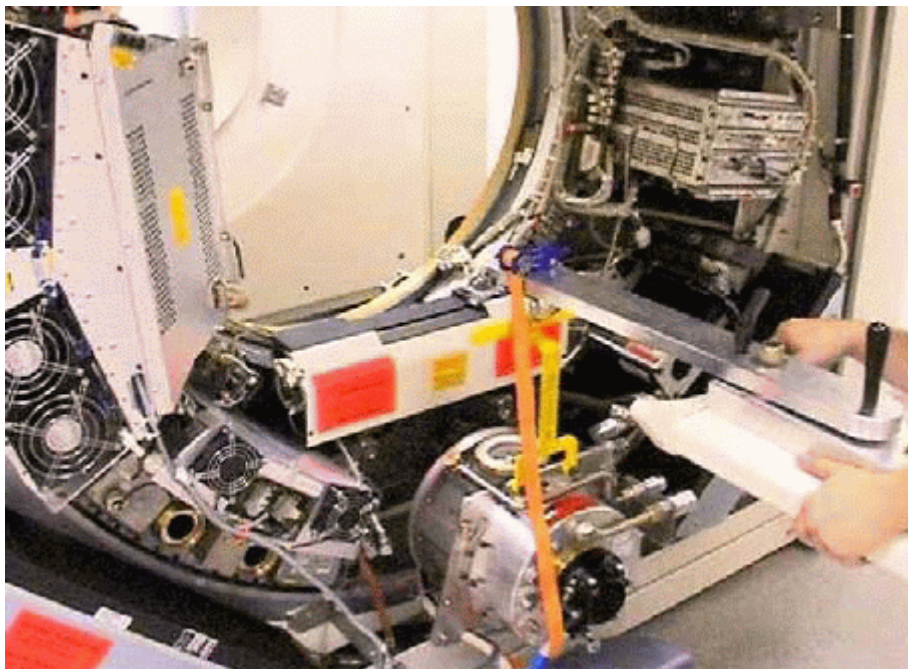


Fig. 24: Moving the bed out.

12. Once clear of the gantry, rotate the lifting device until the X-ray tube is suspended above the transport device.

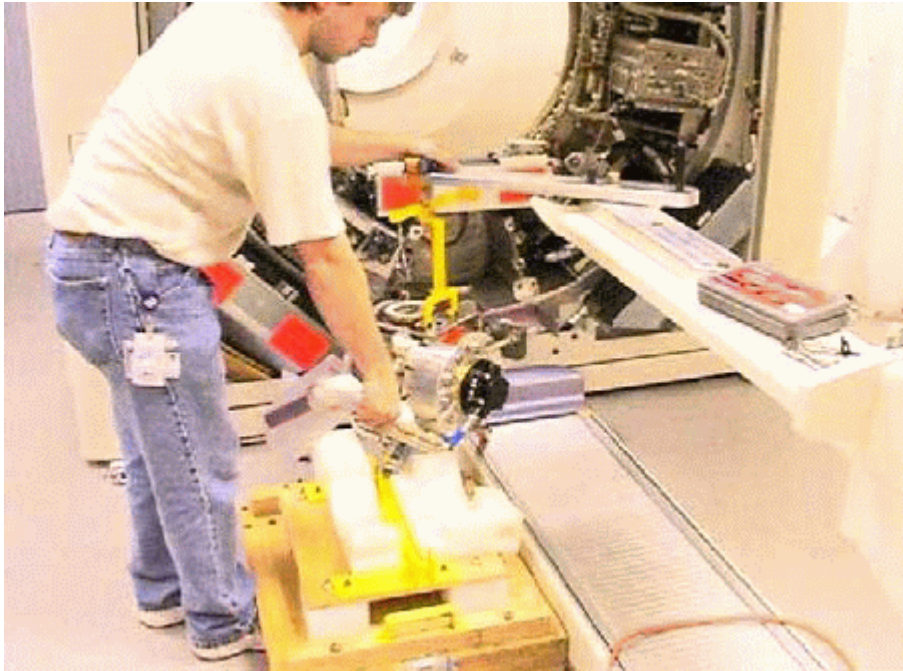


Fig. 25: Swinging the tube for positioning.

13. Lower the bed until the X-ray tube is seated inside the transport device.
14. Remove the strap and secure the X-ray tube to the transport device.
15. Remove the old tube from between the table and the gantry.

Installing

1. Remove the handle from the old tube and attach it and the strapping to the new tube.
2. Remove the oil expansion bottle and X-ray window cover from the new tube assembly.
3. Install the new X-ray tube by reversing the order for removing the old tube.
4. Install the four kitted M10 hexagonal bolts (supplied) and tighten in a crosswise pattern to a torque value of 90 Nm.
5. Connect all cables according the original configuration. (Secure cables with cable ties.)
 - ⇒ Install the high-voltage connectors using new silicon washers, silicon rings, and liquid silicon (supplied).

Final Steps

Adjustment of the expansion membrane

1. Switch on circuit breaker F9 in the PDC.
2. Turn system on at control box.
3. Read out the current tube temperature in the table TA0. In Service select Control -> Table load/modify -> select table TA0-> read Actual tube oil temperature.
4. Determine the membrane position according to the following temperature table (X value in mm).

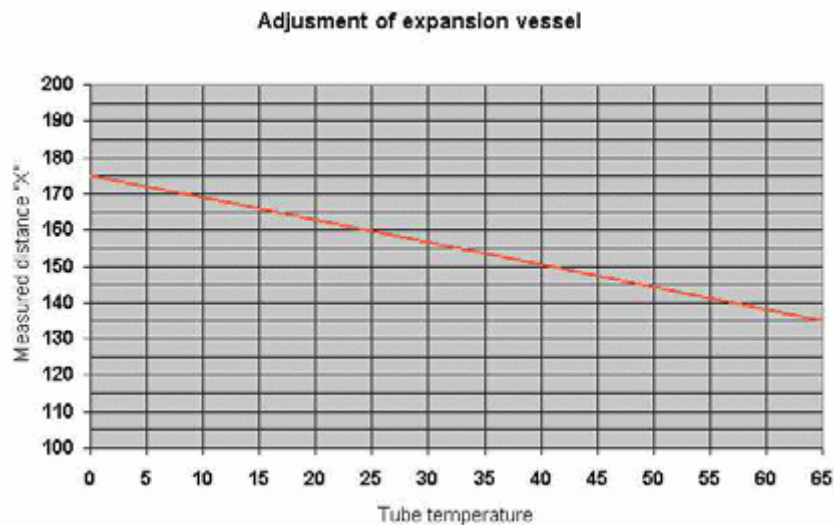


Fig. 26: Tube change (Dura)

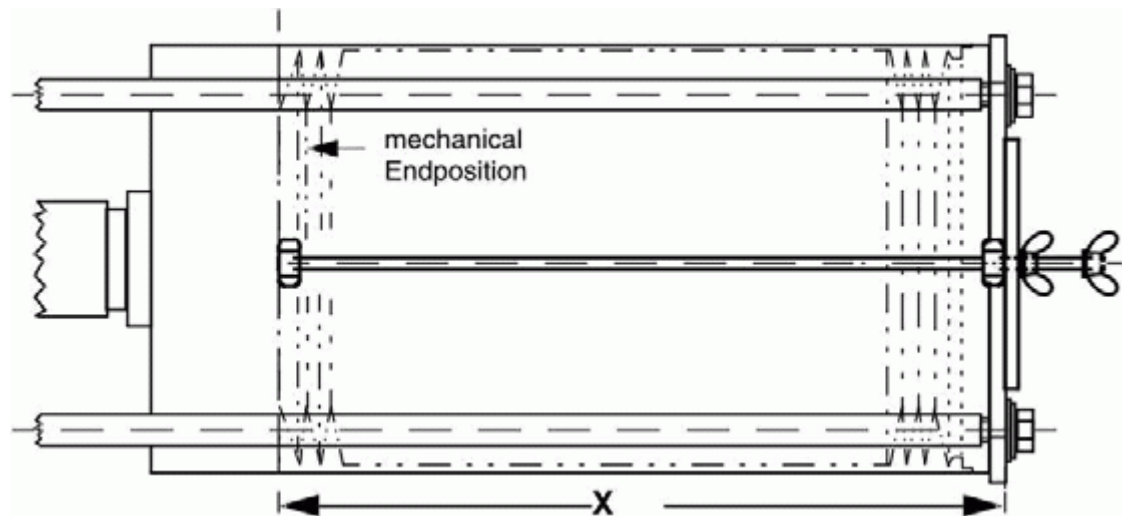


Fig. 27: Membrane position

5. If an adjustment is necessary, switch off the system at the control box and switch off circuit breaker F9 in the PDC. Then check that the LED's on the DPU go off.

NOTE

If no adjustment is necessary, go to 11.

6. Install the adjustment device.
7. Install the expansion vessel to the tube coupling.
8. Adjust the membrane position to the corresponding X value using a measurement device.
9. If the position is correct, remove the expansion vessel from the tube coupling, and remove the adjustment device.
10. After finishing all mechanical work, switch rotation switch S301 to "1".
11. Unlock safety bolt for gantry rotation and close gantry.

Mechanical

1. Re-attach the pallet to the PHS.
2. Re-attach the Gantry Center Panel.
3. Close the gantry cover and install the plexi-ring.
4. Switch on circuit breaker F9 in the PDC.
5. Turn system on at the control box.
6. Remember to install expansion vessel, X-ray window cover, and protective caps to the old tube. Return lifting handle, table pad and strap to the box before sealing it.

Software

1. When exchanging the X-ray tube assembly it is important to store the tube history file on the enclosed floppy disk and send it back along with the defective tube assembly. The applicable reimbursement will be paid only if the tube history file was correctly sent back with the defective tube assembly.
2. Save copy of tube history to floppy (instructions are included with the tube).
3. Go to Service --> Tune-up --> Balance Gantry
Refer to Tune-Up Instructions.
4. Go to Service --> Tune-up --> Installations
Refer to Tune-Up Instructions.
5. Select Replace Tube and give the reason for return using the pull down menu.

- Enter a comment if necessary.
 - Enter the new serial number and revision level of the tube from the label affixed to the tube assembly.
 - Press GO after entering the parameters. A new tube history is generated, and the tube serial number is entered into the configuration.
 - You can view the old tube history by opening an Additional Window and selecting Reports --> Tube.
6. Continue following the tune-up steps in the guided tour.

Testing

1. Run Constancy test.
2. Perform a backup of settings.

Labeling

There is a second label of the X-ray tube is included in tube delivery.

In some countries, Regulatory Affairs requires a second label for the X-ray tube to be located outside the gantry.

To attach them, refer to the location of the labels as described in the Operator Manual (System Owner Manual / location of labels).

Straton Tube

Overview

The figure below shows the location of the XRS parts in the STRATON gantry.

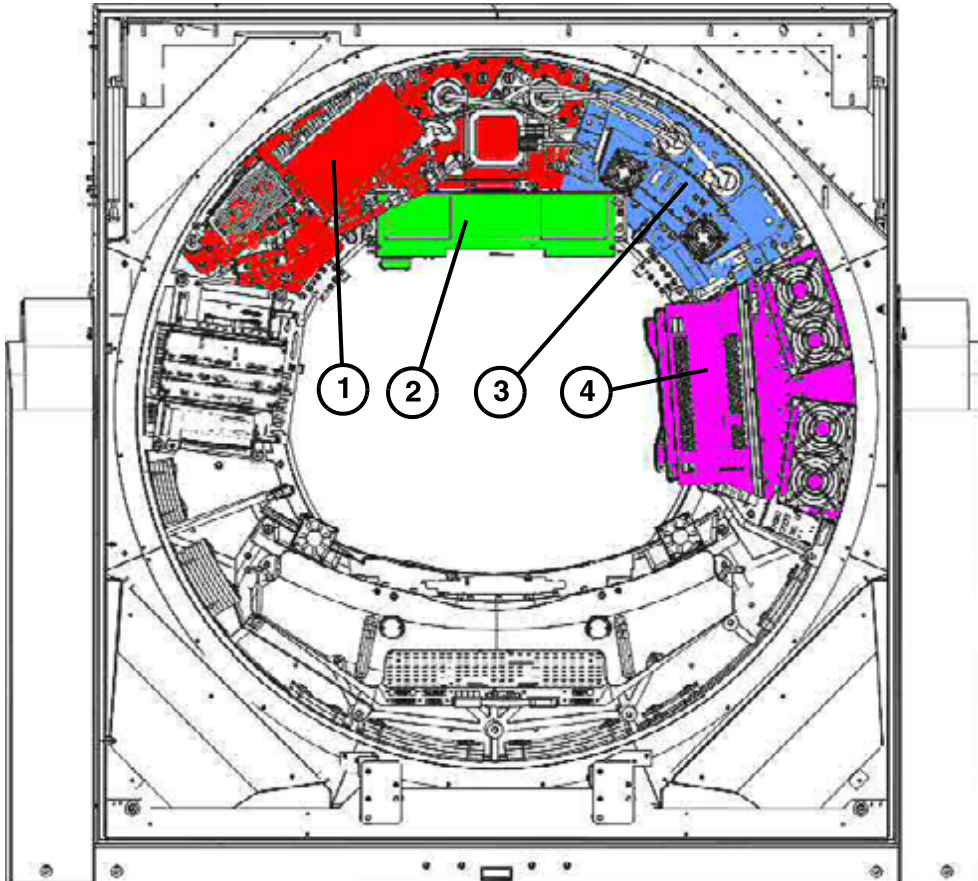


Fig. 28: Front of gantry

- | | |
|--------|--------------------------------|
| Pos. 1 | Straton Tube |
| Pos. 2 | Tube Collimator |
| Pos. 3 | HV Transformer E604 |
| Pos. 4 | Generator Electronics Box E603 |

Prerequisites

Tools and auxiliary equipment

- Standard tool kit

- The following tools are included with the part:
 - One strap with tensioner
 - One pad for the tabletop
 - One set of new screws
 - One handlebar
 - Lifting device to help with the replacement of the tube assembly
 - Protective covers for HV plugs
- Tools required for Straton tube replacement:
 - Lifting device kit: part number 10244855
Contents: PHS Vertical Motor Power Interface, Lifting device, instructions

Time and manpower

- 2 Person
- 1.5 hours (mechanical)

Safety



Avoid accident and injury or damage of parts.

Risk of accident and injury!

- ⇒ **Read and observe the safety information contained in the Biograph Safety Procedure and Biograph Safety Report.**

Preliminary steps



1. Lower the patient table to open the front door of the gantry.
2. Switch **off** the system at the control box.
3. Remove pallet from PHS
4. Open the gantry front door and disable rotation with switch **S 301**.
5. **Switch off circuit breaker F9 in the PDC and ensure that the LEDs on the DPU and DMS go off.**
6. Remove the plastic center panel from the front side of the gantry. Note the location of the large washers.
7. Turn the gantry by hand until the tube is in a 3 o'clock position.

Preparing the transport box

1. Open the transport box and remove the covers.

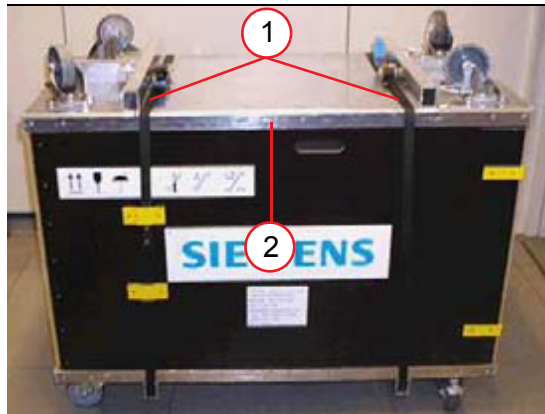


Fig. 29: Straton box

- Remove the 2 straps, unlock the cover wheels, and remove the top cover.
- Take out the service tools and papers.

2. Remove the service tools from the transport box.

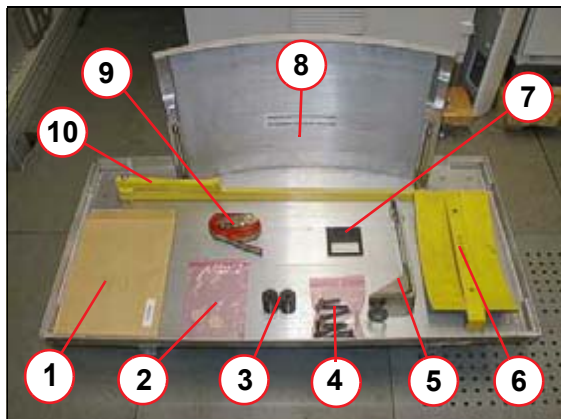


Fig. 30: Contents of the STRATON box

- | | |
|---------|--|
| Pos. 1 | STRATON documentation |
| Pos. 2 | HV insulators |
| Pos. 3 | HV plug protective covers |
| Pos. 4 | Set of new screws |
| Pos. 5 | Lifting device |
| Pos. 6 | Pad for tabletop |
| Pos. 7 | Floppies for tube history and TEA tables |
| Pos. 8 | Tube protection plate |
| Pos. 9 | Lifting strap with tensioner |
| Pos. 10 | Handle bar |

3. Remove covers.



Fig. 31: Removing the side covers

- Lift out the 2 hinged side covers.
- Remove the upper 2 styrofoam parts.

4. Remove protections.

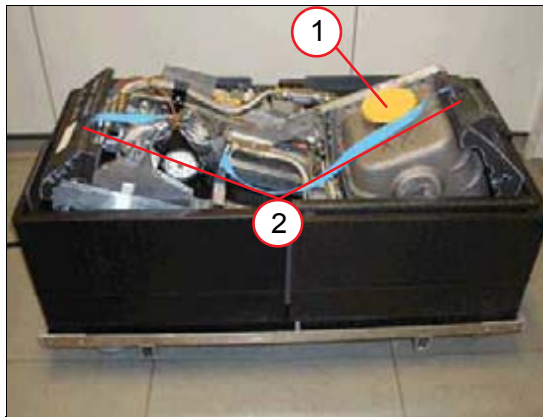


Fig. 32: Unpacked tube assembly

- Remove tube window cover (item 1).
- Remove the 2 styrofoam cushions (item 2).
- Remove the plastic bag from the plugs.

Removing

CAUTION

[hz_serdoc_F13G02U01M01]

Temperature of X-ray tube and/or tube cooling device parts may be above 70 degrees C (with STRATON up to 130 degrees C).

Risk of burns!

- ⇒ Avoid contact with oil and exposed parts of X-ray tube and cooling device.
- ⇒ Wait until X-ray tube and cooling device parts are chilled down.

1. Disconnect cables (tube side).

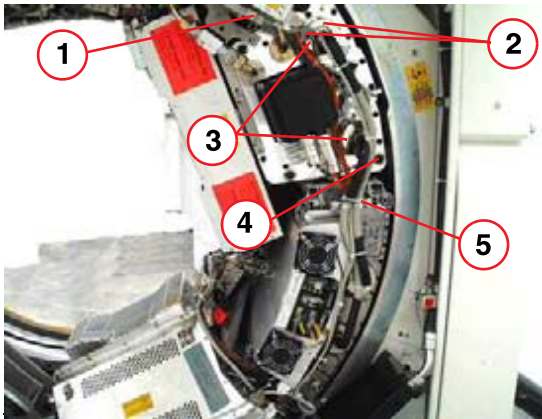


Fig. 33: STRATON plug connections (with newer type of HV cables)

- Disconnect the following cables and connectors from the X-ray tube:
 - Cable W551_X21 at the motor controller (item 1) and cable clamp
 - Loosen the 2 Allen screws on both securing rings (item 2), unscrew the ring, and remove the 2 HV plugs (item 3). Install the protection caps included in the delivery
 - Ground strap (item 4)
 - Remove all tie wraps as needed to release cables from the tube assembly
- NOTE: Mark the positions and quantity of tie wraps removed. When installing the new tube, put the same number of tie wraps in the same positions. This will ensure proper CT balance.

Note: HV cables are delivered in pairs and must also be replaced in pairs. In most systems, a holding bracket (image/item 5) is available at the HV tank. The bracket assures better attachment points for the cables.

2. Disconnect cables (C-box side).

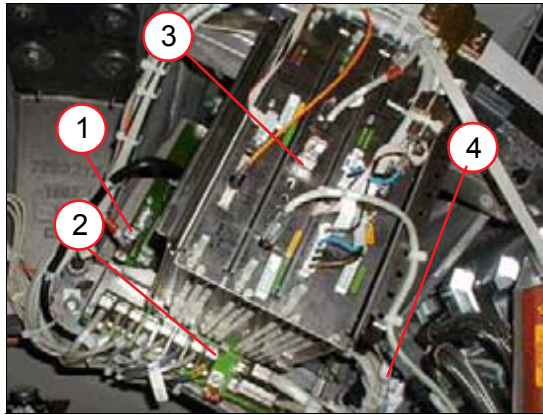


Fig. 34: C-Box connections

- Disconnect the following cable and connectors from the X-ray tube and C-box.
 - Cable C-box D510_X10 (item 1)
 - Cable C-box D510_X12 (item 2)
 - Cable FOC D514_X4 on the FOC board (item 3)
 - Power cable E550_X22 (item 4)
 - Power cable E535_X11 (item 4)
 - Cut all cable ties as needed to release these cables.
- NOTE: Mark the positions and quantity of tie wraps removed. When installing the new tube, put the same number of tie wraps in the same positions. This will ensure proper CT balance.
- Cut the cable ties securing the PSD cable to the tube



3. Move the tube to just past the **6 o'clock** position ensuring that the loose cables do not get trapped.
4. **Secure the rotation with the safety bolt; use the hole between the 149 and 168 degree positions as marked on the edge of the drum.**

5. Remove 6 of 8 retaining screws. -> Only remove the 6 retaining screws as instructed. Do not remove the last 2 retaining screws at this time.

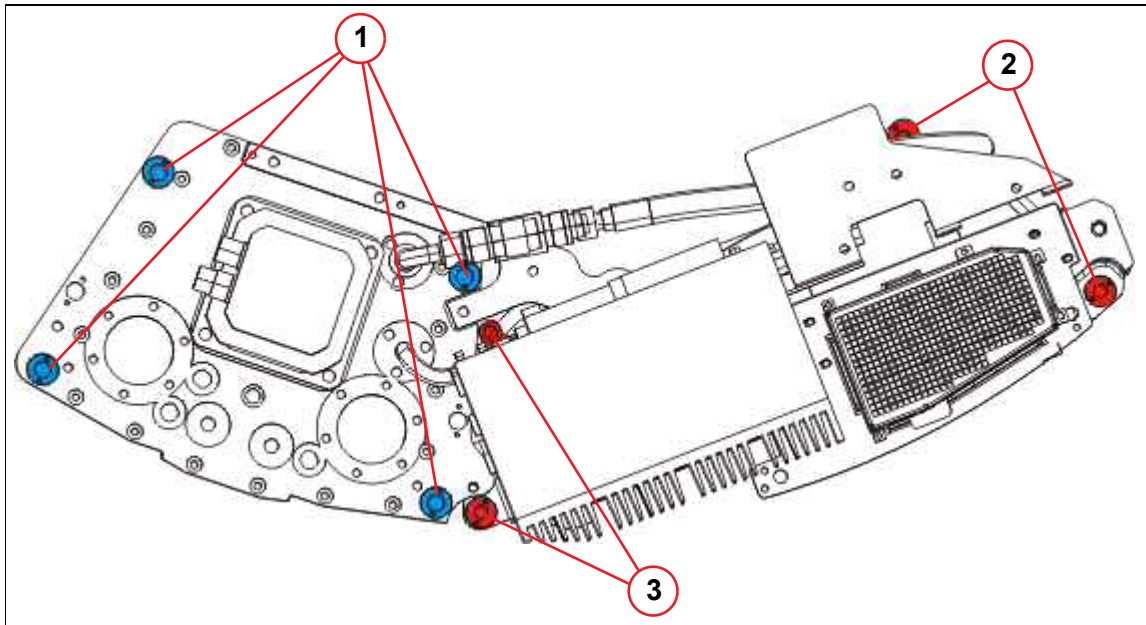


Fig. 35: Straton Tube assembly

Pos. 1	4 x holding screws (10 x 65mm)
Pos. 2	2 x holding screws (10 x 40mm)
Pos. 3	2 x holding screws (10 x 40mm)

- Remove the 4 M10 (65 mm-long) hexagonal bolts (item 1) using an 8 mm hexagonal key.
- Remove the 2 M10 (40 mm-short) hexagonal bolts (item 2) using an 8 mm hexagonal key -> Remove these 6 bolts only. Do not remove the last 2 bolts at this time.

6. Connect the PHS Vertical Motor Service Interface to the bed.

NOTE

Consult the instructions that come with the PHS Vertical Motor Service Interface if needed.

CAUTION

Use care when raising or lowering the PHS with the Vertical Motor Service Interface. The limit switches are not monitored when the system is off.

Limit switches may be damaged if PHS is raised all the way up.

⇒ Only raise PHS high enough to lift tube and extract tube.

7. Installing the lifting device.

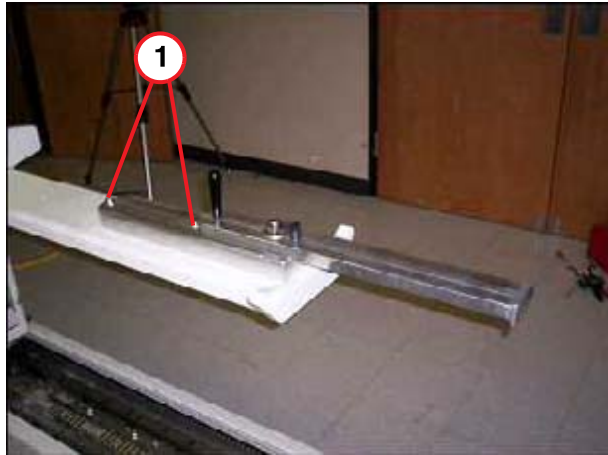


Fig. 36: Biograph Lifting device attached to PHS vertical support

Pos. 1 Bolts

8. Attaching lifting device to CT tube.

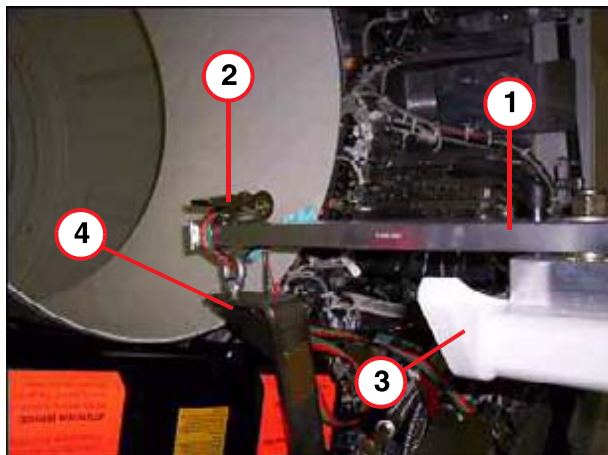


Fig. 37: Straton Assembly with lifting device installed

Pos. 1 Lifting device
Pos. 2 Lifting strap
Pos. 3 PHS vertical support
Pos. 4 Lifting handle

- Install the tube lifting handle on the tube (item 4).
- Secure the tube to the tabletop with the strap and tighten.(item 2).

9. Raise the table enough to take the weight of the tube.

10. Remove the last 2 retaining screws.

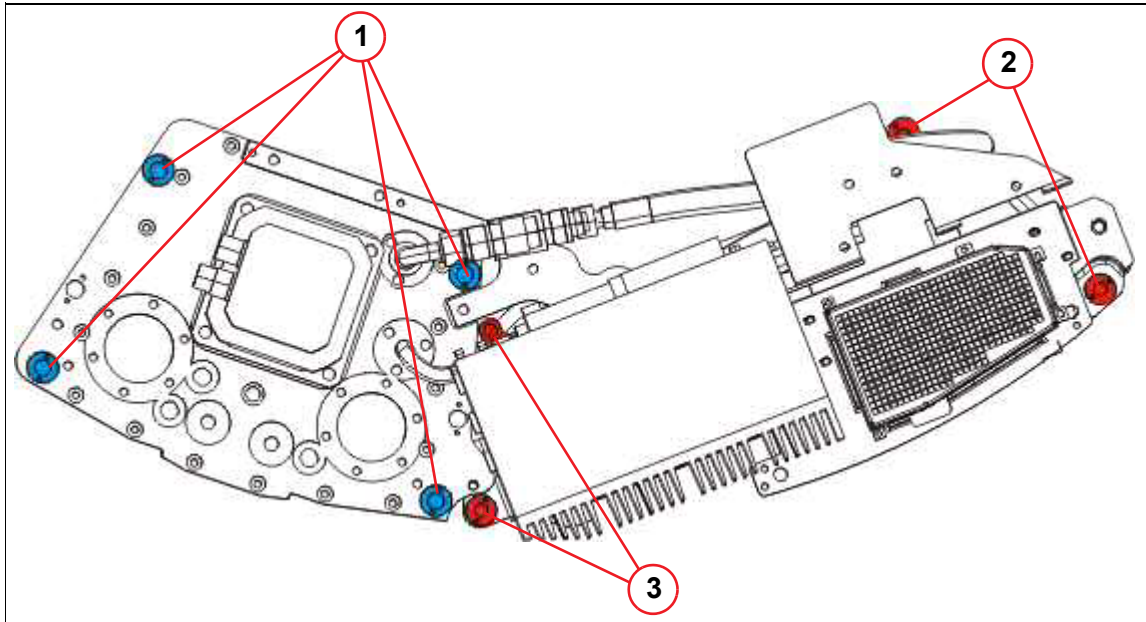


Fig. 38: Straton Tube assembly

- Pos. 1 4 x holding screws (10 x 65mm)
Pos. 2 2 x holding screws (10 x 40mm)
Pos. 3 2 x holding screws (10 x 40mm)

- Remove the final 2 M10 (40 mm-short) hexagonal bolts using an 8 mm hexagonal key (item 3).

11. Remove the old tube.



Fig. 39: Straton assembly with lifting device installed

- Remove the old tube as shown, being careful to avoid damaging the light markers. **Do not pull the tube by the oil connector; use the lifting device.**

12. Move PHS pedestal back to extract the tube.

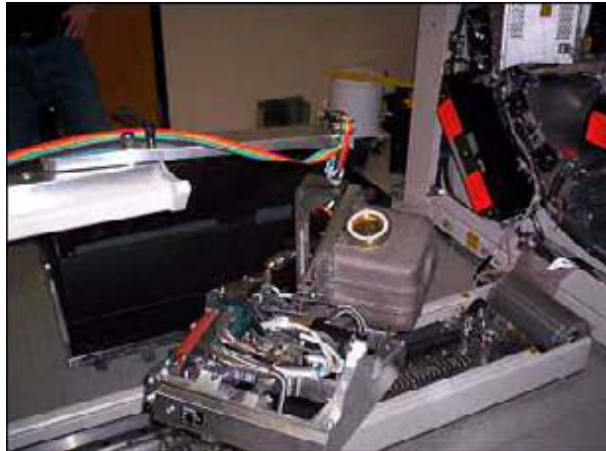


Fig. 40: Extracting tube from gantry

13. Use the swing arm of the lifting device to move the tube to the side of the bed. This allows the tube to be lowered on to the transport device.



Fig. 41: Tube moved to side

14. Install the tube protection plate.



Fig. 42: Installation of the protection plate

- Raise the table slightly to provide access to the bottom of the tube.
- Slide cover into the retaining holes.
- Secure the plate with the two long holding nuts on the left and right side of the tube.

15. Position the transport device.

NOTE

Limited vertical motion makes it necessary to remove all styrofoam pieces from the transport device.

- Roll the transport device under the tube and install the two bottom pieces of styrofoam. It will be necessary to slide them up and under the tube.
- Install the 2 styrofoam pieces. The piece with the notches for the tube frame is placed on the left side .

16. Lower the tube assembly onto the transport device and remove the lifting strap.

17. Use the handle bar to remove the tube.



Fig. 43: Removing the tube assembly

18. Remove the lifting device from the old tube.

Installing

1. Attach the lifting device and strap to the new tube.
2. Place the new X-ray tube to the side of the patient table and attach the lifting strap to the lifting device.
3. Lift the tube, remove the styrofoam pieces, and remove the transport device from under the table.
4. Remove the protection plate from the new tube.
5. Swing the lifting device toward the gantry.
6. Move tube gently towards the gantry until it is positioned correctly over the guide pins, being careful to avoid damage to the light markers.
 - ⇒ It may be necessary to shift the tube slightly to align it with the guide pins.



7. Special Notes regarding the stabilizing frame.

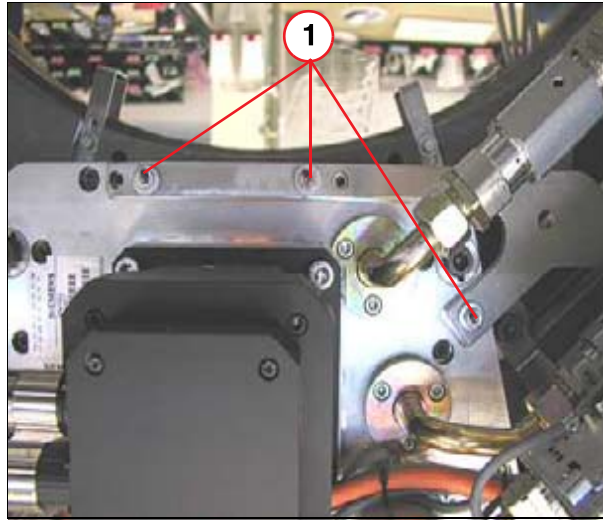


Fig. 44: STRATON Stabilizing frame, detail.

- ⇒ Due to the high rotation speed of the Sensation 40/64/Cardiac 64 system, additional contact points are required with the drum of the gantry to give mechanical support.
- ⇒ To assure that the STRATON in Sensation 40/64/Cardiac 64/Open fits correctly into the gantry with the two extra contact points touching the drum, the cooling part of the STRATON tube assembly can be moved to a limited degree. To ensure this flexibility, the holding screws of the stabilizing frame (item 1) are not tightened. **Do NOT tighten them.**

8. Checking the contact points.

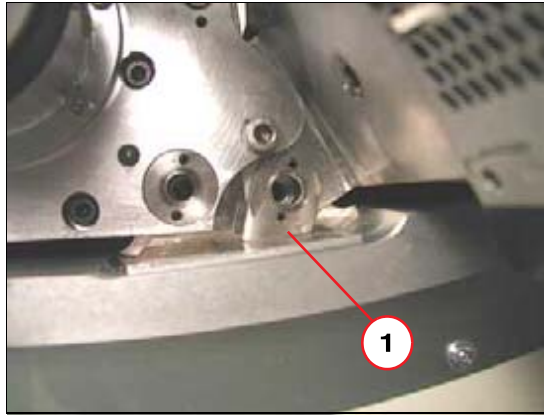


Fig. 45: Contact point at lower center of tube assembly

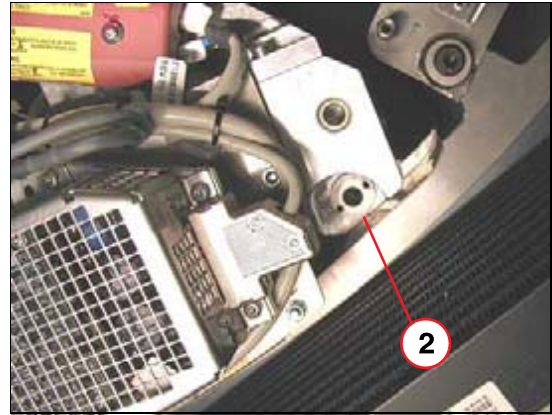


Fig. 46: Contact point at lower right of tube assembly

- After the tube is in place, loosely install the upper left (M10x65mm) and upper right (M10x40mm) new bolts to prevent the tube from coming off the locating pins. Lower the table until the contact points (item 1 near lower center bolt, item 2 near the lower right bolt) of the STRATON tube assembly are touching the gantry drum.
 - A piece of paper can be used to check that the points are in contact with the drum.
9. Install the 8 (remaining 6 for Sensation 40/64/Cardiac64/Open) new M10 hexagonal bolts (supplied). Make sure the longer bolts are installed in the correct position; refer to (Fig. 38 / p. 54).
 10. Hand tighten the eight M10 hexagonal bolts in a crosswise pattern starting with the four central bolts.
 11. Re-check that the contact points are touching the drum.
 12. Tighten the eight M10 hexagonal bolts in a crosswise pattern starting with the four central bolts using **90 Nm**.
 13. Remove the lifting bracket and strap.
 14. Unlock the safety bolt for gantry rotation.
 15. Move the gantry by hand to the 3 o'clock position.
 16. Connect all cables according to the original configuration. Secure the cables with cable ties. Pay special attention to X21 as it easily deforms and can be connected backward.

**NOTE**

Attach tie wraps to the places marked when disconnecting the cables. Be sure to use the same quantity of tie wraps that were removed. This will ensure proper balance of the CT.

17. Tie wrap the PSD cable in its original position.

18. Install the high voltage connectors.

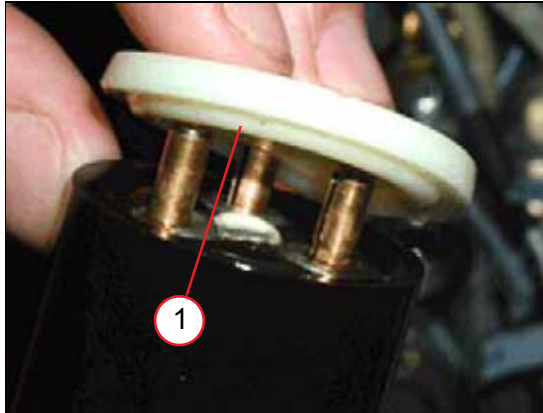


Fig. 47: Removing the tube assembly

Pos. 1 Silicon disk ring

- Check to ensure connector is clean and inspect for damage.
- Install silicon ring.
- **Install the new silicon washers as shown.**
 - ⇒ The shape of the isolation washer is asymmetric. The surface with the more prominent ring has to point to the high voltage plug (item 1). It is no longer necessary to use silicon oil, because silicon oil is already present in the material of the insulation washer.
- Insert the high voltage plugs into the receptacles.
 - ⇒ Verify the position of the notch.
- Hand-tighten the fastening rings and secure them lightly with the two Allen screws on each ring.

Final steps

Prepare tube transport box

1. Install protective covers.

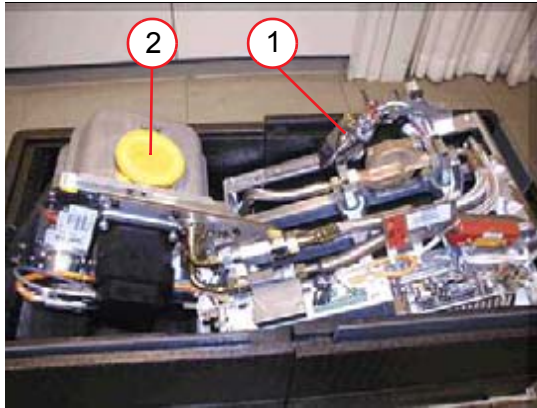


Fig. 48: Protective covers

- Cover the tube plugs (item 1) with the previously removed plastic bag, secure with tie wrap.
- Install tube window cover (item 2).

2. Close the transport box.



Fig. 49: Replacing protective plate in box

- Replace the 2 upper styrofoam parts of the transport box. A small gap between the top and bottom pieces is not unusual. Once the straps are tightened this gap closes and the tube is held securely.
- Place tube protection plate in the transport box.
 - ➡ Make sure that the plate is positioned correctly (item 1).

3. Close transport box.



Fig. 50: Packing the tools

- Reassemble the transport box sides and ensure that all service tools, papers and floppy have been placed inside as shown in the illustration.

4. Close transport box.

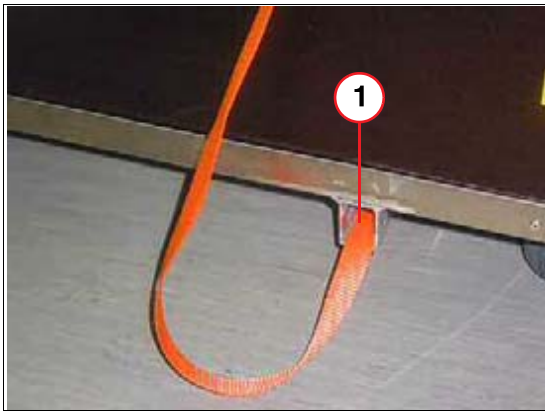


Fig. 51: Correct positioning of strap

- Install the top and secure it with the straps. The straps should be fed inside the special molding of the base (as shown in the illustration) to protect them during shipment.

Mechanical

1. Install plastic center panel on the front of the gantry.
2. Rotate gantry by hand 360 degrees and check for problems.
3. Disconnect the PHS Vertical Motor Service Interface.
4. Switch on circuit breaker F9 in the PDC.
5. Turn on S301.

6. Close the front door of the gantry.
7. Reattach pallet to PHS.
8. Turn on the system at the control box.

Software

1. Copy the TEA table provided on a floppy and delivered with the new tube to **C:\SOMARIS\config\ControlSystemTables**.

NOTE

The TEA table will automatically load directly from the X-ray tube electronics , beginning with software version VB10 and above, and X-ray tubes with revision level 2 and above.

2. Ensure the gantry is switched on and at reset. Go to **Service --> Tuneup-->Tube Replace**. Click **GO**.
 - Give the reason for return using the pull-down menu.
 - Enter a comment if necessary.
 - Enter the revision level of the new tube. (The serial number of the new tube is automatically read from the EPROM).
 - Press **GO** after entering the parameters. A new tube history is generated, and the tube serial number is entered into the config.
 - You can view the old tube history by opening an Additional Window and selecting **Reports --> Tube**.
3. When exchanging the X-ray tube it is important to store the tube history file on the enclosed floppy disk and enclose it with the tube. **The applicable reimbursement will only be paid if the tube history file was sent back correctly.**

To create the floppy:-

- Go to **Service-->Reports-->Tube--> All histories..**
 - Select all history files **for the tube you have just replaced**.
 - Leave the **save to filename** as defaulted.
 - Leave the **filetype** as Gnuzip(.txt.gz)
 - Change the **Target dir** to A:\ via the drop-down list.
 - Insert the tube history floppy into the A: drive and click the **Extract** button.
4. Wait until the floppy has been written and return it with the tube.
 5. Go to **Service --> Tune-up -->X-ray Tube Tune-up**.
Refer to Tune-Up Instructions (MI02-023.817.01.xx.xx).
 6. Continue following the tune-up steps.

Testing

1. Run the Constancy test.
2. Perform a backup of settings.

Labeling

The second label is included in the tube delivery.

It may very well be required by Regulatory Affairs that labeling is shown on the outside of the gantry.

For attachment, refer to the location of labels described in the Operator Manual (also refer to the System Owner Manual / location of labels).

Tube collimator E555

Prerequisites

Tools and auxiliary equipment

- Standard service kit

Time and manpower

- 1 person
- 1 hour (mechanical)

Safety



Avoid accident and injury or damage of parts.

Risk of accident and injury!

- ⇒ **Read and observe the safety information contained in the Biograph Safety Procedure and Biograph Safety Report.**
-

Preliminary steps



1. Lower the patient table in order to open the gantry front door.
2. Remove the pallet from the PHS.
3. Open the gantry front door and disable rotation with switch S 301.
4. Switch **OFF** the system at the control box.
5. **Switch OFF the on-site power using the on-site ON/OFF switch!** Check that the LEDs on the DPU and DMS are off.



6. Turn the gantry by hand until the tube is in the **12 oclock** position.
7. **Secure the rotating gantry with the safety bolt.**

Removing

1. Disconnect the cables

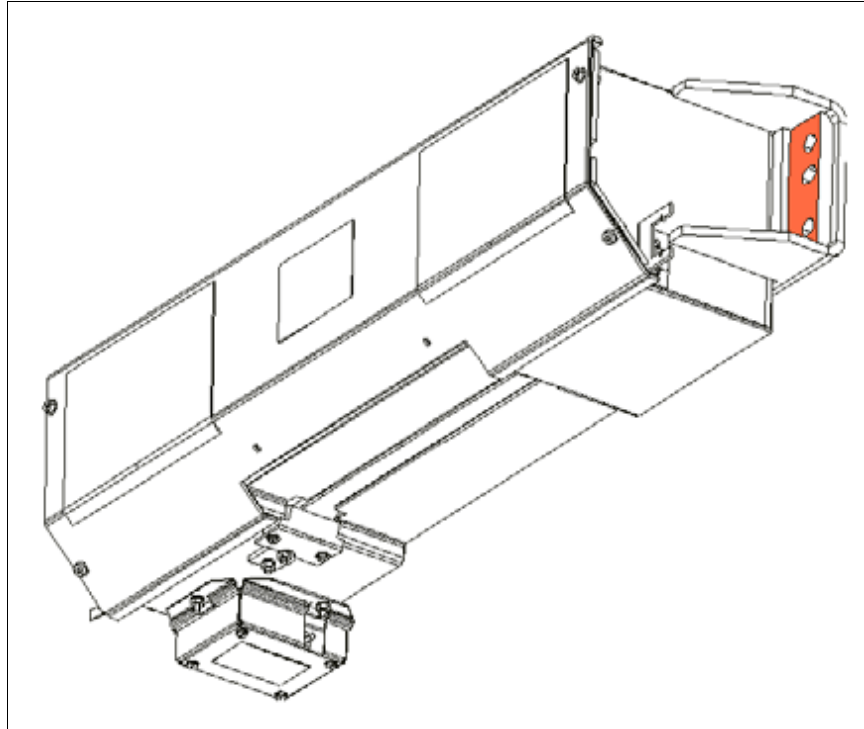


Fig. 52: Diaphragm

- Disconnect the following cables:
 - On the right side, cable E555_X552.
 - On the left side, cable E555_X551.
 - On the ref. Converter, connector J1
 - ⇒ **There is no ref. converter on the Sensation 40/64/Cardiac64.**
 - The ground connection on the left side.
 - Connector D514_X3 on D514 board (C-Box)
 - Connector E510_X15 on backplane (C-Box)
- 2. Remove the cable ties as needed to release the cable from D514_X3 to PHI_Z_PSD.
 - ⇒ **Note the routing of this cable as it is important to follow the same path during reassembly. Do not fold, bend, or roll up the cable.**
- 3. Remove cable ties as needed to release the cable from E510_X15 to the collimator.
- 4. Remove the four 10mm hexagonal screws.
- 5. Remove the collimator from the gantry complete with cables.

Installing



1. Installation is the reverse of removal.
2. Tighten the four screws with a torque of **90 Nm**.
3. Connect all cables according the original configuration.

Final steps

Mechanical

1. After finishing all mechanical work, switch the rotation switch S301 to "1".
2. Unlock safety bolt for gantry rotation.
3. Close the gantry door.
4. Reattach the pallet to the PHS.
5. Switch on the onsite power using the onsite OFF/ON switch.
6. Switch on the system (**SYSTEM/ON**) at the control box.

Software

Load the new calibration table into the FOC controller.

1. Insert the floppy (included with the spare part) into the floppy drive.
2. Open the Windows Explorer on the Navigator console.
3. Check the format of the calibration tables.

The floppy contains two files:

FOC1_xxx.fsh

FOC2_xxx.fsh

xxx is the serial number of the new collimator.

The two files must have the a.m. format before they can be used with the flashload tool.

4. Copy the files from the floppy to the following directory -> **C:/Somaris/Config/Control-SystemFlash**.
5. In case the files have a different format:
Rename file 0xxxx_1.fsh to FOC1_xxx.fsh (the 1 after FOC indicates parameter set 1)
Rename file 0xxxx_2.fsh to FOC2_xxx.fsh (the 2 after FOC indicates parameter set 2).
6. Close the Windows Explorer.
7. Go to Option/Service/Local Service and select **Control -> Flashload**.

8. Select **FOC** by clicking its button.

The field show the previous tube collimator parameters in the field **Param1** and **Param2** (Vxxx).

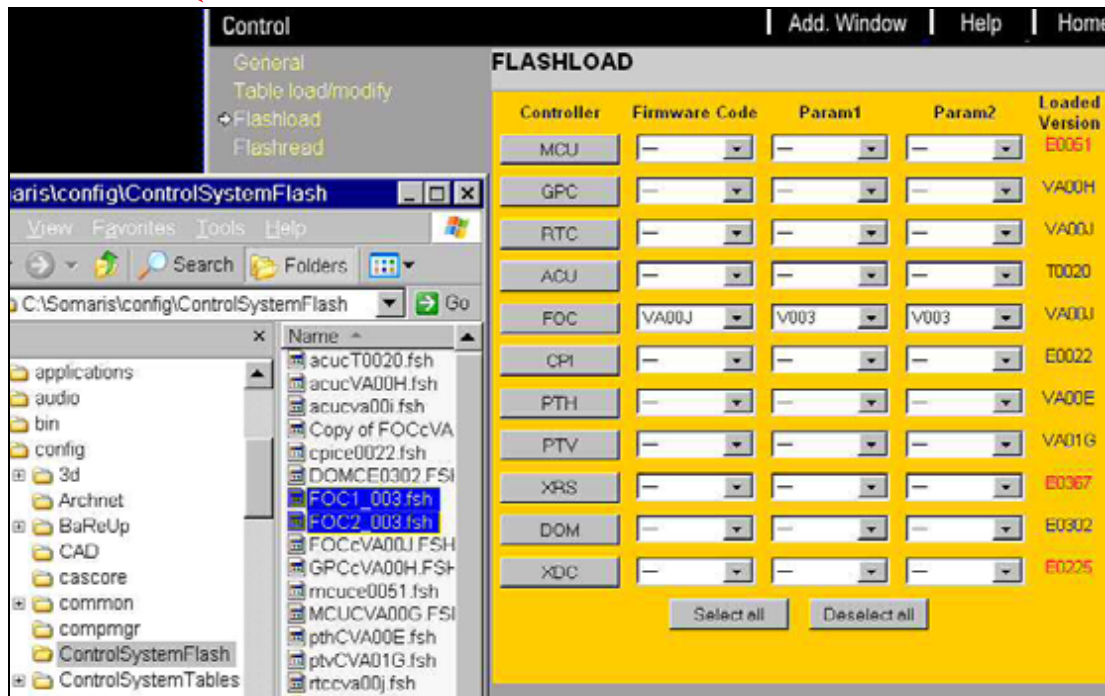


Fig. 53: Flashload of collimator calibration tables

9. Use the drop-down list to select the new tables for Param1 and Param2 (according to the serial number of the collimator).
10. Click "**GO**" to perform the flashload of the parameters.
11. Wait until the message **flashload successfully done**.
12. Continue with the **Tube collimator replacement** guided tour located in Local Service-> Tune-up -> Installation.

Refer to the Tune-Up Instructions (MI12-023.817.01.xx.02).

Testing

1. Perform Quality tests.
2. Back up the system settings.

PHI_Z_MON D554

Prerequisites

Tools and auxiliary equipment

- Standard service kit

Time and manpower

- 1 person
- 1 hour

Safety



Avoid accident and injury or damage of parts.

Risk of accident and injury!

- ⇒ **Read and observe the safety information contained in the Biograph Safety Procedure and Biograph Safety Report.**
-

Preliminary steps

1. In the service platform **Service-> Test Tools->Stat-modes**, display static mode 00 and change body/region to **Head**. Click load and then cancel the static mode platform. This positions the compensator to provide easier access to the screws holding the PHI_ZMON.
2. Lower the patient table in order to open the gantry front door.
3. Remove the pallet from the PHS.
4. Open the gantry front door and disable rotation with switch S 301.
5. Switch **OFF** the system at the control box.
6. **Switch OFF on-site power using the on-site ON/OFF switch!** Check that the LEDs on the DPU and DMS are off.
7. Turn the gantry by hand until the tube is in the **12 oclock** position.
8. **Secure the rotating gantry with the safety bolt.**



Removing

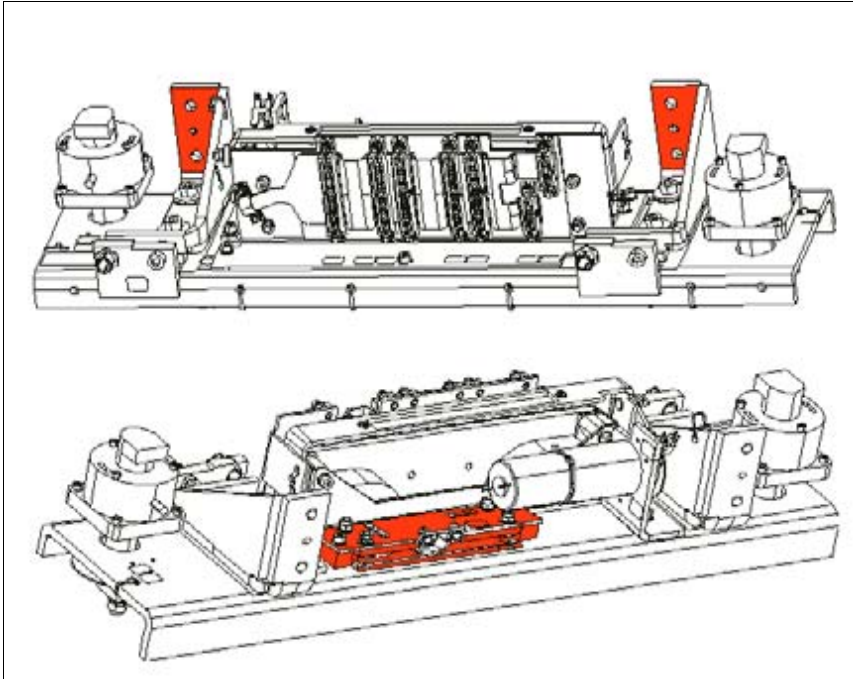


Fig. 54: Diaphragm mechanics

1. Remove the collimator cover.
2. Remove the cables.
 - E555_X552 on the right side of the collimator box and E555_X551 on the left.
 - D514_X3 on D514 board (C-Box)
 - E510_X15 on Backplane (C-Box)
3. Cut cable ties as needed to release the 3 cables.
 - ⇒ **Note the routing of the Z-Mon cable (D514_X3), since it is important to follow the same path during reassembly.**
4. Remove the four bolts holding the mechanics and remove the mechanics from the collimator box. Be careful handling the mechanics of the collimator box. Place it on foam or other soft material. **Mechanically sensitive device!**
5. Remove the screws holding the PHI_Z_Mon board.

Installing



1. Install the new PHI_Z_Mon board.
2. Install the collimator mechanics and tighten the retaining screws with a torque of **24 Nm**. (Be careful not to damage any parts with the tools).
3. Connect all cables according to the original configuration. Secure with cable ties.



4. Close the collimator box and tighten the screws with a torque of **5 Nm**.

Final steps

Mechanical

1. After finishing all mechanical work, switch rotation switch S301 to "1".
2. Unlock safety bolt for gantry rotation.
3. Close the gantry door.
4. Reattach pallet to PHS.
5. Switch on the system.

Software

Perform the required Tune-up steps according to the Tune-up instructions MI12-023.817.01.xx.02.

Testing

1. Perform Quality tests
2. Back up the system settings.

D400 in generator electronics box

Prerequisites

Tools and auxiliary equipment

- Standard service kit
- Torque wrench 1-5 Nm

Time and manpower

- 1 person
- 1 hour

Safety



Avoid accident and injury or damage of parts.

Risk of accident and injury!

- ⇒ **Read and observe the safety information contained in the Biograph Safety Procedure and Biograph Safety Report.**

Preliminary steps

1. Lower the patient table in order to open the gantry front door.
2. Remove the pallet from the PHS.
3. Open the gantry front door and disable rotation with switch S 301.
4. Switch **OFF** the system at the control box.
5. **Switch OFF on-site power using the on-site ON/OFF switch!** Check that the LEDs on the DPU and DMS go off.
6. Turn the gantry by hand until the electronics box of the generator is in the **6 o'clock** position.
7. **Secure the rotating gantry with the safety bolt.**



Removing



1. Before working on the XRS
⇒ **Check that the DCC voltage is discharged**, measure between D440_X1, X2 of the XRS.

2. Disconnect the cable Gen rot D400_X3 and cable Gen rot D400_X5.
3. Remove the 10 screws of the electronics box cover of the generator using a 3mm hexagonal key.
4. Remove the metal cover.
5. Disconnect the following cables:



Fig. 55: D400 generator electronics

- To the right of D400
 - connector X12 and X13
 - At the bottom of D400
 - Connectors X6, X1 and X15 (Ground)
 - On top of D400
 - Connector X7, X8 and X9
6. Remove the six Nylock nuts and remove the board. **Be careful not to damage the cables!**

Installation



1. Remove MCB from the old board and install it in the new board.
2. Remove D400_X4 (dongle) from the old board and connect it to the new board.
3. Install the new board and reconnect all cables. Make sure that all the plugs are connected securely. (You will hear a click when the plug is correctly locked in place). The ground connection is tightened to **4.8 Nm**
4. Tighten the six Nylock nuts to **1.9 Nm**.
5. Attach the metal cover to the electronics box, tighten **all** ten screws with **3 Nm** and reconnect cables Gen rot D400_X3 and Gen rot D400_X5.

Final steps

Mechanical

1. After finishing all mechanical work, switch rotation switch S301 to "1".
2. Unlock the safety bolt for gantry rotation.
3. Close covers.
4. Reattach pallet to PHS.
5. Switch on the system.

Software

1. Continue with tune-up steps as described in the guided tour **Service-> Tuneup-> Installations.**

Refer to Tune-Up Instructions (MI12-023.817.01.XX.02).

Testing

1. Perform the quality test.

STRATON FIL assembly D470

Prerequisites

Tools and auxiliary equipment

- Standard service kit
- Torque wrench 1 - 5 Nm

Time and manpower

- 1 person
- 1 hour

Safety



Avoid accident and injury or damage of parts.

Risk of accident and injury!

- ⇒ **Read and observe the safety information contained in the Biograph Safety Procedure and Biograph Safety Report.**
-

Preliminary steps



1. Lower the patient table to open the front door of the gantry.
2. Remove pallet from PHS.
3. Open the gantry front door and disable rotation with switch S 301.
4. Switch **OFF** the system at the control box.
5. **Switch OFF on-site power using the on-site ON/OFF switch!** Check that the LEDs of the DPU and DMS go off.
6. Turn the gantry by hand until the generator electronic box is in the **12 oclock** position.
7. **Secure the rotating gantry with the safety bolt.**



Removing



1. Before working on the XRS
 - ⇒ **Check that the DCC voltage is discharged** and measure between D440_X1,X2 of the XRS.
2. Disconnect the cable Gen rot D400_ X3, X5.

3. Remove the ten screws of the generator electronic box cover using a 3 mm hexagonal key, and remove the metal cover.
4. Open the D400 tray to gain access to the FIL assembly .
5. Disconnect the following cables:
 - On the right side, cable E603_X4.
 - On the left side, cable E603_X5, and cable E603_X7.
 - Remove the small ribbon cable D470_X3.
6. Remove the FIL assembly.

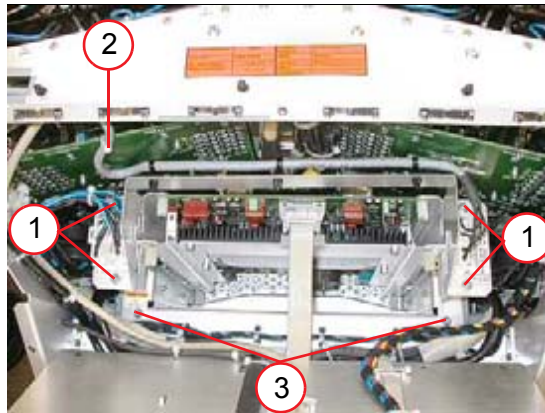


Fig. 56: FIL board in E-box

- Remove the four screws holding the wire bracket assembly (Item 1).
- Cut the tie wrap (item 2) and position the bracket outside of the frame.
- Remove the two screws holding the assembly (item 3).
- Carefully slide the assembly out, paying special attention to the wire looms at the bottom.

⇒ **Be careful not to damage the cables or other boards.**

Installing



1. Carefully install the new assembly and tighten the two screws with a torque of **3 Nm**.
2. Connect all cables according to the original configuration.
3. Reinstall the bracket assembly with a torque of **3 Nm**, tie wrap the cable as before.
4. Move board D400 back to its original position. Reinstall the metal cover for the electronics box, tighten all ten screws with a torque of **3 Nm**, and connect cable Gen rot D400_X3 and cable Gen rot D400_X5. Secure cables with cable ties.

Final steps

Mechanical

1. After finishing all mechanical work, switch rotation switch S301 to "1".
2. Unlock safety bolt for gantry rotation.
3. Close the gantry.
4. Reattach pallet to PHS.
5. Switch on the system.

Testing

1. Perform some scans.

XRS Power Supply E606

Prerequisites

NOTE

There are 3 variants of XRS power supply E606:

Variant 1 (early Sensation 16/Cardiac)- separate plugs for power, sense and inhibit.

Variant 2¹ (later Sensation 16/Cardiac)- equipped with adapter board; separate plugs for power and sense, one plug for inhibit on adapter board used.

Variant 3 (Sensation 40/64/Cardiac64 and Open) Similar to variant 2, however, the inhibit plugs are not used, and slots 4 and 5 have different voltages.

- Identify the type of power supply available (The spare part will be delivered as variant 2 or 3).

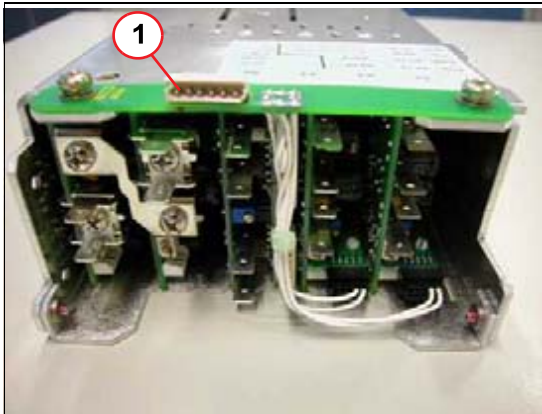


Fig. 57: E606 variant2 with adapter board
Pos. 1 Inhibit connector



Fig. 58: E606 Variant 3 supply

- To improve the assembly and replacement of E606 and to avoid incorrect installation of the inhibit plugs, an adapter board was created that is attached to the power supply. This board connects all 4 inhibit plugs to one single coded plug. The remaining 3 connectors will not be used any more and will be affixed at the cable harness by a cable tie until the cable will be changed to only one inhibit connector.

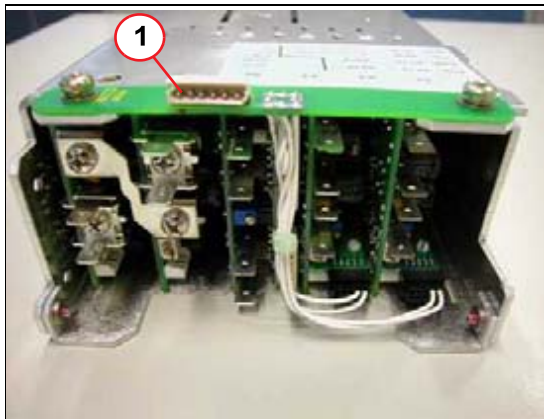


Fig. 59: E606 variant2 with adapter board

Pos. 1 Inhibit connector

Tools and auxiliary equipment

- Standard service kit
- Torque wrench 1-5 Nm
- **Cable W562 (7109254) may be required (variant1).**

Time and manpower

- 1 person
- 1 hour

Safety



Avoid accident and injury or damage of parts.

Risk of accident and injury!

- ⇒ **Read and observe the safety information contained in the Biograph Safety Procedure and Biograph Safety Report.**

Preliminary steps

1. Lower the patient table in order to open the gantry front door.
2. Remove pallet from PHS.
3. Shut down the system.
4. Switch **OFF** the system at the control box.

80 XRS parts, Sensation 10/16/40/64/Cardiac/Cardiac64



5. **Switch OFF on-site power using the on-site ON/OFF switch!** Check that the LEDs on the DPU and DMS go off.
6. Open the gantry front door and disable rotation with switch S 301.
7. Turn the gantry by hand until the generator electronic box is in the **6 oclock** position.

Removing

1. Remove the power connectors, noting the position of wires for reassembly.

NOTE

To remove the power connections, pull on the connector - not on the wires.

2. Disconnect the Inhibit cable

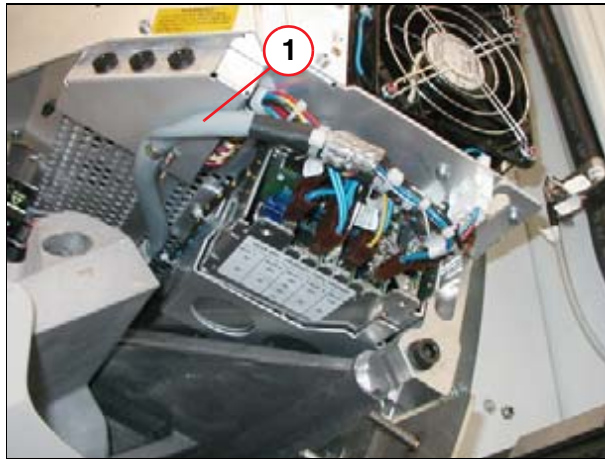


Fig. 60: Variant 1 supply, with old style power connectors
Pos. 1 Inhibit cable

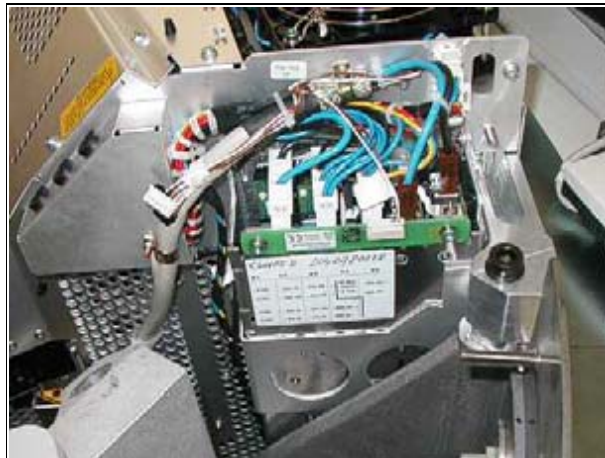


Fig. 61: Variant 2 power supply with new style power connectors

- Disconnect the inhibit cable from the XRS power supply and remove ties and clamp to release the cable (item 1). There are 3 variants of this supply
 - ⇒ **Variant 1** (early Sensation16/Cardiac) 4 plugs connected directly to the supply. The black plugs have a lock in the center of the plug (this makes the procedure difficult), the white ones at each end (which is easier). Currently this cable is difficult to remove without damaging the black plugs, however, the replacement part should be a variant 2 supply which only uses the white INHIP +15v plug. Cable W562 (71-09-254) is also available for order. Contact UPTIME for the latest information on this issue.
 - ⇒ **Variant 2** (later Sensation 16/Cardiac) A single plug connected to an adapter board mounted on the supply. Remove the single inhibit plug (INHIP +15v)
 - ⇒ **Variant 3** (Sensation 40/64/Cardiac64, Open) This variant does not use the inhibit plugs at all and has an extra 24V supply for the fans.

3. Remove the Nylock nut holding the ground connection.

Release the supply (Sensation 10 / 16 / Cardiac)

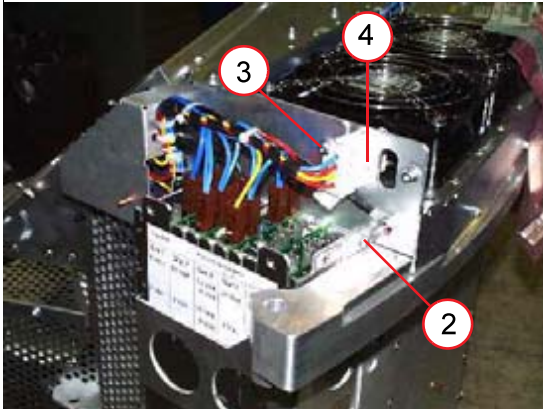


Fig. 62: XRS power supply

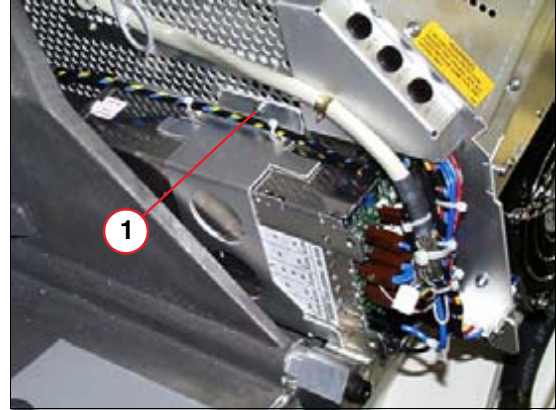


Fig. 63: XRS power supply

- Remove the screw (item 1) from the support bracket of the power supply.
- Remove the screw holding the supply (item 2).
- Unplug E603_X6 (item 4)
- Use a screwdriver and release plug X6 from the frame. Push it back far enough to allow you to slide out the power supply.
- Remove the left-hand screw holding the fan frame (item 3)
- Slide the supply out removing cable ties as needed until you can remove Plug E606_AC in back of the supply. After that continue sliding the supply out.

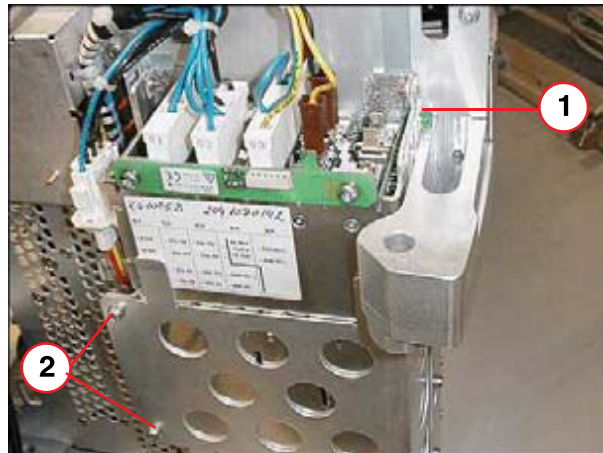
Release the supply (Sensation 40/64 / Cardiac 64 ONLY)

Fig. 64: Variant 3 power supply

- Remove the screw of the bracket holding the power supply (item 1).
- Remove the 2 screws holding the middle of the supply (item 2).
- Slide the supply out removing cable ties as needed until you can remove Plug E606_AC in back of the supply. After that continue sliding the supply out.

Installing

1. Reconnect E606_ AC as you slide in the supply.
2. Insert the screw holding the supply (item 2) and tighten with a torque of **1.3 Nm**.
3. Tighten the center screw of the supply and the fan frame screws to **3 Nm**.
4. Tighten the ground connector Nylock nut to **4.8 Nm**.
5. Reconnect the inhibit connections.

Variant 1: 4 inhibit plugs are used.

Variant 2: Only one inhibit plug **INHIP +15V** is used; attach the remaining 3 plugs with a cable tie to the cable harness.

Variant 3: No inhibit connectors are used.

6. Reconnect supply.



Fig. 65: XRS power supply connectors



Fig. 66: Details of variant 1 inhibit plugs

Tab. 1 XRS power supply connections (Variant 1 and 2)

Slot 1	Slot 2		Slot 3	Slot 4	Slot 5
*INHP +5V	*INHP 15V	*INHP -15	S3a_15V	Not used	Not used
S1_5V	S2b_15V		S3a_0V	Link Wire	E606_48V+
Not used	S2a_15V		No connector	Not used	Not used
S1_0V	S2b_0V	*INHP +15	S3b_15V	Not used	Not used
Not used	S2a_0V		S3b_0V	E606_48V-	Link Wire

Tab. 2 XRS power supply connections (Variant 3))

Slot 1	Slot 2	Slot 3	Slot 4	Slot 5
Not used	Not used	S3a_15V	S4a_24V	No connector
Not used	S2b_15V	S3a_0V	S4a_0V	E606_48V+
Not used	S2a_15V	No connector	No connector	E606_48V+
Not used	S2b_0V	S3b_15V	S4b_5V	E606_48V-
Not used	S2a_0V	S3b_0V	S4b_0V	E606_48V-

- Connect all cables according to the original configuration.

Final steps

System check

1. Turn power to **ON**.
2. Switch rotation switch S301 to **1**
3. Check the function of the system
4. Close the front door
5. Reattach pallet to PHS.
6. Test system.

Fuses F4, F5 generator electronics

Prerequisites

Tools and auxiliary equipment

- Standard service kit

Time and manpower

- 1 person
- 1 hour

Safety



Avoid accident and injury or damage of parts.

Risk of accident and injury!

- ⇒ **Read and observe the safety information contained in the Biograph Safety Procedure and Biograph Safety Report.**

Preliminary steps

1. Lower the patient table in order to open the gantry front door.
2. Remove pallet from PHS.
3. Switch **OFF** the system at the control box.
4. **Switch OFF on-site power using the on-site ON/OFF switch!** Check that the LEDs of the DPU and DMS go off .
5. Open the gantry front door and disable rotation with switch S 301.
6. Turn the gantry by hand until the generator electronic box is in the **9 o'clock** position.
7. **Secure the rotating gantry with the safety bolt.**



Removing



1. Before working on the XRS
 - ⇒ **Check that the DCC voltage is discharged** and measure between D440_X1,X2 of the XRS



2. To access the fuses: Remove the 12 screws from the metal cover underneath the electronics box of the generator. You should now be able to access the fuses (arrowed).

⇒ **Measure UDC voltage to make sure that there is no voltage present before replacing the fuses!**

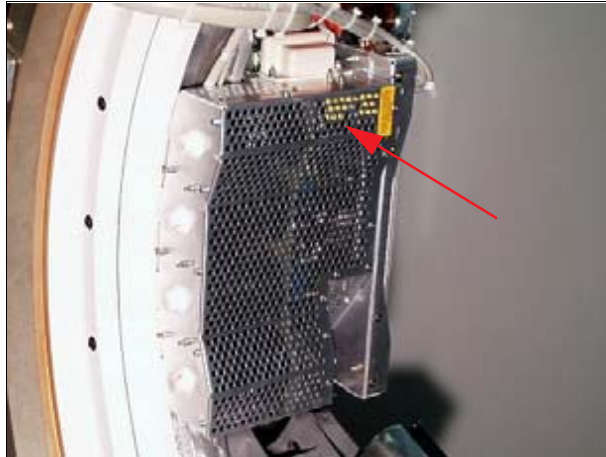


Fig. 67: Back view of generator electronics

Installing



1. After replacing the fuses, reinstall the metal cover and install all screws. Tighten to **3 Nm**.

Final steps

Mechanical

1. After finishing all mechanical work, switch rotation switch S301 to "1".
2. Unlock safety bolt for gantry rotation.
3. Close the gantry front door.
4. Reattach pallet to PHS.
5. Switch on the system.

Testing

1. Perform some test scans.

Fan of generator electronic box

Prerequisites

Tools and auxiliary equipment

- Standard service kit
- Torque wrench 1-5 Nm

Time and manpower

- 1 person
- 1 hour

Safety



Avoid accident and injury or damage of parts.

Risk of accident and injury!

- ⇒ **Read and observe the safety information contained in the Biograph Safety Procedure and Biograph Safety Report.**
-

Preliminary steps



1. Lower the patient table in order to open the gantry front door.
2. Switch **OFF** the system at the control box.
3. **Switch OFF on-site power using the on-site ON/OFF switch!** Check that the LEDs of the DPU and DMS go off.



4. Open the gantry front door and disable rotation with switch S 301.
5. Turn the gantry by hand until the generator electronic box is in the **3 oclock** position.
6. **Secure the rotating gantry with the safety bolt.**

Removing

1. Unplug the power cable from the defective fan, (E603_M1,M2,M3,M4)
2. Remove the 4 screws.

3. Remove the defective fan.

⇒ **Note the voltage of the fan and ensure that the replacement is the correct type** (Sensation 40/64/Cardiac64/Open have 24 V fans, earlier models have 220 V fans)

Installing



1. Install the new fan in reverse order. Apply a drop of Loctite 221 and tighten to **1.5 Nm**.

Final steps

Mechanical

1. After finishing all mechanical work, switch rotation switch S301 to "1".
2. Unlock safety bolt for gantry rotation.
3. Switch on the system.

Testing

1. Check that the fan is running.
2. Close the gantry front door.
3. Reattach pallet to PHS.

Overview

The left stand includes the following components:

- Rotation converter
- Tilt converter
- Tilt sensor
- MCU/RTC board
- CPI board
- Balancing board

LUST Rotation Converter

Prerequisites

Tools and auxiliary equipment

- Standard service kit.

Time and manpower

- 1 person
- 1 hour

Safety



Avoid accident and injury or damage of parts.

Risk of accident and injury!

- ⇒ **Read and observe the safety information contained in the Biograph Safety Procedure and Biograph Safety Report.**
-

Preparation

1. Switch the system to **COMP/ON**.
2. **Switch off circuit breakers F8 and F9 in the PDC**
3. Remove the upper covers for the left side of the gantry.

Removing



Fig. 68: Rotation converter

1. Disconnect connectors E301, X1. E301, X11
2. Remove the 4 nuts securing the rotation converter (arrows).
3. Remove the defective converter.
4. Remove the CAN-COM unit by pushing the red lock button and pulling it away from the converter. Transfer it to the new converter
5. Transfer the connectors from X2 (including jumpers) to the new converter.

Installing

1. Install the new rotation converter in the reverse order.

Final steps

Mechanical

1. Install the upper cover for the left side of the gantry.
2. Switch on circuit breakers **F8** and **F9** inside the PDC.
3. Switch the system back to **SYSTEM/ON**

Tests

1. Perform different rotation modes in service to ensure that the rotation converter is functioning properly.

LUST Tilt Converter

NOTE

The tilt motor is disabled for Biograph 40/64 systems.

Tilt Sensor

Prerequisites

Tools and auxiliary equipment

- Standard service kit.

Time and manpower

- 1 person
- 1 hour

Safety



Avoid accident and injury or damage of parts.

Risk of accident and injury!

- ⇒ **Read and observe the safety information contained in the Biograph Safety Procedure and Biograph Safety Report.**
-

Preparation

1. Switch the system to **COMP/ON**.
2. **Switch off circuit breakers F8 and F9 in the PDC**
3. Remove the upper cover for the left side of the gantry.

Removing



Fig. 69: Tilt sensor

1. Disconnect connector B301 X1.
2. Remove the cable clamp.
3. Remove the 2 screws securing the sensor to its mounting plate (arrows).
4. Remove the defective sensor.

Installing

1. Install the new tilt sensor in the reverse order.

Final steps

Adjustments

1. Remove the cover from the MCU/RTC board.
2. Switch on circuit breakers **F8** and **F9** inside the PDC.
3. Switch the system back to **SYSTEM/ON**.
4. Switch the RTC (rotation tilt control) to SERVICE. Switch S6 on the MCU/RCT board (Service RTC) upward.
5. Press S10 (Service STOP on, yellow) to switch on the FC.
6. Tilt the gantry with Tilt Pos and Tilt Neg (S3 and S4 on MCU/RTC) until it is vertical. Check with the spirit level.
7. Loosen the mounting screws.

8. Rotate the tilt sensor on the upper mounting screw until the LED (Tilt 0°) on the MCU comes on.
9. Fix both mounting screws. Make sure the tilt sensor is not misadjusted in the process.
10. Switch the system back to **COMP/ON**.

Mechanical

1. Install the cover for the MCU/RTC board.
2. Install the covers for the left side of the gantry.
3. Switch the system back to **SYSTEM/ON**

Tests

NOTE

Tilt is disabled on Biograph 40/64 systems.

1. Test the tilting of the system to ensure it is functioning properly.

MCU-RTC Board

Prerequisites

Tools and auxiliary equipment

- Standard service kit.

Time and manpower

- 1 person
- 1 hour

Safety



Avoid accident and injury or damage of parts.

Risk of accident and injury!

- ⇒ **Read and observe the safety information contained in the Biograph Safety Procedure and Biograph Safety Report.**
-

Preparation

1. Switch the system to **COMP/ON**.
2. **Switch off circuit breakers F8 and F9 in the PDC**
3. Remove the covers for the left side of the gantry (upper cover first).

Removing



Fig. 70: MCU/RTC board

1. Remove the metal cover from board D301(MCU/RTC).
2. Disconnect all connectors from the upper and lower row.
3. Remove the twelve screws (circled) and two screws found below both MCB's D2.
4. Remove the defective board.

⇒ **Observe ESD precautions**

Installing

1. Install the new board with 12 screws.
2. Ensure that connectors X1-X18 are connected properly and secure.
3. Don't forget to transfer connector X21 (without cables) to the new board.
4. Install the metal cover for board D301.

Final steps

Mechanical

1. Install the covers for the left side of the gantry.
2. Switch on circuit breakers **F8** and **F9** inside the PDC.
3. Switch the system back to **SYSTEM/ON**

Software

1. The controllers on MCU/RTC must now be flashed.
2. Go to Local Service and select **Control -> Flashload**. Then select **MCU** and **RTC** by clicking the respective buttons.
3. The fields show the **firmware** available on disk in the firmware field. Click the drop-down menu and select the latest/valid version; then click "**GO**" to perform the flashload.
4. After you get the message **flashload successfully done**, continue with the next step.

Tests

1. Perform some test scans.

CPI Unit

Prerequisites

Tools and auxiliary equipment

- Standard service kit.

Time and manpower

- 1 person
- 0.5 hours

Safety



Avoid accident and injury or damage of parts.

Risk of accident and injury!

- ⇒ **Read and observe the safety information contained in the Biograph Safety Procedure and Biograph Safety Report.**
-

Preparation

- Switch the system to **COMP/ON**.
- **Switch off circuit breakers F8 and F9 in the PDC**
- Remove the lower cover from the left side of the gantry.

Removing

1. Remove all cables from the CPI unit.
2. Remove the cable clamp (if available).
3. Remove the 2 retaining nuts.
4. Remove the CPI unit.

Installing

1. Install the CPI unit in the reverse order.
2. Make sure that there is a washer under the cable clamp (if available) of the upper holder.

Final steps

Software

1. The controllers on CPI must now be flashed.
2. Go to Local Service and select **Control -> Flashload**. Then select **CPI** by clicking the buttons.
3. The fields show the **firmware** available on disk in the firmware field. Click the drop-down menu and select the latest/valid version; then click "**GO**" to perform the flashload.
4. After you get the message **flashload successfully done**, continue with the next step.

Mechanical

1. Install the covers for the left side of the gantry.

Tests

1. Check the function of the CPI unit (check function of connected cardio option and/or bolus injector and/or respiratory gating...).

Balancing unit

Prerequisites

Tools and auxiliary equipment

- Standard service kit.

Time and manpower

- 1 person
- 1 hour

Safety



Avoid accident and injury or damage of parts.

Risk of accident and injury!

- ⇒ **Read and observe the safety information contained in the Biograph Safety Procedure and Biograph Safety Report.**
-

Preparation

- Switch the system to **COMP/ON**.
- **Switch off circuit breakers F8 and F9 in the PDC**
- Remove the lower cover from the left side of the gantry.

Removing



Fig. 71: Balancing unit

- Remove all plugs to the unit.
- Remove the 4 retaining nuts.

2. Remove the balancing unit.

Installing

1. Install balancing unit in reverse order.

Final steps

Mechanical

1. Install the covers for the left side of the gantry.

Tests

1. Check the balancing function.

Refer to Tune-Up Instructions (MI12-023.817.01.xx.xx).

Balancing Sensors

Prerequisites

Tools and auxiliary equipment

- Standard service kit

Time and manpower

- 1 person
- 1 hour

Safety



Avoid accident and injury or damage of parts.

Risk of accident and injury!

- ⇒ **Read and observe the safety information contained in the Biograph Safety Procedure and Biograph Safety Report.**
-

Preparation

- Switch the system to **COMP/ON**.
- **Switch off circuit breakers F8 and F9 in the PDC.**
- Remove the lower cover from the left side of the gantry.
- Open the front or back cover as appropriate.

Removing

1. There are two sensors, one located at the front of the gantry on the right side and the other at the back of the gantry inside the back cover mounting bracket just below X350 (the plug for the back panels).

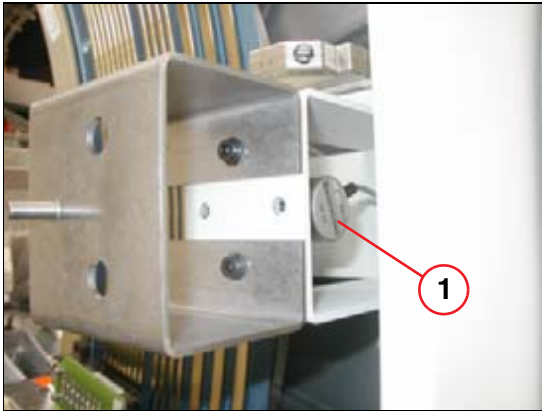


Fig. 72: Balancing sensor, back

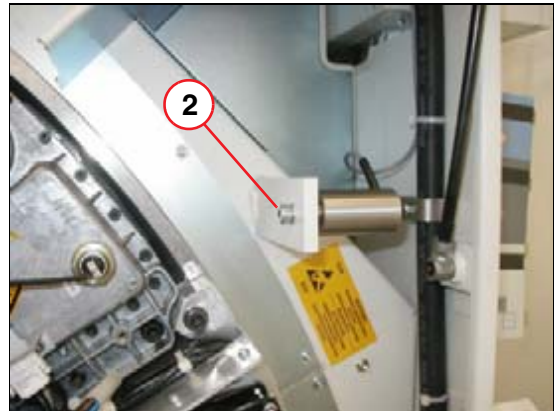


Fig. 73: Balancing sensor, front
Pos. 1 Retaining bolt

- Remove all cable ties to release the sensor cable and unplug cable from balancing unit.
 - Remove the retaining bolt (item 2).
2. Remove the balancing sensor.

Installing

1. Install sensor in reverse order.

Final steps

Mechanical

1. Install the covers of the gantry.
2. Switch on F8 and F9 and switch to **SYSTEM/ON** at control panel.

Tests

1. Check the balancing function.

Refer to Tune-Up Instructions and perform a “**Calibration measurement**” (MI12-023.817.01.XX.XX).

Overview

The front side includes the following components:

- Heat exchangers
- Fans
- Front door
- Control panels
- Static markers
- Display panel

Heat Exchanger

Prerequisites

Tools and auxiliary equipment

- Wrench M32
- Adjustable wrench
- Instructions:
 - “Replacement of Parts: WCS - Water/Water“ (CT02-023.841.05.xx.xx)
 - “Replacement of Parts: WCS - Water/Air“ (CT02-023.841.06.xx.xx)
- Standard service kit

Time and manpower

- 1 person
- 1 hour

Safety



Avoid accident and injury or damage of parts.

Risk of accident and injury!

- ⇒ **Read and observe the safety information contained in the Biograph Safety Procedure and Biograph Safety Report.**
-

Preparation

1. Move the table to its lowest position.
2. Switch **OFF** the system at the control box.
3. **Switch OFF onsite power using the onsite ON/OFF switch!**
4. Remove the pallet from the PHS.
5. Open the gantry front door.
6. Drain the system as described in the instructions:
 - “Replacement of parts: WCS - Water/Water“, Section “Draining and filling the cooling system“
 - “Replacement of Parts: WCS - Water/Air“, Section “Draining and filling the cooling system“

Removing

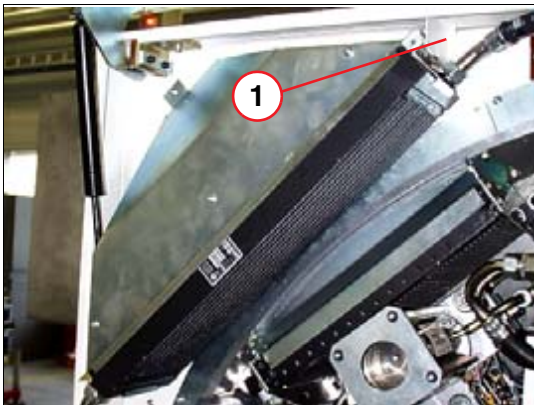


Fig. 74: Heat exchanger

1. All 3 heat exchangers are similar, top left is shown.
2. Loosen both cap nuts and remove the cooling hoses.
⇒ **Hold the hexagonal nut on the heat exchanger with a wrench whilst removing cap nuts.**
3. Remove the four mounting nuts (e.g. item 1) from the heat exchanger.
4. Remove the defective heat exchanger.

Installing

1. Install the new heat exchanger in the reverse order.
2. Ensure that the cooling hose connections are sealed (max. **40 Nm**).
3. Refill the system as described in the instructions:
“Replacement of Parts: WCS - Water/Water“, Section “Draining and filling the cooling system“
“Replacement of Parts: WCS - Water/Air“, Section “Draining and filling the cooling system“

Final steps

Mechanical

1. Ensure that the heat exchanger and cooling system are functioning properly.
2. Close the gantry front door and carefully insert the plexiglas ring.

3. Reattach pallet to PHS.

Tests

1. Perform a test scan

Fan for the heat exchanger

Prerequisites

Tools and auxiliary equipment

- Standard service kit.

Time and manpower

- 1 person
- 1 hour

Safety



Avoid accident and injury or damage of parts.

Risk of accident and injury!

- ⇒ **Read and observe the safety information contained in the Biograph Safety Procedure and Biograph Safety Report.**
-

Preparation

1. Move the table to its lowest position.
2. Switch **OFF** the system at the control box.
3. **Switch OFF onsite power using the onsite ON/OFF switch!**
4. Remove pallet from PHS.
5. Open the gantry front door.
6. Move gantries into service position. (MI12-001.813.01.xx.xx)
7. Remove the gantry rear cover.

Removing

1. Lower right fan

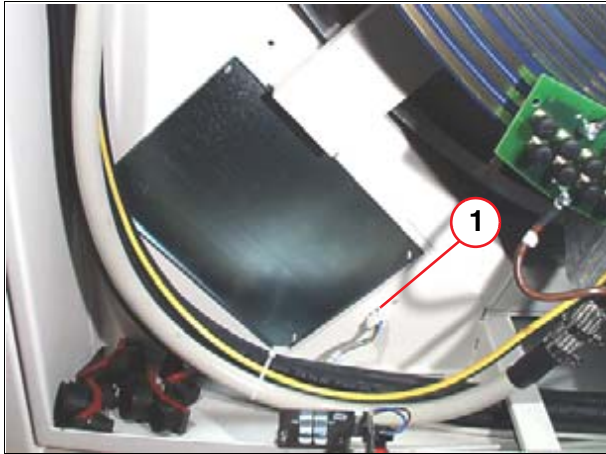


Fig. 75: Lower right fan

- Remove the screws from the rear fan cover.
- Disconnect M306 at cable connection (item 1) and release the plug from the gantry frame

2. Lower left fan



Fig. 76: Brake assembly

- Remove the brake assembly. Instructions can be found at ([Brake assembly / p. 259](#))
- Disconnect M305 at cable connection and release the plug from the gantry frame

3. Upper left fan



Fig. 77: Upper left fan

- Remove the connectors D344 X1, X3 and X4 from the D344 NOISE_CAN board, tie the cables back
 - Remove the screws from the rear fan cover.
 - Disconnect M304 at cable connection and release the plug from the gantry frame
4. Remove the screws from the front deflector and remove the deflector.
 5. Remove the 4 Hexagonal screws holding the fan and remove the defective fan through the opening at the back.

Installing

1. Install the new fan in the reverse order.

Final steps

Mechanical

1. Switch on the system.
2. Ensure that the fan and heat exchanger are functioning properly.
3. Replace covers
4. Move gantries back into scan position.
5. Reattach pallet to PHS.
6. If NOISE-CAN or rotation brake were removed, ensure they are functioning properly

Tests

1. Perform a test scan

Gantry front door

Prerequisites

Tools and auxiliary equipment

- Standard service kit.

Time and manpower

- 2 people
- 1 hour

Safety



Avoid accident and injury or damage of parts.

Risk of accident and injury!

- ⇒ **Read and observe the safety information contained in the Biograph Safety Procedure and Biograph Safety Report.**
-

Preparation

1. Move the table to its lowest and far back position.
2. Remove pallet from PHS.
3. Switch **off** the system.
4. **Switch OFF onsite power using the onsite ON/OFF switch!**
5. Open the gantry front door by loosening the 2 screws at the bottom of the front cover.



Removing

1. Loosen all cables necessary to remove the front door

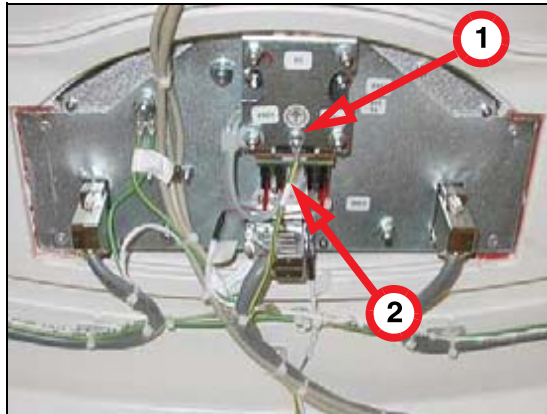


Fig. 78: Center cables

- Ground wire (pos. 1)
- Cable D2_X2 (pos. 2)

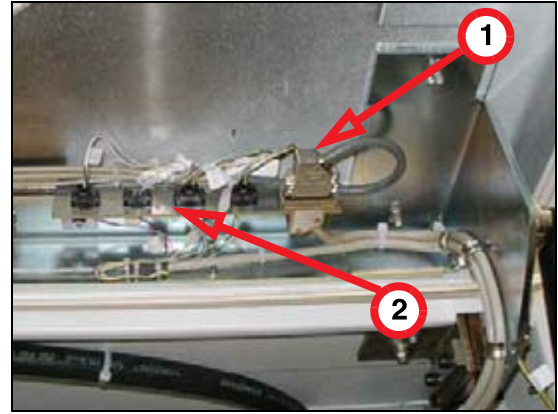


Fig. 79: Cables on right side of door

- E352_X1 (pos. 1)
- X327_a,b,c,d (pos. 2)

2. Mark the position of the left and right frames.

⇒ **This assures that the new gantry front door can be attached to the previous position during reinstallation.**

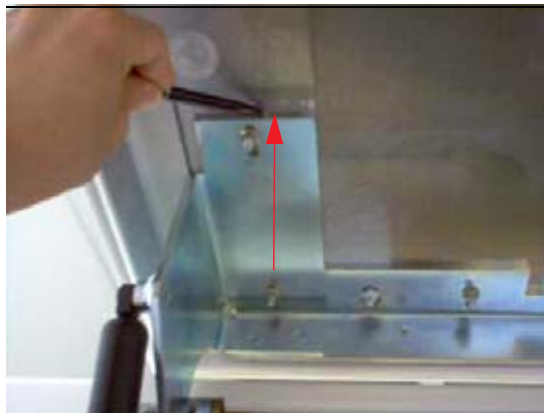


Fig. 80: Front cover marking, left

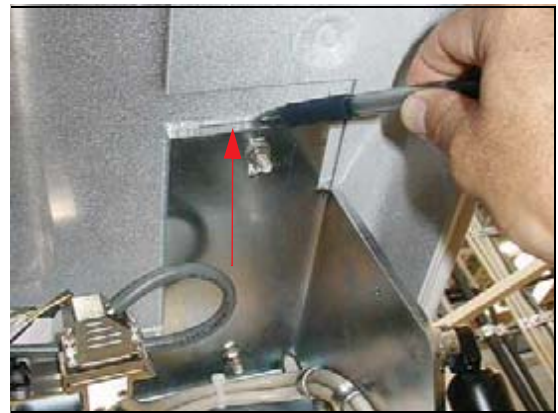


Fig. 81: Front cover marking, right

3. Loosen (do not remove) the 2 gantry door holding nuts by approximately 2mm.
 ⇨ These will help to support the door until it is ready to be removed.



Fig. 82: Door holding nut, left side

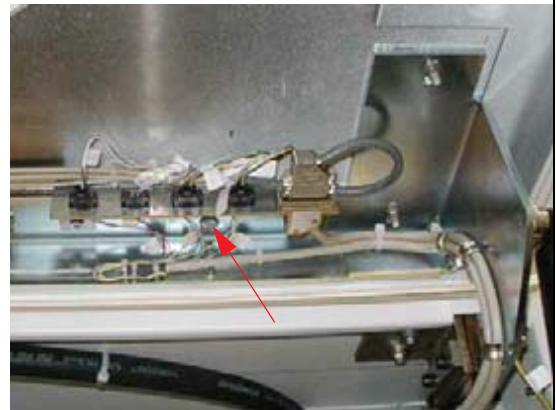


Fig. 83: Door holding nut, right side

4. Remove the rest of the nuts while **lightly** supporting the door. (the holding nuts will take most of the weight).
 ⇨ **You need a second person to do this!**
5. Lift the door then push it slightly towards the gantry to allow the 2 holding nuts to pass through the large part of their slots, and remove the door.

Installing

1. Install the 2 holding nuts on the new door.
2. Mount the door by passing the holding nuts through the large slots and pull the door away from the gantry to engage them.
3. Install the rest of the nuts and adjust the front door to the previously marked position.
4. Tighten the screws with a torque of **10 Nm**
5. Reinstall all cables in reverse order. Make sure that the cables are mounted as before using cable ties and cable clamps.



Final steps

Adjustments

1. Ensure that the door fits correctly by closing it. Adjust it if necessary, using the holding nuts.

Only when this does not help should the adjusting screws on the left and right side of front door (arrows) be used.



Fig. 84: Door adjustment screws, left



Fig. 85: Door adjustment, right side



2. Tighten the holding nuts with a torque of **10 Nm**.
3. If the door fits correctly, pay attention that switches S304 and S305 on the gantry lower left and right side are correctly adjusted gantry front door is closed (rotation of the gantry must stop if screw is opened).
4. Reattach pallet to PHS.

Tests

1. Perform a test scan.

Gantry Front Panels

Prerequisites

Tools and auxiliary equipment

- Standard service kit

Time and manpower

- 1 person
- 1 hour

Safety



Avoid accident and injury or damage of parts.

Risk of accident and injury!

- ⇒ **Read and observe the safety information contained in the Biograph Safety Procedure and Biograph Safety Report.**
-

Preparation

1. Move the table to its lowest position.
2. Switch **OFF** the system at the control box.
3. **Switch OFF onsite power using the onsite ON/OFF switch!**
4. Remove pallet from PHS.
5. Open the front door of the gantry.

Removing

1. Remove the panel



Fig. 86: Control panel

- Disconnect the plug at the display panel and cut ties to release cable.
- Remove the 6 nuts and remove the panel.

Installing

1. Install in the reverse order.

Final steps

Mechanical

1. Switch on the system.
2. Ensure that all of the functions are working properly.
3. Replace covers
4. Reattach pallet to PHS.

Tests

1. Perform a test scan

Static light markers Sensation 10/16/Cardiac/40/64/Cardiac 64

Prerequisites

Tools and auxiliary equipment

- Standard service kit

Time and manpower

- 1 person
- 1/2 hour

Safety



Avoid accident and injury or damage of parts.

Risk of accident and injury!

- ⇒ **Read and observe the safety information contained in the Biograph Safety Procedure and Biograph Safety Report.**
-

Preparation

1. Move the table to its lowest position.
2. Switch **OFF** the system at the control box.
3. **Switch OFF onsite power using the onsite ON/OFF switch!**
4. Remove pallet from PHS.
5. Open the front door of the gantry.

Removing

- ⇒ **Replacement of all 3 markers is similar**

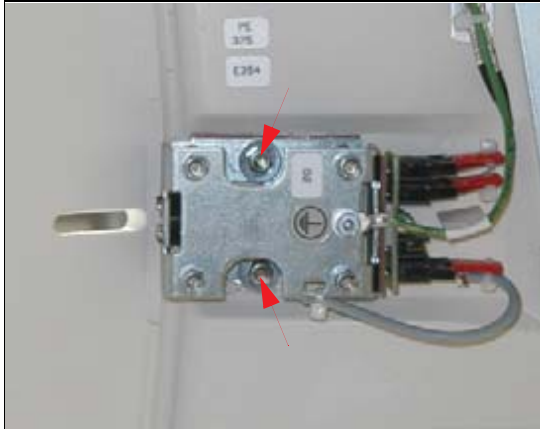


Fig. 87: Location of securing nuts

- Disconnect cable connection (pos 1)
- Disconnect ground cable (pos 2)
- Remove the 2 securing nuts (pos 3)

Installing

1. Installation is in the reverse order of removal.
2. Tighten the screws.
3. Connect all cables according to the original configuration.

Final steps

Mechanical

1. Replace covers.
2. Switch on the system.
3. Ensure that the markers are functioning properly.
4. Reattach pallet to PHS.

Tests

1. Perform a test scan

Adjustments

1. Adjusting markers

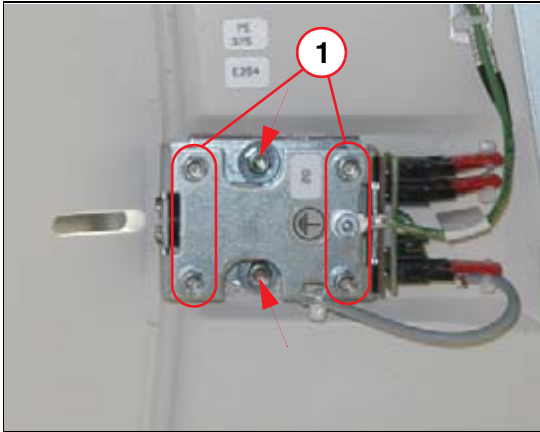


Fig. 88: Adjustment of position

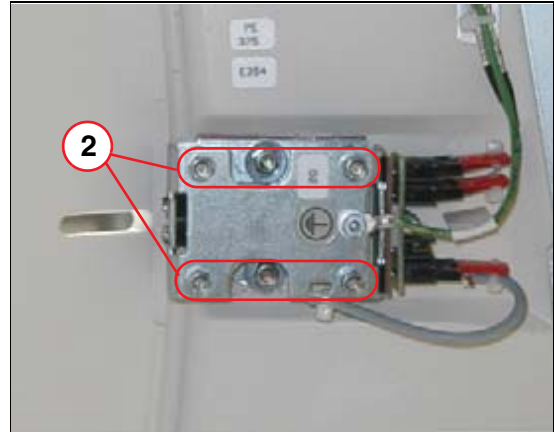


Fig. 89: Adjustment of skew

1. Adjust the angle of the markers by loosening the 2 securing nuts (arrows).
2. Adjust the position of the marker via the 4 adjusting nuts. Each pair (circle 1) should be adjusted equally and opposite to each other. For example, when turning one pair once in clockwise direction, turn the other pair once in counter-clockwise direction. This adjustment applies only to the top marker.
3. Adjust the skew of the markers via the 4 adjusting nuts. Again, each pair (circle 2) should be adjusted equally and opposite to each other. For example, when turning one pair once in clockwise direction, turn the other pair once in counter-clockwise direction.

2. Adjusting side markers (opening/closing of the gantry door is not described)

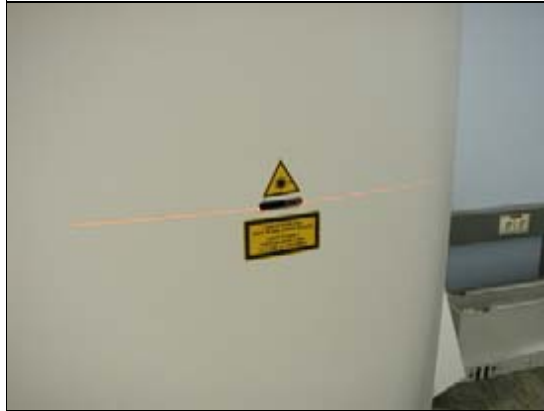


Fig. 90: Initial setting of the side marker

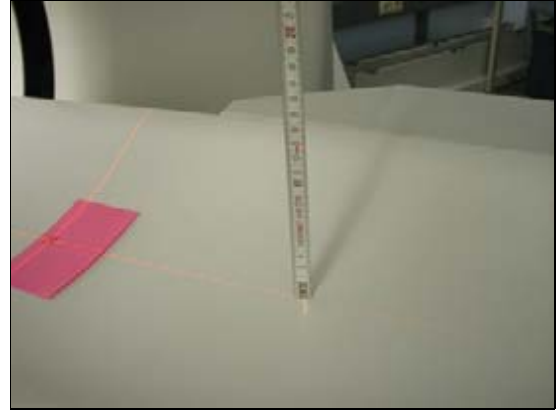


Fig. 91: Checking the side marker

1. Adjust the angle of the markers; then tighten the 2 securing nuts. The markers should be aimed at the opposite marker slot.
2. Close the gantry door, raise the table height to 125, and feed the tabletop into the scan field. Measure the distance from the tabletop to the marker light at each end of the marker line.
3. Adjust the skew of the markers with the 4 adjusting nuts to make these measurements equal. Use the adjusting nuts in pairs (circle 1)([Fig. 88 / p. 126](#))
4. Re-check the previous settings and adjust as necessary.

3. Adjusting the top marker (opening/closing of the gantry door is not described)

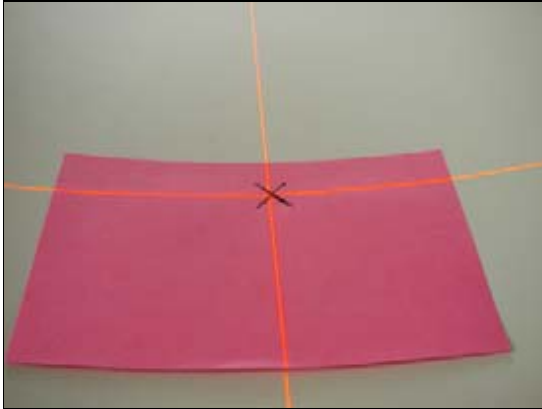


Fig. 92: Adjusting top marker

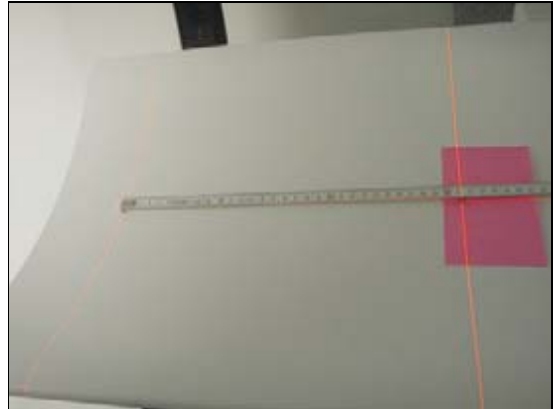


Fig. 93: Measuring distance

1. Close the gantry door, raise the table height to 125, and feed the tabletop into the scan field. Place some tape on the tabletop and mark the center of the tabletop at exactly 31cm from the rotating markers.
2. Adjust the angle of the marker. The marker line along the length of the table should be aimed at the mark on the table. Tighten the 2 securing nuts.
3. Adjust the position of the marker so that the line across the table hits the mark on the table. Use the adjusting nuts in pairs (circle 2)([Fig. 89 / p. 126](#))
4. Measure the distance from the rotating markers at either side of the table and adjust the skew of the markers; use the adjusting nuts in pairs (circle 1).
5. Re-check the previous settings and adjust as necessary.

Display Panel

Prerequisites

Tools and auxiliary equipment

- Standard service kit

Time and manpower

- 1 person
- 1 hour

Safety



Avoid accident and injury or damage of parts.

Risk of accident and injury!

- ⇒ **Read and observe the safety information contained in the Biograph Safety Procedure and Biograph Safety Report.**
-

Preparation

1. Move the table to its lowest position.
2. Switch **OFF** the system at the control box.
3. **Switch OFF onsite power using the onsite ON/OFF switch!**
4. Remove pallet from PHS.
5. Open the front door of the gantry.

Removing

1. Display Panel



Fig. 94: Display panel

- Disconnect the 3 cable connectors.
- Remove the 3 nuts (circled).
- The panel comes out from the front; carefully close the gantry door until you can reach the panel to withdraw it.

Installing

1. Install in the reverse order.

Final steps

Mechanical

1. Switch on the system.
2. Ensure that the panel is functioning properly.
3. Close covers
4. Reattach pallet to PHS.

Tests

1. Perform a test scan, ensure that all display functions are operational.

Overview

- The first figures show an overview of all parts involved in the Data Measurement System DMS and its location inside the gantry.

The next sections contain detailed descriptions of how to exchange all major parts of the DMS.

NOTE

Follow the C-MOS guidelines when touching the DMS

NOTE

UHR comb may also be named UHR-phi comb

UHR-plus comb may also be named advanced UHR comb

The overview shows all parts which are involved in the data acquisition.

This image shows a Sensation 40/64 or Cardiac 64. The Open does not include an UHR

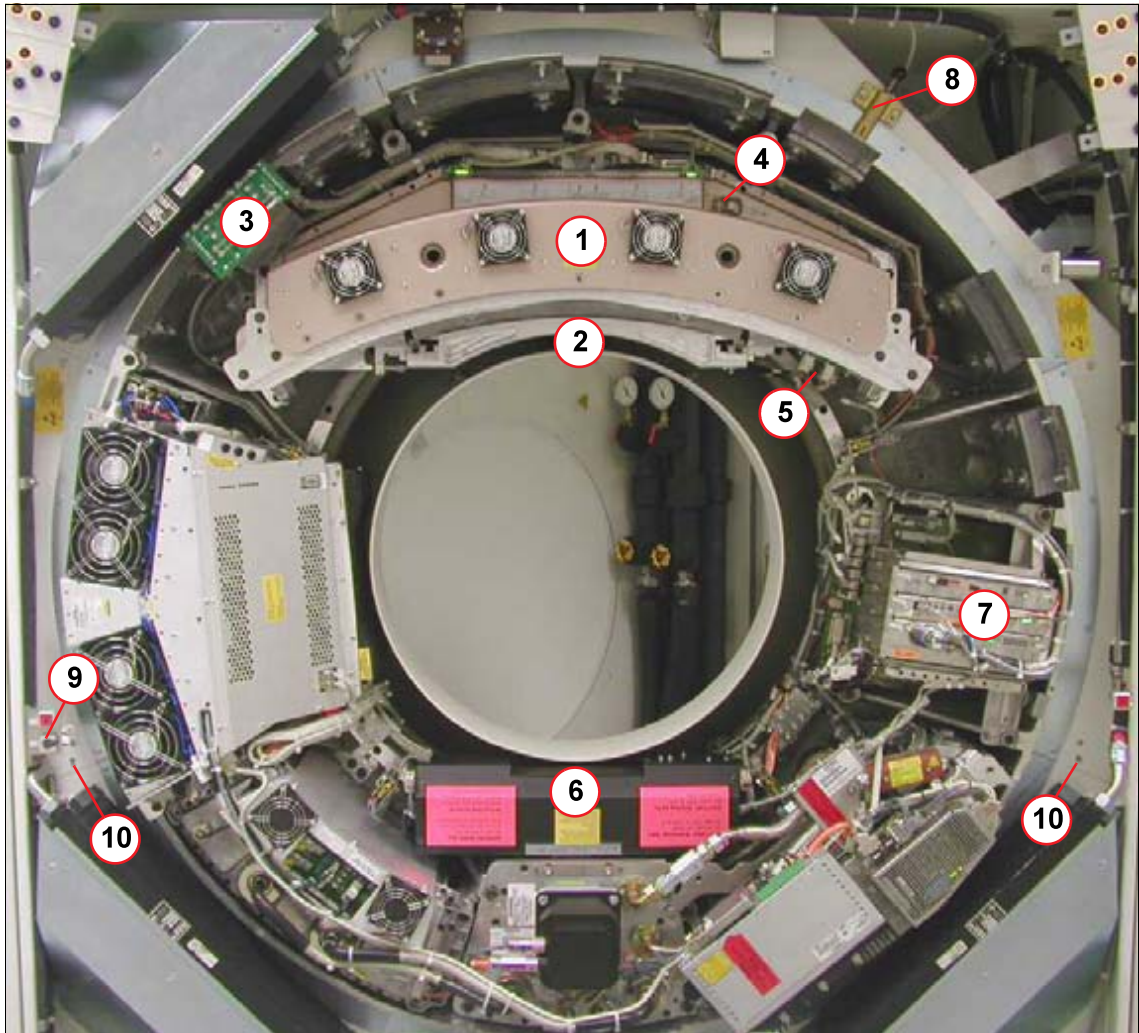


Fig. 95: Overview

- Pos. 1 DMS
- Pos. 2 UHR Phi or UHR Phi-z comb
- Pos. 3 DMS Power supply
- Pos. 4 Reference monitor board
- Pos. 5 UHR Control board
- Pos. 6 Tube collimator
- Pos. 7 C-Box
- Pos. 8 Safety bolt
- Pos. 9 Safety switch S301
- Pos. 10 All possible ground connections

- The complete DMS, positioned at 12 o' clock. Channel 1 is on the left side.
The 2 inner screws are of type M10 x 55, the outer 4 are M10 x 65

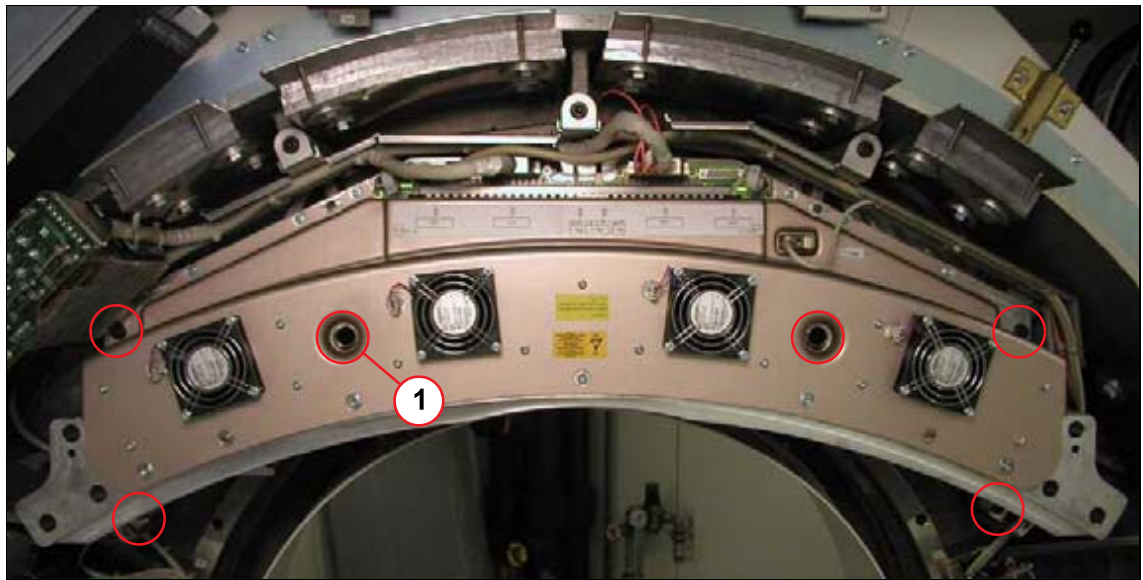


Fig. 96: DMS
Pos. 1 6 x DMS screws

- Be careful when tightening the cover. There are two different torques used at the same type of screws.

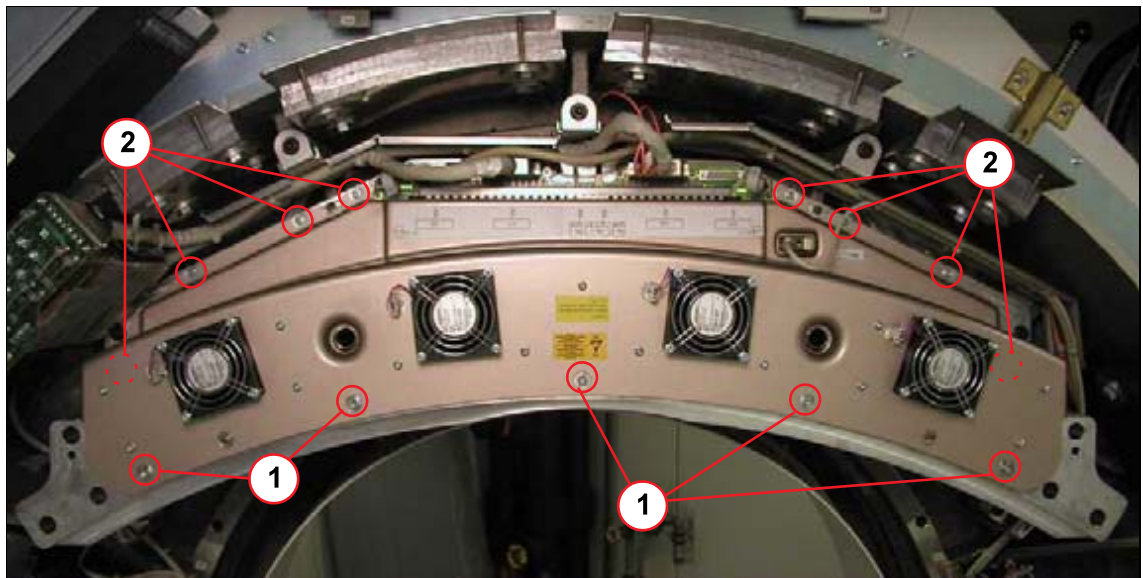


Fig. 97: DMS Cover
Pos. 1 5 x screws tightened with 10 Nm
Pos. 2 8 x screws tightened with 3 Nm

- All replaceable parts inside the DMS

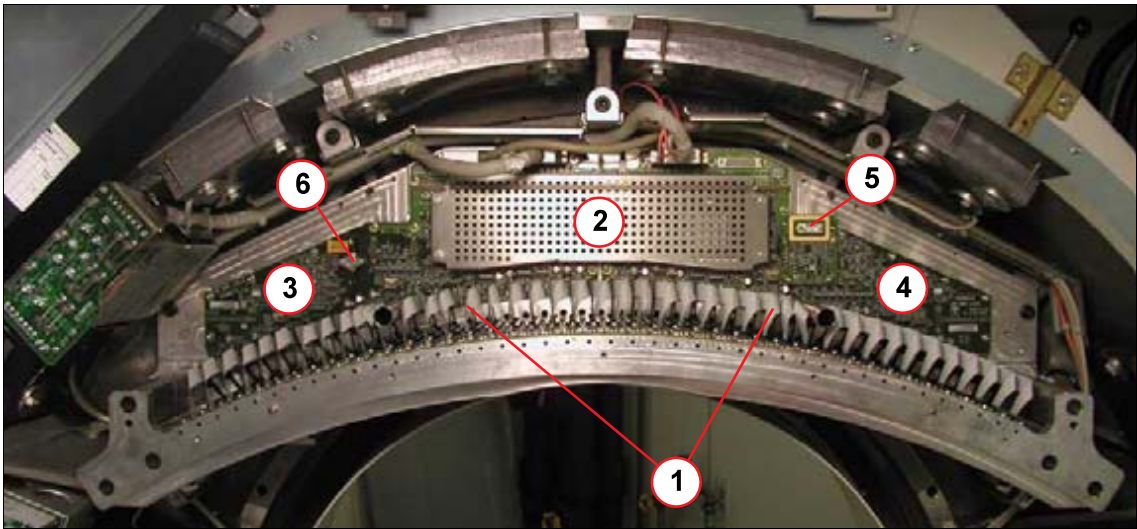


Fig. 98: DMS without cover

- | | |
|--------|-------------------------|
| Pos. 1 | Modules |
| Pos. 2 | DAS controller |
| Pos. 3 | Right backplane |
| Pos. 4 | Left backplane |
| Pos. 5 | Reference monitor board |
| Pos. 6 | Temperature switch F1 |

- This figure shows the cabling of the DMS

In addition the following connection exists:

- X3 Ref. mon. is connected to FOC D514 via W567 and X14 at the c-box backplane
- X80 UHR control is connected to FOC D514 via W594 and X5 at the c-box backplane

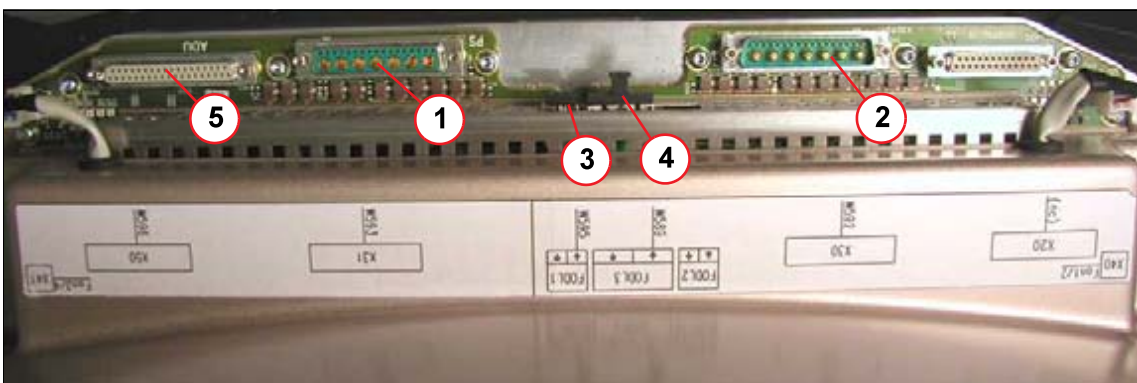


Fig. 99: DMS cabling

- | | |
|--------|---|
| Pos. 1 | Plug X31 - connected to DMS power supply X200 via cable W593 |
| Pos. 2 | Plug X30 - connected to DMS power supply X100 via cable W592 |
| Pos. 3 | Plug FODL1 - connected to slip ring transmitter via cable W595 |
| Pos. 4 | Plug FODL3 - connected to ADU via cable W589 |
| Pos. 5 | Plug X50 - connected to ADU D513 via W596 and X1 at c-box backplane |

- All parts for the UHR comb

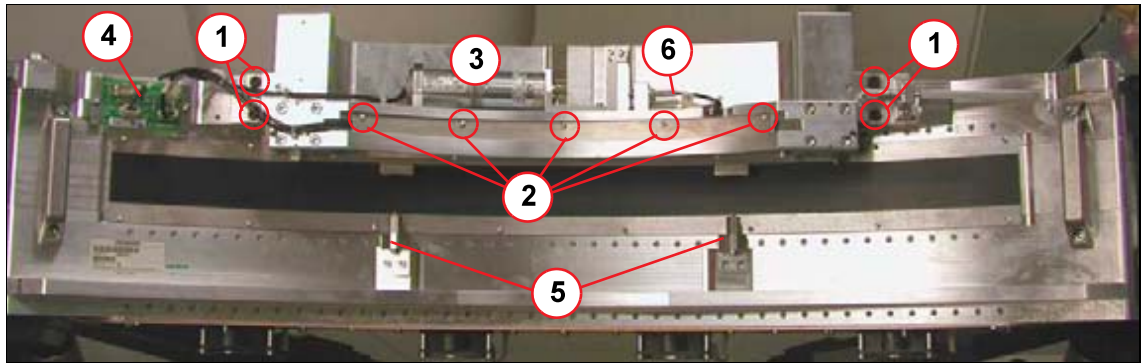


Fig. 100: UHR comb

- Pos. 1 4 screws holding the complete UHR
- Pos. 2 Screws of protective cover
- Pos. 3 UHR Motor
- Pos. 4 UHR control board
X80 is connected to FOC board D514 via W594 X5 at the C-box backplane
- Pos. 5 UHR guide pins
- Pos. 6 Limit switches (not installed in newest systems)

Modules

Prerequisites

Tools and auxiliary equipment

- Standard tool kit
- Torx tool kit with adapter for torque wrench
- Torque wrench 1.0 Nm, 1.4 Nm, 3 Nm and 10 Nm

Time and manpower

- Estimated time to repair: 2 - 3 hours depending on the position of the module
- The delivery of a module order has to be released by CS HSC 221

Safety



Avoid accident and injury or damage of parts.

Risk of accident and injury!

- ⇒ **Read and observe the safety information contained in the Biograph Safety Procedure and Biograph Safety Report.**

Preliminary steps

1. Move patient down.
2. Remove pallet from PHS.
3. Open gantry front door and disable rotation with S301
4. Switch off the gantry at the control box (CompOn)
5. Switch off DMS power supply using F9 in PDC and check that the LED's at the DMS go off
6. Turn DMS into the 12 o'clock position by hand
7. Lock safety bolt to prevent unintentional gantry movement
8. Apply ground connection between stator and rotor



Removing

NOTE

Follow the C-MOS guidelines when touching the DMS

1. Unplug X3 at the reference monitor board
2. Unplug the connectors for fans X40 and X41
3. Remove 13 screws from the DMS cover using a 5mm Allen key
4. Remove DMS cover
5. Locate defective module by means of the engraved numbers at the cradle

NOTE: If DMS is in the 12 o'clock position, channel 1 is on the left side.

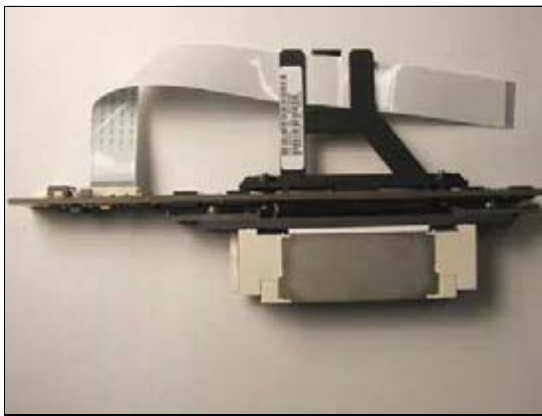


Fig. 101: Module

Center modules

If the problem is in the inner slots (slots 19 - 24) the defective slot will be filled with adjacent modules of the detector, and the spare module will be placed at least 4 slots away from the center. Refer to the appendix for details. ([Replacement of a center module / p. 188](#))

The allocation of channels and slots is shown in this table:

Channels	Slot count	Channels	Slot count
1 - 16	1	337 - 352	22
17 - 32	2	353 - 368	23
33 - 48	3	369 - 384	24
49 - 64	4	385 - 400	25
65 - 80	5	401 - 416	26
81 - 96	6	417 - 432	27
97 - 112	7	433 - 448	28

Channels	Slot count	Channels	Slot count
113 - 128	8	449 - 464	29
129 - 144	9	465 - 480	30
145 - 160	10	481 - 496	31
161 - 176	11	497 - 512	32
177 - 192	12	513 - 528	33
193 - 208	13	529 - 544	34
209 - 224	14	545 - 560	35
225 - 240	15	561 - 576	36
241 - 256	16	577 - 592	37
257 - 272	17	593 - 608	38
273 - 288	18	609 - 624	39
289 - 304	19	625 - 640	40
305 - 320	20	641 - 656	41
321 - 336	21	657 - 672	42



6. Remove module cable from DMS backplane

Hint: **To unlock the plug, pull the locking device. To close the plug press down on the locking device**

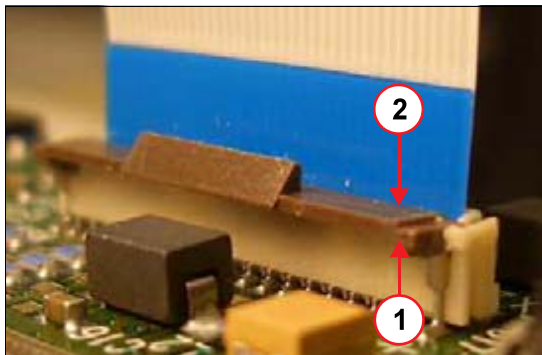


Fig. 102: Connection of module cable

Pos. 1 to open press here

Pos. 2 to close press here

1.

7. Remove the 3 screws holding the module.



Fig. 103: Screws of module

Pos. 1 1 x M3 x 8, tightened with a torque of 1.0 Nm using a T10 tool

Pos. 2 2 x M4 x 20, tightened with a torque of 1.4 Nm using a T20 tool

In case of slots 14 to 29

Remove the DAS controller for easier access

In case of slots 11,12 or 31,32

Remove the protective tube before working on the modules. The protective tube can be removed by turning it counterclockwise. Use a screwdriver to turn it.

When tightening the protective tube, use a torque of **5 Nm**

In case of slots 31,32 and 33

Remove the reference monitor for easier access

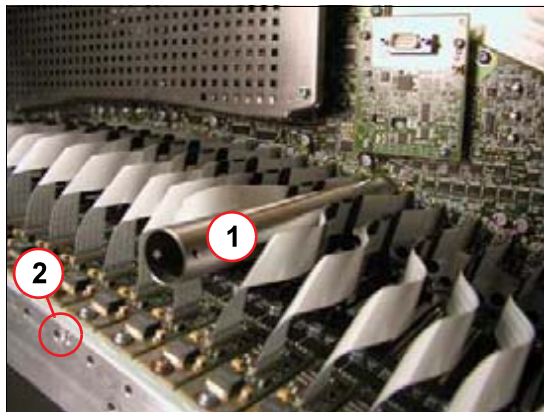


Fig. 104: Protecting tube

Pos. 1 Protecting tube

Pos. 2 Engraved position of slot

8. Remove the old module by pulling on the plastic handle.

Hint: Do not stress the module.

? Loosen also the adjacent module if necessary

Do not touch any part of the PCB



Fig. 105: Removing a module

Installing



1. Insert new module and align it toward the rotating plane
2. While tightening the 3 screws, press the module toward the rotating plane
Tighten the screws next to the handle with **1.4 Nm**, and the one in front with **1.0 Nm**
([Fig. 103 / p. 139](#))
3. Connect the module cable at the backplane
4. Reinstall if necessary: DAS controller, protective tube, or reference monitor
5. Install the DMS cover
6. Reconnect fans X40 and X41
7. Tighten the screws for the DMS cover with **3 and 10 Nm** according to overview
([Fig. 97 / p. 133](#))
8. Reconnect reference monitor X3

Final steps

- Remove ground connection and unlock safety bolt
- Release all service switches and switch on gantry
- Close gantry and power up
- Reattach pallet to PHS.

- Perform a Z-Position check to ensure that the alignment of the new module is ok
([Z-position check of modules / p. 191](#))
(beginning with software version VB20X, this test will be implemented in the tune-up)
- Depending on the position of the defective module, perform a tune-up as described in the guided tours

Options > Local Service > Tuneup > Module (periph.)
or **Module (central)**

Complete DMS

Prerequisites

Tools and auxiliary equipment

- Standard toolkit
- Long version of 8 mm Allen key
- Torque wrench 80 Nm
- Service box
- Vertical motor service interface
- Tools required for DMS replacement:
 - Lifting device kit: part number 10244855
 - Contents: PHS Vertical Motor Power Interface, Lifting device, instructions

Time and manpower

Estimated repair time: 7 hours (2.5 hours for mechanical work)

Safety



Avoid accident and injury or damage of parts.

Risk of accident and injury!

⇒ **Read and observe the safety information contained in the Biograph Safety Documents.**

Preliminary Steps

1. Move patient table out
2. Remove pallet from pedestal
3. Open gantry front door
4. Switch off gantry at control box (CompOn) and disable S301
5. Switch off DPU using F9 in the PDC
6. Turn DMS to 12 o'clock position by hand
7. Lock safety bolt to prevent unintentional gantry movement(
8. Apply ground connection between stator and rotor
9. Remove the center plastic cover of the gantry

Preparing the transport boxes

1. Open both DMS transport boxes and remove the service tools

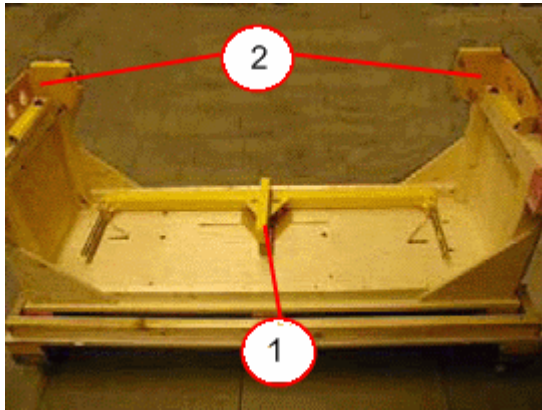


Fig. 106: DMS transport box

Pos. 1 DMS lifting bar

Pos. 2 DMS transportation brackets

2. Install the supported wheels on both transport boxes. Use the supplied jack and crank to lift the DMS transport box for easy installation of wheels.



Fig. 107: DMS transport box with inst. wheels

Removing

1. Disconnect all cables located at the backplane of DMS. Mark optical fiber cables W589 and W590 before removal.

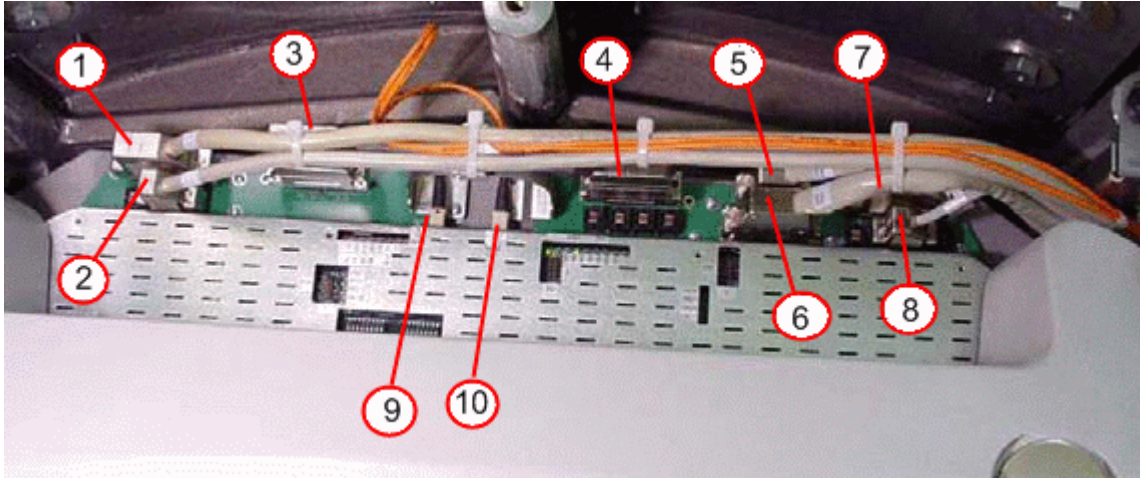


Fig. 108: DMS cabling

- | | |
|---------|---|
| Pos. 1 | Plug J6 - connected to ADU with cable W553-M via C-box backplane X1 |
| Pos. 2 | Plug J9 - connected to C-box backplane X19 via cable W554-M (fan power) |
| Pos. 3 | Plug P45 - connecting left and right motherboard |
| Pos. 4 | Plug J5 - connecting left and right motherboard |
| Pos. 5 | Plug DMS PS2 - connected to DPU with cable W555-M via C-box backplane X18 |
| Pos. 6 | Plug DMS PS1 - connected to DPU with cable W563-M via C-box backplane X17 |
| Pos. 7 | Plug J3 - connected to reference conversion box P1 via cable W564-M |
| Pos. 8 | Plug J4 - connected to C-box backplane X19 via cable W554-M (fan power) |
| Pos. 9 | Plug J8 - connected to ADU X4 via fiber-optical cable W589 |
| Pos. 10 | Plug J7 - connected to transmitter at slip ring X4 via fiber optical cable W590 |

2. Disconnect the ground cable at the right side of the DMS
3. Remove the 2 screws attaching the cable holder.

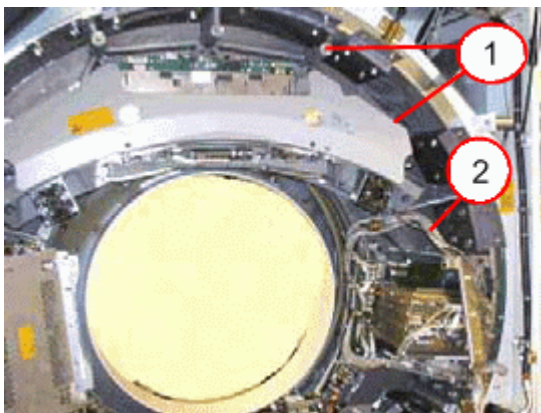


Fig. 109: DMS with cable holder removed

- | | |
|--------|---|
| Pos. 1 | Position of the 2 screws fastening the cable holder |
| Pos. 2 | Cable tree removed |

4. Carefully remove the bundle of DMS cables.
Disconnect the connector from the light marker to avoid damage.
5. Remove the two outermost M8 DMS screws (once the lifting bar is attached these screws may no longer be accessed).

Do not yet touch the inner 2 DMS screws

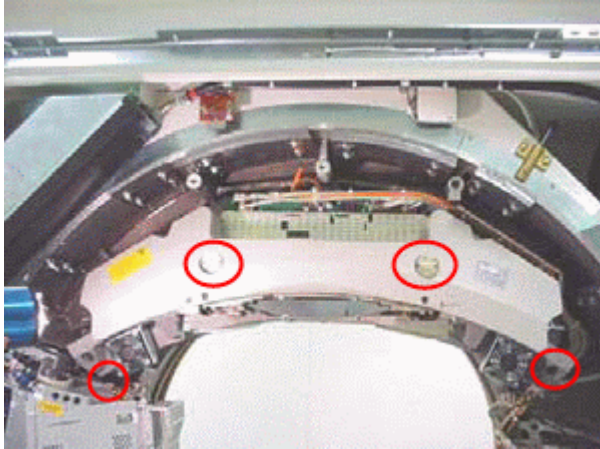


Fig. 110: DMS screws (torque 80 Nm)

6. Unlock the safety bolt and disconnect the ground cable
7. Carefully rotate the DMS by hand into a 6 o'clock position (refer to cable tree).
8. Lock the safety bolt and reapply the ground connection.
9. Unplug X5 at C-box backplane and remove all necessary cable ties to free cable W407
10. Mount lifting bar to handle the DMS via the patient table. Secure the lifting bar on both sides with the security pin.



Fig. 111: DMS lifting bar for installation

Pos. 1 Security pin

11. Connect the PHS Vertical Motor Service Interface to the bed.

NOTE

Consult the instructions that come with the PHS Vertical Motor Service Interface if needed.

CAUTION

Use care when raising or lowering the PHS with the Vertical Motor Service Interface. The limit switches are not monitored when the system is off.

Limit switches may be damaged if PHS is raised all the way up.

⇒ Only raise PHS high enough to lift and extract DMS.

12. Installing the lifting device.

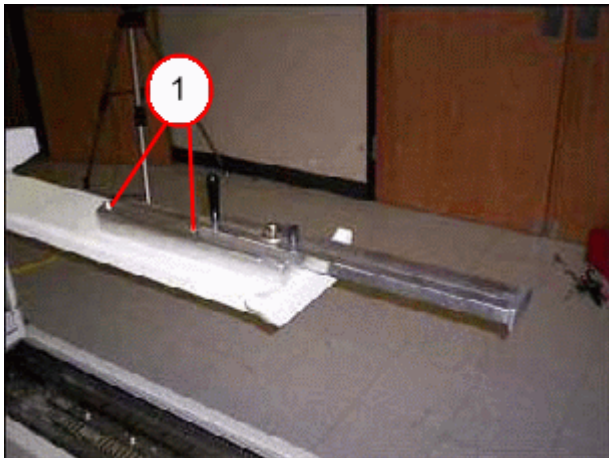


Fig. 112: Biograph Lifting device attached to PHS vertical support

Pos. 1 Bolts

13. Position the lifting device just above the lifting bar and secure with strap.

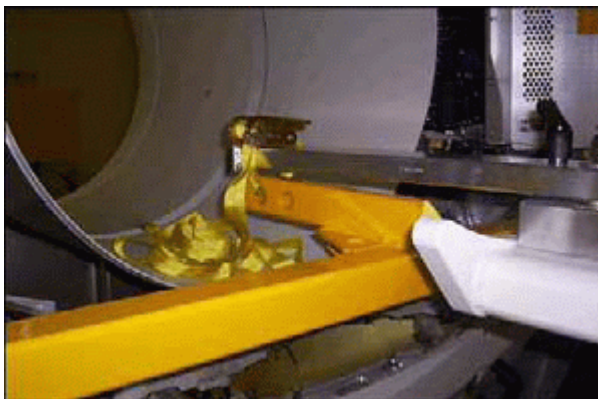


Fig. 113: Attaching lifting bar to lifting device

14. Position the service box to the side of the PHS to pick up the removed DMS.

15. Remove the two inner screws (M10, 8 mm Allen key) while observing the tension of belt

To access the inner 2 screws (M10), first remove the 2 round protective covers by loosening the 2 screws M3 x 8 mm.

NOTE**Use a long version of the 8 mm Allen key!**

Fig. 114: DMS protective cover (torque 100 Ncm)

16. Gently pull the DMS away from the rotating frame while observing the tension of belt. Left side first and then the right side.
17. Once the DMS is free, move the tabletop out of the gantry.
18. Use the lifting device to rotate the DMS over the service box then lower the table using the Vertical Motor Service Interface.
19. Mount transportation brackets and attach old DMS in transportation box.
(For reference, refer to the new DMS in its box).
20. Dismount lifting bar
21. Remove transportation box with old DMS

Installing

1. Attach lifting bar to new DMS
2. Position transportation box with new DMS beside the PHS
3. Attach belt at lifting bar. Apply tension to the belt.
4. Remove the two screws holding the brackets on the transport box.
5. When the DMS is free, remove the transport brackets from the DMS.
6. Rotate the DMS toward the gantry.
7. Move tabletop towards the gantry.
8. Remove the DMS transport box
9. Carefully move DMS into gantry.

To adjust the height use the Vertical Motor Service Interface.

Once in position, push firmly into the pins, left side first and then the right side.

10. Insert the 2 inner DMS screws and tighten with a torque of 80Nm
11. Remove lifting device.
12. Unlock safety bolt and disconnect ground cable
13. Turn DMS to 12 o'clock position by hand
14. Lock safety bolt and reapply ground connection
15. Insert the two outermost DMS screws and tighten with a torque of 80Nm
16. Reattach the cable holder and tighten with a torque of 80Nm
17. Reconnect all cables in reverse order

Tighten the ground cable at DMS (right side) with a torque of 7Nm

Final Steps

- Remove ground connection and unlock safety bolt
- Release all service switches, close gantry, remove lifting device, reattach pallet to PHS, power up and wait at least 15 minutes to warm up the DMS
- Balance the gantry

Options > Local Service > Tuneup > Adjustments > Balance Gantry

Perform this adjustment according the manual (MI12-023.817.01.xx.xx).

- Perform a tune-up as described in the guided tours

Options > Local Service > Tuneup > Installation / Detector

Perform all adjustments according the manual (MI12-023.817.01.xx.xx).

NOTE

Ensure that all entries were removed from the bad channel table before starting the tune-up.

For details, refer to testing and troubleshooting manual (MI12-023.840.01.xx.xx).

- Perform a quality check and backup
- Fill out the return form delivered with the detector
- Reattach all service supports to the transport boxes including the wheels.

DAS Controller

Prerequisites

Tools and auxiliary equipment

- Standard tool kit
- Torx tool kit with adapter for torque wrench
- Torque wrench 1.3 Nm, 3 Nm and 10 Nm

Time and manpower

- Estimated time to repair: 0.7 hour

Safety



Avoid accident and injury or damage of parts.

Risk of accident and injury!

- ⇒ **Read and observe the safety information contained in the Biograph Safety Procedure and Biograph Safety Report.**
-

Preliminary steps



1. Move patient table down
2. Remove pallet from PHS.
3. Open gantry front door and disable rotation with S301
4. Switch off gantry at control box (CompOn)
5. Switch off DMS power supply using F9 in the PDC
6. Turn the DMS to the 12 o'clock position by hand
7. Lock safety bolt to prevent unintentional gantry movement
8. Apply ground connection between stator and rotor

Removing

NOTE

Follow the C-MOS guidelines when touching the DMS

1. Unplug X3 at the reference monitor board
2. Unplug the connectors for fans X40 and X41
3. Remove 13 screws from the DMS cover using a 5 mm Allen key
4. Remove DMS cover
5. Disconnect the 2 fiber optical cables labeled FODL1 and FODL3 at the DAS Controller
6. Remove 4 screws from the DAS Controller cover using a 3 mm Allen key

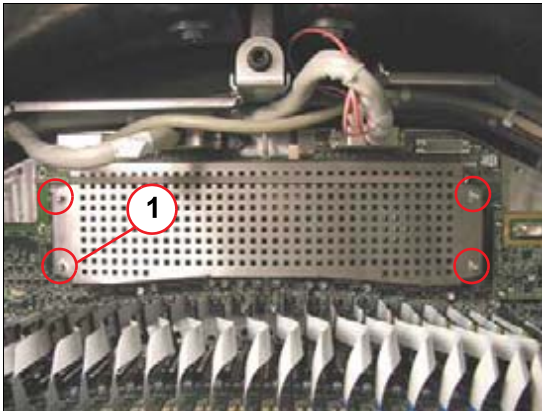


Fig. 115: DAS controller with cover

Pos. 1 Tighten all screws with 3 Nm

7. Remove the cover
8. Remove the 8 screws holding the DAS controller at the backplanes

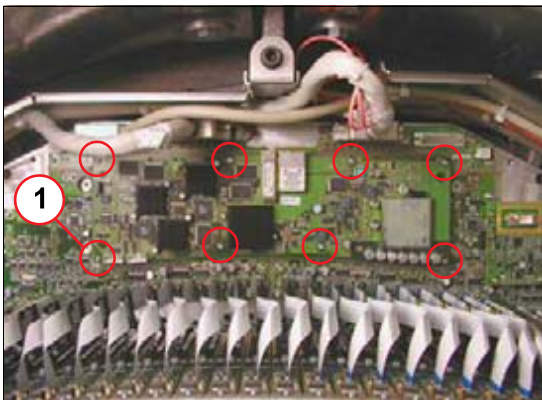


Fig. 116: DAS controller without cover

Pos. 1 Tighten all screws with 1.4 Nm

9. Remove the DAS Controller by pulling both sides at once

Installing

1. To align the controller board and the two backplanes loosen the 18 Torx screws T20 of the left backplane
2. Insert the DAS Controller board
3. Tighten all accessible screws of the left backplane with a torque of **1.4 Nm**
4. Remove the DAS Controller board
5. Tighten the remaining screws of the left backplane with a torque of **1.4 Nm**.
6. Install the DAS Controller board
-  7. Tighten the 8 screws for attaching the DAS Controller with **1.3 Nm**
8. Tighten the 4 screws for attaching the DAS controller cover with **3 Nm**
9. Tighten the screws for the DMS cover with **3 and 10 Nm** according to the overview ([Fig. 97 / p. 133](#))

Final steps

- Remove the ground connection and unlock the safety bolt
- Release all service switches and switch on gantry
- Perform a disk test
- Close covers.
- Reattach pallet to PHS.
- Use spiral and sequence modes to test the system

DMS backplanes

Prerequisites

Tools and auxiliary equipment

- Standard tool kit
- Torx tool kit with adapter for torque wrench
- Torque wrench 1.4 Nm, 3 Nm and 10 Nm

Time and manpower

- Estimated time to repair: 1.5 hours

Safety



Avoid accident and injury or damage of parts.

Risk of accident and injury!

- ⇒ **Read and observe the safety information contained in the Biograph Safety Procedure and Biograph Safety Report.**

Preliminary steps

1. Move patient table out and down.
2. Remove pallet from PHS.
3. Open gantry front door and disable rotation with S301
4. Switch off the gantry at the control box (CompOn)
5. Switch off the DMS power supply using F9 in PDC and check that the LED's at DMS go off
6. Turn the DMS into the 12 o'clock position by hand
7. Lock the safety bolt to prevent unintentional gantry movement
8. Apply ground connection between stator and rotor



Removing

NOTE

Follow the C-MOS guidelines when touching the DMS

1. Unplug X3 at the reference monitor boards
2. Unplug the connectors for fans X40 and X41
3. Remove 13 screws from the DMS cover using a 5 mm Allen key
4. Remove DMS cover
5. Disconnect the 2 fiber optical cables labeled FODL1 and FODL3 at the DAS Controller
6. Remove 4 screws from the cover of the DAS Controller using a 3 mm Allen key
7. Remove 8 screws attaching the DAS controller to the backplane
8. Remove the DAS Controller by pulling both sides simultaneously

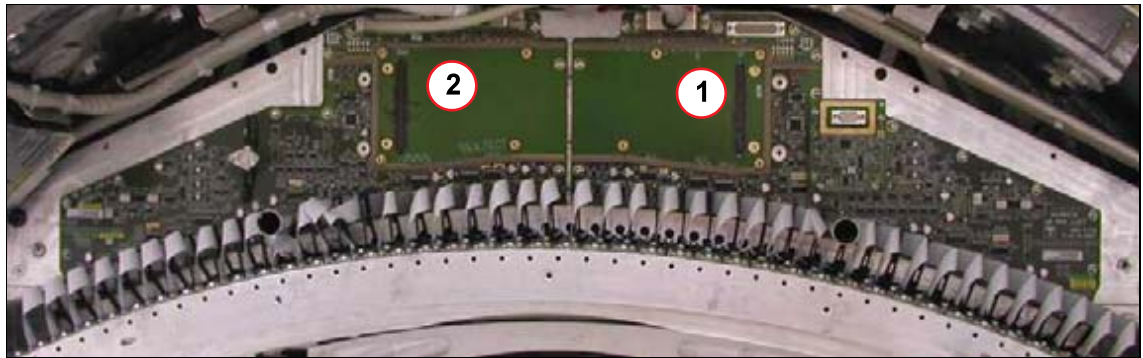


Fig. 117: Backplanes

Pos. 1 Left backplane
Pos. 2 Right backplane

9. Disconnect all module cables from the defective backplane
10. Remove the 18 Torx screws T20 of the defective backplane
11. Take out the backplane

Installing

1. Insert the new backplane, but do not tighten the 18 screws at this point
2. Insert the DAS Controller and attach some of the 17 screws to align the position of backplane and DAS controller
3. Tighten all accessible screws with a torque of **1.4 Nm**
4. Remove the DAS Controller and tighten the remaining screws with a torque of **1.4 Nm**.
5. Reinstall in reverse order
6. Tighten the 8 screws of the DAS Controller with **1.4 Nm**
7. Tighten the 4 screws of the DAS Controller cover with **3 Nm**
8. Tighten the screws of the DMS cover with **3 and 10 Nm** according to overview (Fig. 97 / p. 133)



Final steps

- Remove the ground connection and unlock the safety bolt
- Release all service switches and switch on gantry
- Close covers and reattach pallet to PHS.
- Check system by performing spiral and sequence modes

Reference monitor board

Prerequisites

Tools and auxiliary equipment

- Standard tool kit
- Torx tool kit with adapter for torque wrench
- Torque wrench 1.3 Nm, 3 Nm and 10 Nm

Time and manpower

- Estimated time to repair: 0.5 hours

Safety



Avoid accident and injury or damage of parts.

Risk of accident and injury!

- ⇒ **Read and observe the safety information contained in the Biograph Safety Procedure and Biograph Safety Report.**
-

Preliminary steps



1. Move patient table out and down.
2. Remove pallet from PHS.
3. Open gantry front door and disable rotation with S301
4. Switch off the gantry at the control box (CompOn)
5. Switch off the DMS power supply using F9 in PDC and check that the LED's at the DMS go off
6. Turn the DMS to the 12 o'clock position by hand
7. Lock the safety bolt to prevent unintentional gantry movement
8. Apply a ground connection between stator and rotor



Removing

NOTE

Follow the C-MOS guidelines when touching the DMS

1. Unplug X3 at the reference monitor board
2. Remove 13 screws from the DMS cover using a 5 mm Allen key
3. Unplug the connectors for fans X40 and X41
4. Remove the DMS cover
5. Loosen the 4 screws attaching the reference monitor board

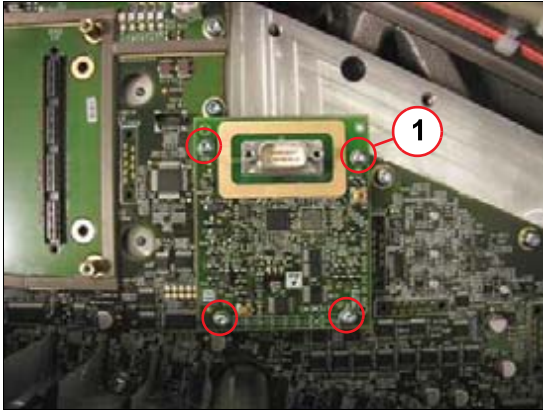


Fig. 118: Reference monitor

Pos. 1 Tighten all screws with 3 Nm

6. Remove the reference monitor board

Installing



1. Install the new reference monitor board
2. Tighten the 4 screws with a torque of **1.3 Nm**
3. Install all parts in reverse order
4. Tighten the screws for the DMS cover with **3 and 10 Nm** according to overview ([Fig. 97 / p. 133](#))

Final steps

- Remove the ground connection and unlock the safety bolt
- Release all service switches and switch on gantry
- Close all covers and reattach pallet to PHS.
- Perform all necessary tune-up steps according to the guided tours in **Options > Local Service > Tuneup > Reference Monitor**
(An Aircal has to be performed)

Complete UHR assembly

Prerequisites

Tools and auxiliary equipment

- Standard tool kit
- Torx tool kit with adapter for torque wrench
- Torque wrench 20 Nm

Time and manpower

- Estimated time to repair: 1.5 hours

Safety



Avoid accident and injury or damage of parts.

Risk of accident and injury!

- ⇒ **Read and observe the safety information contained in the Biograph Safety Procedure and Biograph Safety Report.**
-

Preliminary steps

1. Move patient table out and down.
2. Remove pallet from PHS.
3. Open gantry front door and disable rotation with S301
4. Move UHR out using the service switch
Use service switch S1 at FOC D514 to enable the service function
Use S3 (UHR out) and S4 (UHR in)
5. Switch off the gantry at the control box (CompOn)
6. Switch off the DMS power supply using F9 in PDC and check that the LED's at the DMS go off
7. Turn the DMS into the 6 o'clock position by hand
8. Lock the safety bolt to prevent unintentional gantry movement
9. Apply a ground connection between stator and rotor



Removing

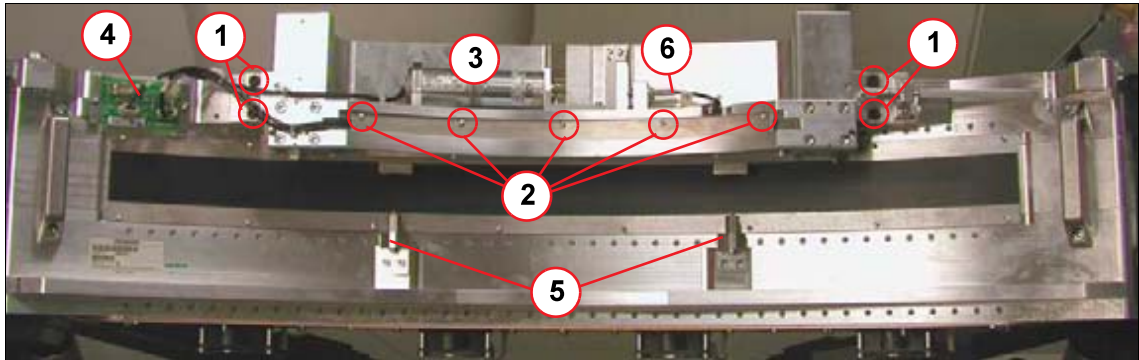


Fig. 119: UHR comb

- Pos. 1 4 screws holding the complete UHR
- Pos. 2 Screws of protective cover
- Pos. 3 UHR Motor
- Pos. 4 UHR control board
X80 is connected to FOC board D514 via W594 X5 at the C-box backplane
- Pos. 5 UHR guide pins
- Pos. 6 Limit switches (not installed in newest systems)

1. Remove plug X80 at the UHR control board
2. Loosen all necessary cable ties
3. Remove the 4 screws M6 x 20 mm holding the UHR assembly (Item 1)
4. Remove the UHR assembly

Installing

1. Install the new UHR assembly in reverse order
Use the new mounting materials supplied with the part
2. The micrometer is **preadjusted**
Push the UHR carefully toward the frame (in the direction of the micrometer) before attaching the screws

Note: The value of the micrometer is preadjusted to 5.00 mm



3. Tighten the 4 screws for the UHR with **20 Nm**

Final steps

- Remove the ground connection and unlock the safety bolt
- Release all service switches and switch on gantry
- Close all covers and reattach pallet to PHS.

- Check the function by loading modes using UHR
- Perform the tuneup steps described in the guided tours
Options > Local Service > Tuneup > UHR assembly
Hint: If not yet available, follow this sequence:
 - Balance the gantry
Options > Local Service > Tuneup > Expert Mode> Balance Gantry
Follow the software instructions
 - Adjust the comb
Options > Local Service > Tuneup > Expert Mode> CombAli
Follow the software instructions
- Perform in **Options > Local Service > Tuneup > Expert Mode**
 - **Aircal**
 - **Waterscaling**
- Perform Constancy test and backup

Complete UHR-plus assembly

Prerequisites

Tools and auxiliary equipment

- Standard tool kit
- Torx tool kit with adapter for torque wrench
- Torque wrench 20 Nm

Time and manpower

- Estimated time to repair: 1.5 hours

Safety

**WARNING**

Avoid accident and injury or damage of parts.

Risk of accident and injury!

- ⇒ **Read and observe the safety information contained in the Biograph Safety Procedure and Biograph Safety Report.**
-

Preliminary steps

1. Move patient table out and down
2. Remove pallet from PHS.
3. Open gantry front door and disable rotation with S301
4. Move UHR out using the service switch
Use service switch S1 at FOC D514 to enable the service function
Use S3 (UHR out) and S4 (UHR in)
5. Switch off the gantry at the control box (CompOn)
6. Switch off the DMS power supply using F9 in PDC and check that the LED's at the DMS go off
7. Turn the DMS into the 6 o'clock position by hand
8. Lock the safety bolt to prevent unintentional gantry movement
9. Apply a ground connection between stator and rotor



Removing

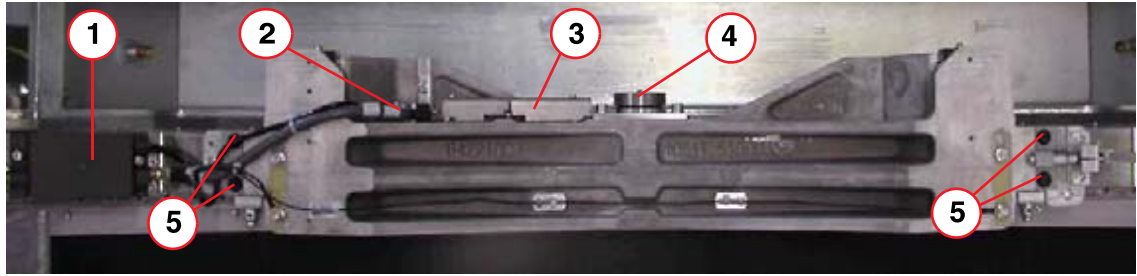


Fig. 120: UHR-plus comb

- | | |
|--------|---|
| Pos. 1 | Cover, control board |
| Pos. 2 | Home sensor |
| Pos. 3 | Cover for adapter D587 (connection for motor) |
| Pos. 4 | UHR motor |
| Pos. 5 | 4 screws, holding the complete UHR |

1. Remove the cover of UHR control board
2. Remove plugs J2 (motor) J3 (end switch) J4 (home sensor) at the UHR control board
3. Loosen all necessary cable ties
4. Remove the 4 screws M6 x 35 mm holding the UHR assembly (Item 5)
5. Remove the UHR assembly

Installation

1. Install the new UHR assembly in reverse order
Use the new mounting materials supplied with the part
2. The micrometer is **preadjusted**
Push the UHR carefully toward the frame (in the direction of the micrometer) before attaching the screws



Note: The value of the micrometer is preadjusted to 5.00 mm

3. Tighten the 4 screws for the UHR with **20 Nm**
4. Install the cover of the control board. Tighten the 2 screws with a torque of **3 Nm**

Final steps

- Remove the ground connection and unlock the safety bolt
- Release all service switches and switch on gantry
- Close all covers and reattach pallet to PHS.
- Check the function by loading modes using UHR

- Perform the tuneup steps described in the guided tours

Options > Local Service > Tuneup > UHR assembly

⇒ Follow the tuneup steps as shown in the guided tours “UHR assembly”.
see also MI12-023.817.01.xx.xx.

- Perform Constancy test and backup

Information for UHR-plus adjustment

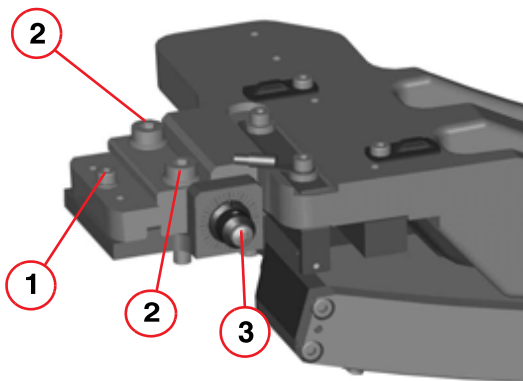


Fig. 121: UHR-plus comb, left side

- Pos. 1 Protection screw for z-direction
Pos. 2 Attachment for UHR-plus comb
Pos. 3 Adjustment for z-direction

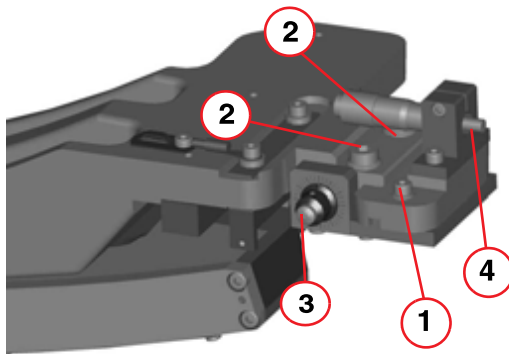


Fig. 122: UHR-plus comb, right side

- Pos. 1 Protection screw for z-direction
Pos. 2 Attachment for UHR-plus comb
Pos. 3 Adjustment for z-direction
Pos. 4 Adjustment for phi- direction

NOTE

Loosen the screws (1,2) for z-adjustment
(1,2/[Fig. 121 / p. 162](#))(1,2/[Fig. 122 / p. 162](#))

Loosen only the screws (2) for phi-adjustment
(2/[Fig. 121 / p. 162](#)) (2/[Fig. 122 / p. 162](#))

UHR motor

Prerequisites

Tools and auxiliary equipment

- Standard tool kit
- Torx tool kit with adapter for torque wrench
- Torque wrench 3.0 Nm

Time and manpower

- Estimated time to repair: 1.0 hours

Safety



Avoid accident and injury or damage of parts.

Risk of accident and injury!

- ⇒ **Read and observe the safety information contained in the Biograph Safety Procedure and Biograph Safety Report.**

Preliminary steps



1. Move patient table out and down
2. Remove pallet from PHS.
3. Open gantry front door and disable rotation with S301
4. Switch off the system
5. Switch off the on-site power and open the fuse holders F100 and F200 in the PDC
6. Remove the gantry rear door
7. Disconnect X325 and X312 and remove the 4 screws holding the funnel
8. Remove the funnel
9. Turn the DMS into the 6 o'clock position by hand
10. Lock the safety bolt to prevent unintentional gantry movement
11. Apply a ground connection between stator and rotor



Removing

1. Remove plug X81 at the UHR control board
2. Loosen all necessary cable ties
3. Remove the 4 screws M4 x 16 mm holding the motor, using a 3 mm Allen key

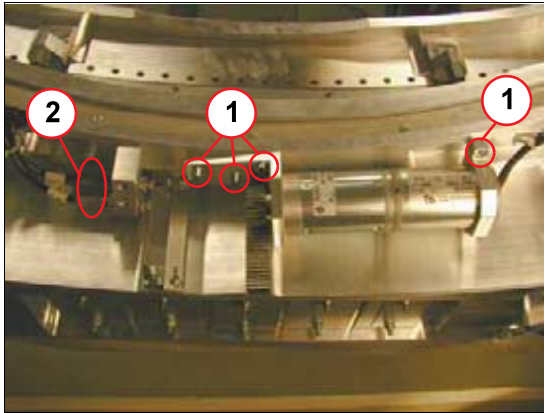


Fig. 123: UHR motor and limit switches

Pos. 1 Screws holding the UHR motor

Pos. 2 Limit switches

4. Remove the motor by pulling it about. 5 mm to the right and then to the top

Installing



1. Insert new motor
2. Connect plug X 81 at the UHR control board and secure cable with cable tie
3. Tighten the 4 screws for the motor with **3 Nm**.
4. Reattach the funnel and close the gantry rear side
5. Close the fuse holders F100 and F200 in the PDC

Final steps

- Remove the ground connection and unlock the safety bolt
- Release all service switches and switch on gantry
- Close all covers and reattach pallet to PHS.
- Check the function by loading modes using UHR

UHR-plus motor

Prerequisites

Tools and auxiliary equipment

- Standard tool kit
- Torx tool kit with adapter for torque wrench
- Torque wrench 3.0 Nm

Time and manpower

- Estimated time to repair: 1.0 hours

Safety



Avoid accident and injury or damage of parts.

Risk of accident and injury!

- ⇒ **Read and observe the safety information contained in the Biograph Safety Procedure and Biograph Safety Report.**

Preliminary steps



1. Move patient table out and down
2. Remove pallet from PHS.
3. Open gantry front door and disable rotation with S301
4. Switch off the system
5. Switch off the on-site power and open the fuse holders F100 and F200 in the PDC
6. Remove the gantry rear door
7. Disconnect X325 and X312 and remove the 4 screws holding the funnel
8. Remove the funnel
9. Turn the DMS into the 6 o'clock position by hand
10. Lock the safety bolt to prevent unintentional gantry movement
11. Apply a ground connection between stator and rotor



Removing

Remove the UHR-plus comb motor

1. Remove the cover
2. Remove the plug X1
3. Loosen the 4 screws
4. Pull out the motor

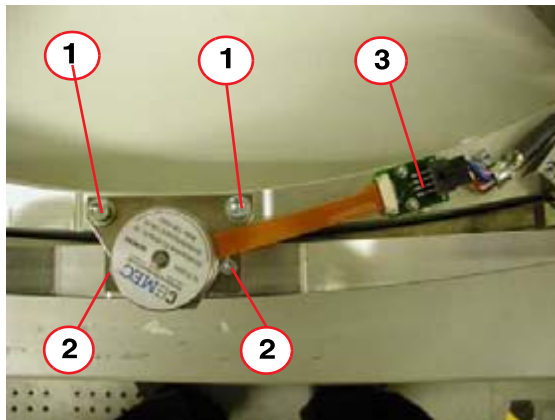


Fig. 124: UHR phi-z comb motor

- Pos. 1 Screws, holding the motor
Pos. 2 Screws, holding the comb
Pos. 3 X1 (adapter D587)

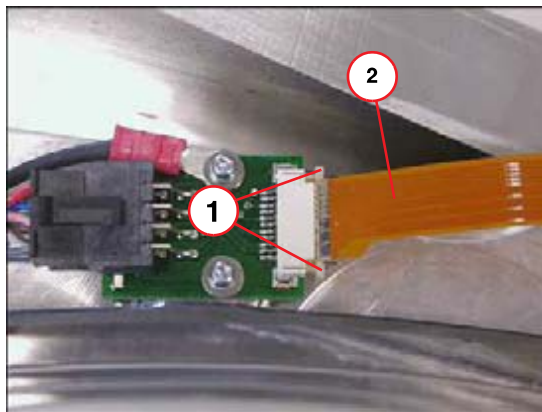


Fig. 125: How to disconnect the printed cable

- Pos. 1 Pull out the interlocks
Pos. 2 Pull out the printed cable

Installation

1. Insert the motor carefully, don't touch or damage the thread
2. Connect plug X1 and lock it (at adapter D587)

NOTE

The spindle must be installed centrally. Ensure smooth running by rotating the motor by hand (1/Fig. 126 / p. 168)



3. Tighten the 2 screws for the motor with **10 Nm**. Use Loctite 243
Tighten the 2 screws for the comb with **3 Nm**. Use Loctite 243

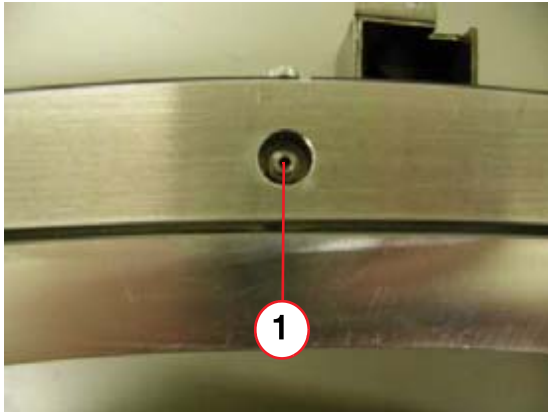


Fig. 126: Spindle of UHR-plus comb motor

Pos. 1 Spindle

4. Install the cover, tighten the screws with **1.3 Nm**. Use Loctite 243
5. Reattach the funnel and close the back of the gantry
6. Close the fuse holders F100 and F200 in the PDC

Final steps

- Remove the ground connection and unlock the safety bolt
- Release all service switches and switch on gantry
- Close all covers and reattach pallet to PHS.
- Check the function by loading modes using UHR

UHR limit switch

Prerequisites

Tools and auxiliary equipment

- Standard tool kit
- Torx tool kit with adapter for torque wrench
- Torque wrench 1.3 Nm and 5 Nm

Time and manpower

Estimated time to repair: 1.0 hour

Safety



Avoid accident and injury or damage of parts.

Risk of accident and injury!

- ⇒ **Read and observe the safety information contained in the Biograph Safety Procedure and Biograph Safety Report.**
-

Preliminary steps



1. Move patient table out and down
2. Remove pallet from PHS.
3. Open gantry front door and disable rotation with S301
4. Switch off the system
5. Switch off the on-site power and open the fuse holders F100 and F200 in the PDC
6. Remove the gantry rear door
7. Disconnect X325 and X312 and remove the 4 screws holding the funnel
8. Remove the funnel
9. Turn the DMS into the 6 o'clock position by hand
10. Lock the safety bolt to prevent unintentional gantry movement
11. Apply a ground connection between stator and rotor



Removing

1. Remove the 5 screws that hold the protective cover and remove the cover
2. Depending on the defect, remove plug X82 or X83 at the UHR control board ([Fig. 128 / p. 175](#))
3. Remove the nut from the corresponding switch
4. Loosen all necessary cable ties
5. Remove the defective switch

Installing



1. Insert the new switch
2. Position the switch so that the LED in the shaft is visible
3. Adjust a gap of 0.25 mm +/- 0.05 mm between the limit switch and the flag at the moving part ([Adjustment of UHR limit switch / p. 190](#))
Use **Loctite 243** at both nuts of the switch when tightened with **5 Nm**
4. Reattach the protective cover and tighten the 5 screws with **1.3 Nm**. Use **Loctite 243**
5. Reattach all cable ties
6. Reattach the funnel and close the gantry rear side
7. Close the fuse holders F100 and F200 in the PDC

Final steps

- Remove the ground connection and unlock the safety bolt
- Release all service switches and switch on gantry
- Close all covers and reattach pallet to PHS.
- Check the function by loading modes using UHR
- Check whether MTF constancy measurement is OK. Only if MTF is out of tolerance, should a Comb Ali adjustment be performed. For details, refer to the adjustment instructions (MI12-023-817.02.xx.02).

UHR-plus limit switch

Prerequisites

Tools and auxiliary equipment

- Standard tool kit
- Torx tool kit with adapter for torque wrench
- Torque wrench 3 Nm and 5 Nm

Time and manpower

Estimated time to repair: 1.0 hour

Safety



Avoid accident and injury or damage of parts.

Risk of accident and injury!

- ⇒ **Read and observe the safety information contained in the Biograph Safety Procedure and Biograph Safety Report.**
-

Preliminary steps



1. Move patient table out and down
2. Remove pallet from PHS.
3. Open gantry front door and disable rotation with S301
4. Switch off the system
5. Switch off the on-site power and open the fuse holders F100 and F200 in the PDC
6. Remove the gantry rear door
7. Disconnect X325 and X312 and remove the 4 screws holding the funnel
8. Remove the funnel
9. Turn the DMS into the 6 o'clock position by hand
10. Lock the safety bolt to prevent unintentional gantry movement
11. Apply a ground connection between stator and rotor



Removing

1. Remove the union nut and pull out the home sensor
2. Remove the cover of control board
3. Remove the plug (Home) on the control board
4. Loosen all necessary cable ties
5. Remove the defective switch

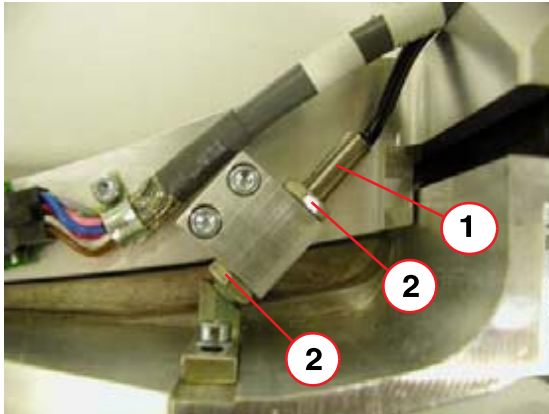


Fig. 127: Home sensor of UHR-plus comb

Pos. 1 Home sensor

Pos. 2 Nut

Installation

1. Insert the new switch
2. Position the switch so that the LED in the shaft is visible
3. Adjust a gap of 0.25 mm +/- 0.05 mm between the limit switch and the flag at the moving part ([Fig. 127 / p. 172](#))



Use **Loctite 243** at both nuts of the switch when tightened with **5 Nm**

4. Reattach the protective cover and tighten the 2 screws with **3 Nm**. Use **Loctite 243**
5. Reattach all cable ties
6. Reattach the funnel and close the back of the gantry
7. Close the fuse holders F100 and F200 in the PDC

Final steps

- Remove the ground connection and unlock the safety bolt
- Release all service switches and switch on gantry
- Close all covers and reattach pallet to PHS.

- Check the function by loading modes using UHR
- Check whether the MTF constancy measurement is OK. Perform a Comb Ali adjustment only if MTF is out of tolerance. For details, refer to the adjustment instructions (MI12-023-817.02.xx.02).

UHR control board

Prerequisites

Tools and auxiliary equipment

- Standard tool kit
- Torx tool kit with adapter for torque wrench
- Torque wrench 1.4 Nm

Time and manpower

Estimated time to repair: 0.5 hour

Safety



Avoid accident and injury or damage of parts.

Risk of accident and injury!

- ⇒ **Read and observe the safety information contained in the Biograph Safety Procedure and Biograph Safety Report.**
-

Preliminary steps

1. Move patient table out and down
2. Remove pallet from PHS.
3. Open gantry front door and disable rotation with S301
4. Switch off the gantry at the control box (CompOn)
5. Turn the DMS into the 6 o'clock position by hand
6. Lock the safety bolt to prevent unintentional gantry movement
7. Apply a ground connection between stator and rotor



Removing



Fig. 128: UHR control board

1. Remove the plugs X80, X81, X82 and X83 at the UHR control board D584
2. Remove the 2 screws M4 x 10 mm using a T20 Torx tool
3. Remove the UHR control board.

Installing

1. Insert the new UHR control board
2. Tighten the 2 screws with a torque of **3 Nm**

Final steps

- Remove the ground connection and unlock the safety bolt
- Release all service switches and switch on gantry
- Close all covers and reattach pallet to PHS.
- Check the function by loading modes using UHR

UHR-plus control board

Prerequisites

Tools and auxiliary equipment

- Standard tool kit
- Torx tool kit with adapter for torque wrench
- Torque wrench 1.3 Nm; 3 Nm

Time and manpower

Estimated time to repair: 0.5 hours

Safety



Avoid accident and injury or damage of parts.

Risk of accident and injury!

- ⇒ **Read and observe the safety information contained in the Biograph Safety Procedure and Biograph Safety Report.**

Preliminary steps

1. Move patient table out and down
2. Remove pallet from PHS.
3. Open gantry front door and disable rotation with S301
4. Switch off the gantry at the control box (CompOn)
5. Turn the DMS into the 6 o'clock position by hand
6. Lock the safety bolt to prevent unintentional gantry movement
7. Apply a ground connection between stator and rotor



Removing

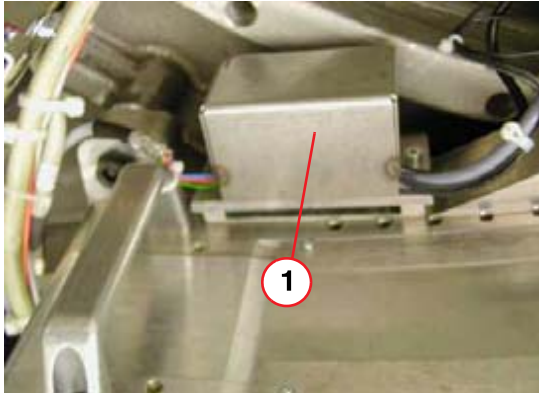


Fig. 129: UHR-plus comb control board

Pos. 1 Cover of control board

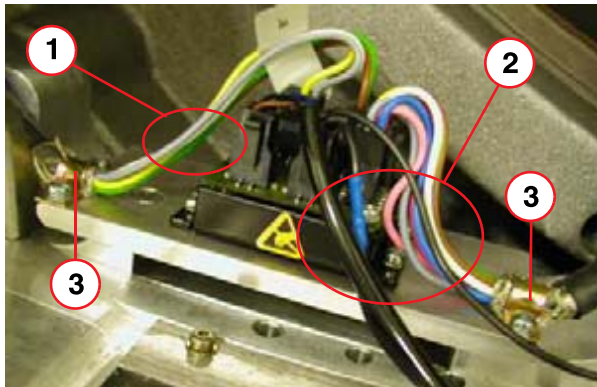


Fig. 130: UHR-plus control board, D586

Pos. 1 Input cable J1 (power) J6 (RS232)

Pos. 2 Output cable J2 (motor) J3 (end switch)

Pos. 3 Cable fastening

1. Remove the cover of the control board, remove the plugs J1 (power), J2 (motor), J3 (end switch) J4 (Home) and J6 (Rs232) at control board D586.
2. Remove cable fastenings (item 3)
3. Remove the 2 screws M4 x 10 mm using an Allen key 2.5 mm
4. Remove the UHR -plus control board.

Installation

1. Insert the new UHR-plus control board
2. Tighten the 2 screws with a torque of **3 Nm**. Use Loctite 243
3. Install plugs J1 (power), J2 (motor), J3 (end switch), J4 (Home), and J6 (Rs232) at control board D586.

4. Install cable fastenings (item 3). Tighten with a torque of **1.3 Nm** . Use Loctite 243.
5. Install the cover of the control board. Tighten the 2 screws with a torque of **3 Nm**.

Final steps

- Remove the ground connection and unlock the safety bolt
- Release all service switches and switch on gantry
- Close all covers and reattach pallet to PHS.
- Check the function by loading modes using UHR

DMS power supply

Prerequisites

Tools and auxiliary equipment

- Standard tool kit
- Torx tool kit with adapter for torque wrench
- Torque wrench 10.2 Nm

Time and manpower

- Estimated time to repair: 1.0 hours

Safety



Avoid accident and injury or damage of parts.

Risk of accident and injury!

- ⇒ **Read and observe the safety information contained in the Biograph Safety Procedure and Biograph Safety Report.**

Preliminary steps

NOTE

There are 2 different power supplies in the field (adjustable and non adjustable). For more advice, pls. refer to the following.

The output voltages indicated in the table are valid for both power supplies.



1. Move patient table out and down
2. Remove pallet from PHS.
3. Open gantry front door and disable rotation with S301
4. Switch off the gantry at the control box (CompOn)
5. Switch off the DMS power supply using F9 in PDC and check that the LED's at the DMS go off
6. Turn the DMS into the 6 o'clock position by hand
7. Lock the safety bolt to prevent unintentional gantry movement
8. Apply a ground connection between stator and rotor



Removing

1. Remove all necessary cable ties and disconnect the mains plug X1 W597



Fig. 131: DMS power supply

2. Remove plug X100 and X200 using an Allen key, installing Fig.30

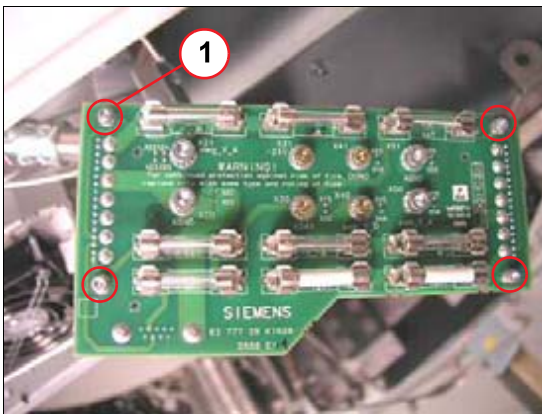


Fig. 132: DMS power supply distribution board, non-adjustable.

Pos. 1 Remove this screws to disconnect X100 and X200

3. Loosen the 4 nuts M6 holding the power supply at the gantry (Item1)

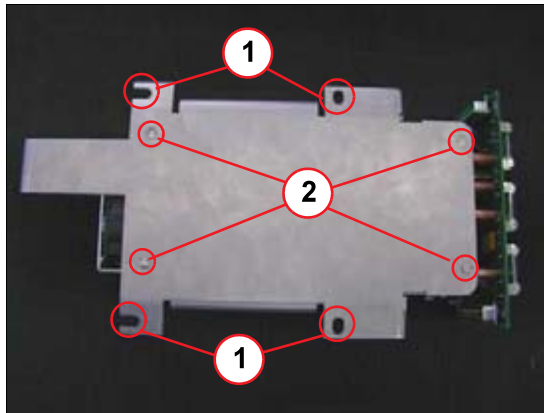


Fig. 133: DMS power supply mounting plate

Pos. 1 10.2 Nm connection to gantry

Pos. 2 1.8 Nm connection to power supply

4. Remove the DMS power supply.

Do not loosen the screws on the power supply - mounting plate (Item 2)

Installing

1. Move the new power supply back into the gantry and tighten the 4 nuts with a torque of **10 Nm**
2. Reconnect X1, X100 and X200
3. Reattach all cable ties

4. Check / adjust the power supply

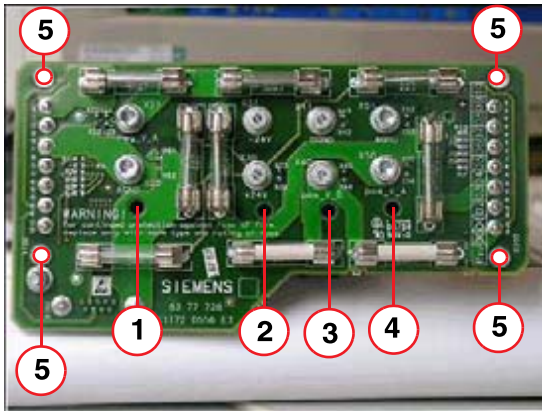


Fig. 134: DMS power supply distribution board, adjustable.

- Pos. 1 Adjustment for -3.7 Volt
 Pos. 2 Adjustment for +24 Volt
 Pos. 3 Adjustment for +3.3 Volt
 Pos. 4 Adjustment for +4.5 Volt
 Pos. 5 Remove the screws to disconnect X100 and X200

⇒ The power supply is adjustable.
 Check / adjust the power supply after replacement.

Important: The tolerances are valid only if you measure at the DMS backplanes !

Test point	Ground	Description	Tolerance
+24V	AGND	24V / 6A controller / fan Permanently on	+/- 10% 100mV ripple
pos_V_D	DGND	3.4V / 11A module digital	+/- 5% 30mV ripple
pos_V_A	AGND	4.6V / 21A module analog	+/- 4% 20mV ripple
neg_V_A	AGND	-3.8V / 18A module analog	+/- 4% 20mV ripple

Final steps

- Remove the ground connection and unlock the safety bolt
- Release all service switches and switch on gantry
- Close all covers and reattach pallet to PHS.
- Perform test scans

Fans

Prerequisites

Tools and auxiliary equipment

- Standard tool kit
- Torx tool kit with adapter for torque wrench
- Torque wrench 1.0 Nm

Time and manpower

- Estimated time to repair: 0.5 hours

Safety



Avoid accident and injury or damage of parts.

Risk of accident and injury!

- ⇒ **Read and observe the safety information contained in the Biograph Safety Procedure and Biograph Safety Report.**
-

Preliminary steps



1. Move patient table out and down
2. Remove pallet from PHS.
3. Open gantry front door and disable rotation with S301
4. Switch off the gantry at the control box (CompOn)
5. Switch off the DMS power supply using F9 in PDC and check that the LED's at the DMS go off
6. Rotate the DMS to the 12 o'clock position by hand
7. Lock the safety bolt to prevent unintentional gantry movement
8. Apply a ground connection between stator and rotor



Removing

1. Unplug the connector of the corresponding fan
2. Remove the four screws using a torx T20 tool

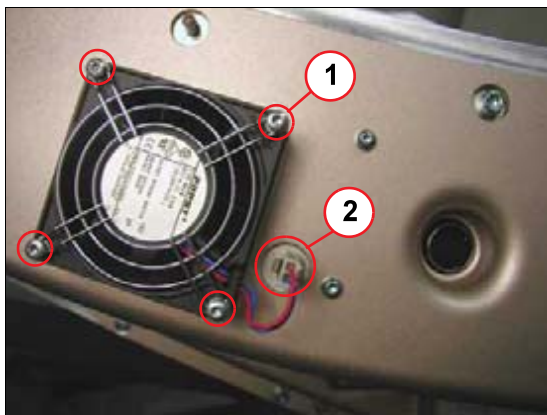


Fig. 135: Fan

- Pos. 1 4 x screws tightened with 0.6 Nm
Pos. 2 Connector

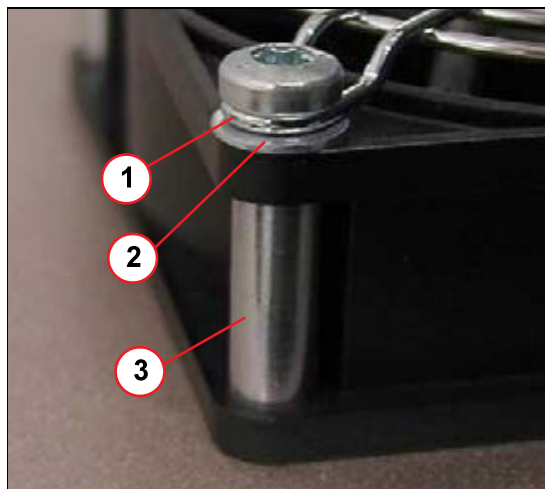


Fig. 136: Details of fan

- Pos. 1 Grid
Pos. 2 Washer
Pos. 3 Spacer

3. Take out the old fan
4. Remove the protective grid and the spacers from old fan

Installing



1. Install new fan in reverse order using the old grid, spacers and washers
2. Tighten the four screws of the fan with a torque of **1.0 Nm**

Final steps

- Remove the ground connection and unlock the safety bolt
- Release all service switches and switch on gantry
- Check if the fans are running, at least at a slow speed after the power is switched on
- Close the gantry
- Reattach pallet to PHS.

Temperature switch

Prerequisites

Tools and auxiliary equipment

- Standard tool kit
- Torx tool kit with adapter for torque wrench
- Torque wrench 1.3 Nm, 3 Nm and 10 Nm

Time and manpower

- Estimated time to repair: 1.0 hours

Safety



Avoid accident and injury or damage of parts.

Risk of accident and injury!

- ⇒ **Read and observe the safety information contained in the Biograph Safety Procedure and Biograph Safety Report.**
-

Preliminary steps



1. Move patient table out and down
2. Remove pallet from PHS.
3. Open gantry front door and disable rotation with S301
4. Switch off the gantry at the control box (CompOn)
5. Switch off the DMS power supply using F9 in PDC and check that the LED's at the DMS go off
6. Rotate the DMS to the 12 o'clock position by hand
7. Lock the safety bolt to prevent unintentional gantry movement
8. Apply a ground connection between stator and rotor



Removing

NOTE

Follow the C-MOS guidelines when touching the DMS

1. Unplug X3 at the reference monitor board
2. Unplug the connectors for fans X40 and X41
3. Remove 13 screws from DMS cover using an 5 mm Allen key
4. Remove DMS cover
5. Unplug the 2 wires from the temp switch F1, located on the right backplane



Fig. 137: Temperature switch

6. Remove the 2 screws holding the switch
7. Remove switch F1

Installing



1. Install the new switch in reverse order
2. Tighten the nuts M3 for the temp switch with a torque of **1.3 Nm**
3. Tighten the screws for the DMS cover with **3 and 10 Nm** according to overview ([Fig. 97 / p. 133](#))

Final steps

- Remove the ground connection and unlock the safety bolt
- Release all service switches and switch on the gantry
- Close all covers and reattach pallet to PHS.
- Check function by performing test scans

Appendix

Channel list

Channels	Slot count	Channels	Slot count
1 - 16	1	337 - 352	22
17 - 32	2	353 - 368	23
33 - 48	3	369 - 384	24
49 - 64	4	385 - 400	25
65 - 80	5	401 - 416	26
81 - 96	6	417 - 432	27
97 - 112	7	433 - 448	28
113 - 128	8	449 - 464	29
129 - 144	9	465 - 480	30
145 - 160	10	481 - 496	31
161 - 176	11	497 - 512	32
177 - 192	12	513 - 528	33
193 - 208	13	529 - 544	34
209 - 224	14	545 - 560	35
225 - 240	15	561 - 576	36
241 - 256	16	577 - 592	37
257 - 272	17	593 - 608	38
273 - 288	18	609 - 624	39
289 - 304	19	625 - 640	40
305 - 320	20	641 - 656	41
321 - 336	21	657 - 672	42

Replacement of a center module

In case of a defective center module (slot 19 - 24) we will not replace the whole six pack, but just the defective module. To ensure best image quality we will not place the spare module in the center slots, but rather outside the center (outside the inner six slots). The defective slot will be filled with a functioning adjacent one. This measure keeps the center of the detector homogeneous.

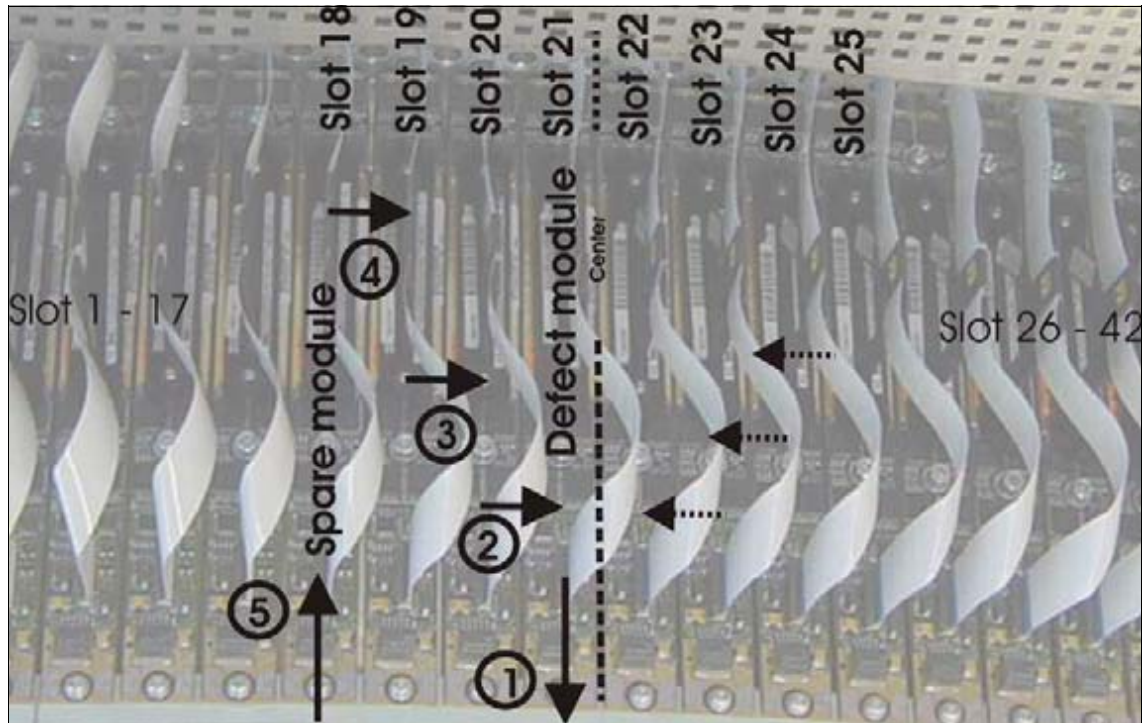


Fig. 138: Center module replacement

The example shows a defective module in the center slot 21.

The following steps have to be performed:

- 1 remove defective module in slot 21
- 2 move slot 20 to 21
- 3 move slot 19 to 20
- 4 move slot 18 to 19
- 5 insert spare module in slot 18

According to this scheme, the exchange of slot 19 to 24 has to be performed.

Overview of all used screws

Part	Quantity	Size and type	Tool	Torque	Loctite
DMS (inner screws)	4	M10 x 65 mm	Allen key 8 mm	110 Nm	no
	2	M10 x 55 mm	Allen key 8 mm	110 Nm	no
DMS Cover	8	M6 x 12 mm	Allen key 5 mm	3 Nm	no
	5	M6 x 12 mm	Allen key 5 mm	10.2 Nm	no
Module	2	M4 x 20 mm	Torx T20	1.4 Nm	no
	1	M3 x 8 mm	Torx T10	1.0 Nm	no
DAS controller	8	M3 x 8 mm	Torx T10	1.3 Nm	no
DAS controller cover	4	M4 x 45 mm	Allen key 3 mm	3.0 Nm	no
Backplanes	18	M4 x 10 mm	Torx T20	1.4 Nm	no
Reference monitor	4	M3 x 8 mm	Torx T10	1.3 Nm	no
UHR complete	4	M6 x 25 mm	Allen key 5 mm	20 Nm	no
UHR-plus complete	4	M6 x 30			
UHR motor	4	M4 x 16 mm	Allen key 3 mm	3.0 Nm	no
UHR limit switch	2	fine thread nut	open-end wrench 13 mm	5.0 Nm	243
Home sensor	2				
UHR control board	2	M4 x 10 mm	Torx T20	1.4 Nm	no
UHR-plus control board					
UHR protective cover	5	M3 x 6 mm	Torx T10	1.3 Nm	243
UHR-plus protective cover (motor)	2	M3 x 6	Torx T10	1.3 Nm	No
UHR guide pin holder	4	M4 x 10 mm	Allen key 3 mm	3.0 Nm	No
UHR guide pin	4	M4 x 12 mm	Allen key 3 mm	3.0 Nm	No
Fans	4	M4 x 35 mm	Torx T20	1.0 Nm	No
Temperature switch	2	M3 x 8 mm	Torx T10	1.3 Nm	No
DMS power supply	4	M4 x 6 mm	Torx T20	1.8 Nm	270
PS at gantry	4	Nut M6	Open-end wrench	10.2 Nm	No
PS ground wire	1	M4 x 6 mm	Allen key 3 mm	1.8 Nm	242
DMS ground wire	2	M6 x 12 mm	Allen key 5 mm	10.2 Nm	No

Adjustment of UHR limit switch

After replacing the limit switch, the position of the limit switch has to be adjusted.

Adjustment of the limit switches

1. To be able to move the UHR without power, remove the UHR motor by loosening the 4 screws M4 x 6mm using an 3 mm Allen key.

After reattaching the UHR motor, tighten the screws with **3.0 Nm**.

2. Use a feeler gauge to measure the distance between limit switch and flag.

The distance should be approximately 0.25mm +/- 0.05mm. Use **Loctite 243** when tightening the screw with **5 Nm**.

Flag for limit switches

The flag which is activating the limit switch is adjusted in the factory.

Functional description:

While moving the UHR comb in, the LED at the shaft of the limit switch 1 has to be activated 1.2 mm before the mechanical end stop.

While moving the UHR comb out, the LED at the shaft of the limit switch 2 has to be activated 4.5mm before the mechanical end stop.

Adjustment of UHR guide pins

The following adjustment has to be performed when Comb/AlI has to be done.

The adjustment assures that the guide pins align properly, when the comb is in the inner position. For this purpose, either the part mounted at the moving part or the counter part can be adjusted.

Sequence of alignment:

- Remove both guide pin holders (Item 1) by removing the 2 x 2 screws M4 x 10mm, using a 3mm Allen key

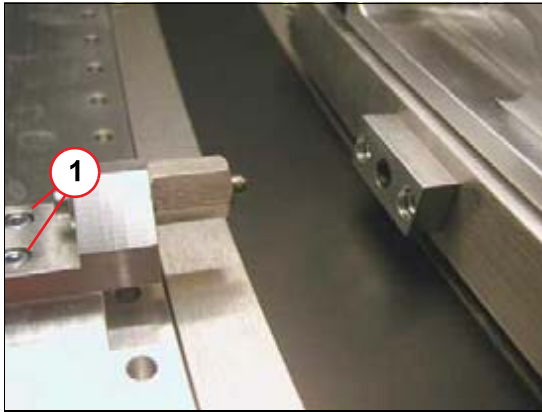


Fig. 139: UHR guide pin

Pos. 1 Screws of guide pin holder (not installed in newest systems)

- Perform Comb/Alf and shift the comb in the PHI direction according to the software using the micrometer

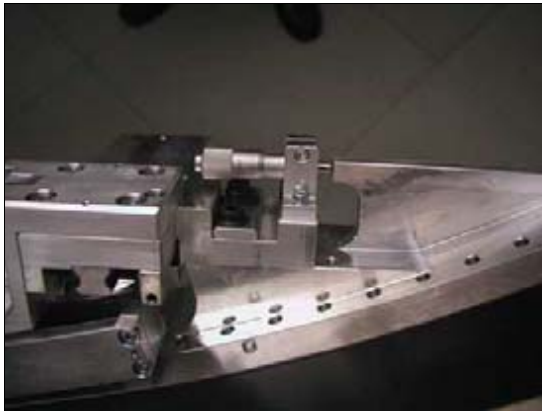


Fig. 140: UHR micrometer

- Repeat the steps of Comb/Alf if necessary
- After Comb/Alf adjustment is finished, install the guide pin holder again.
- Use service switch S1 at FOC D514 to enable the service function.

To position the guide pins properly, use S3 (UHR out) and S4 (UHR in) to move the comb several times in and out the guide pins before tightening the screws of the guide pin holder with **3 Nm**

Z-position check of modules

To check the Z-position after replacing a DMS module, use analyze

NOTE

Beginning with VB20x, this function will be implemented in the tune-up sequence.

- Remove all absorbers, use a static mode and select the following parameters:
120 kV, 300 mA, 20 x 1.2 mm, 1 sec and all processing steps
(the checkbox for PHIRPP and Focus large is also selected)
It is important is to select the "save raw data" checkbox
- Press **Go** to load the selected mode
- Wait for the Press **START key** to appear and then press the Start button
- Load the same mode again by pressing **Go**
- Open an **Additional Window** and select **Control** and then **General**
- Select **Collimator** and change the opening from 9.8mm to 8.0mm and press **Go**
- Press the Start button
- Open **Analyze** and load the first raw data set

- Select **Arithmetics**, **Subtract** and under **ICS-Rawdata** the raw data set from the second scan

In addition select for both data sets:

Frame 1 - 1160

Channel 1 - 672

Slice 1-20

FFS: Phi:1, Z: 1

and press **Go**

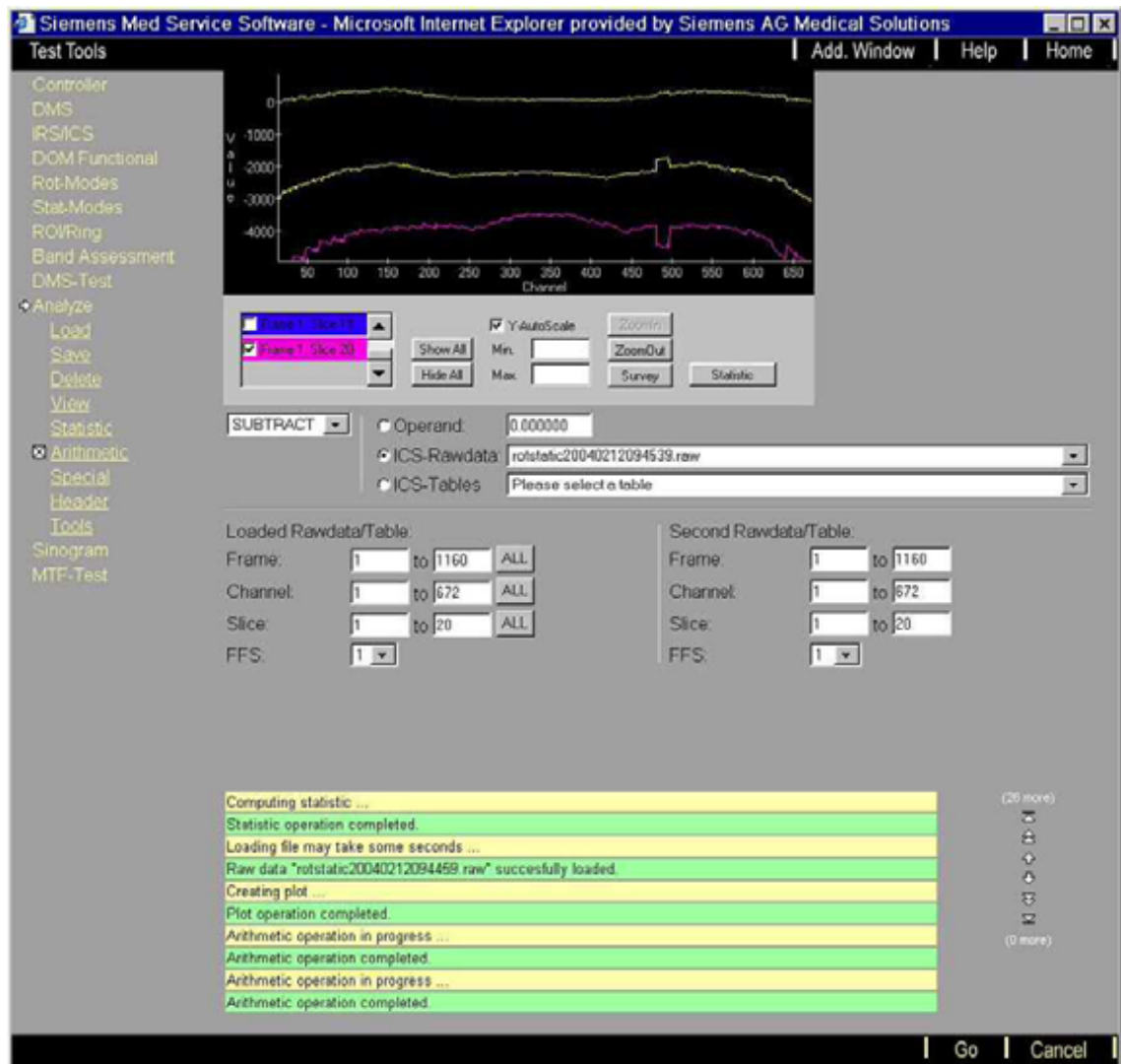


Fig. 141: Unsatisfactory Z position of module

- ⇒ In this example two slots are not aligned properly.
You see the drop in the graph in the first and last slice.

- Check if there is a drop of 16 channels, especially at the position where you performed the repair.
If yes, loosen the module exchanged before and press it toward the rotating plane before fixing it again.
Repeat the test.

Firmware download for DAS controller

The firmware of the DAS controller is not downloadable via CAN bus.

Both a special procedure and a special cable are necessary.

A cabel has to be connected between a parallel interface port of the Navigator and the connector X50 (normally connected to ADU) at the DMS.

If necessary, the exact procedure will be described in the relevant update procedure.

Overview

The figure below shows the C-box of the rotating part of the gantry.

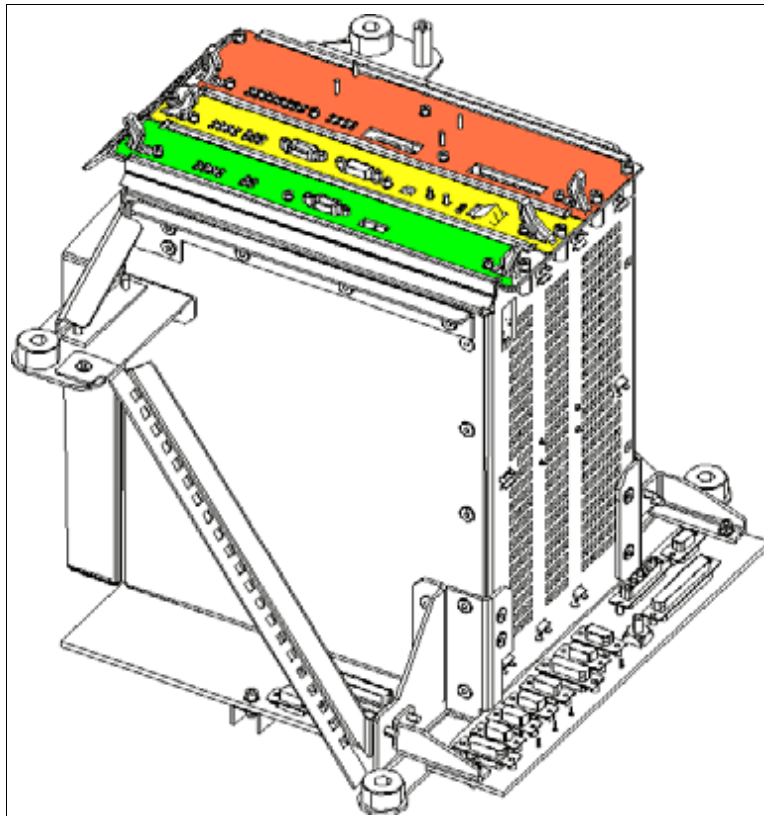


Fig. 142: C-box

- The C-box (E510) includes the following components:
 - D540 **DPU**
 - D514 **FOC**
 - D513 **ADU**

C-Box

Prerequisites

Tools and auxiliary equipment

- Standard service kit.
- 25 Nm to 130 Nm torque wrench with 7mm hexagonal attachment 1/2" and 5 mm Allen attachment 1/2">
- Extension 1/2", 300 mm long
- Self-locking nuts M4 and safety bolts M6 as required

Time and manpower

- 1 person
- 1 hour

Safety



Avoid accident and injury or damage of parts.

Risk of accident and injury!

- ⇒ **Read and observe the safety information contained in the Biograph Safety Procedure and Biograph Safety Report.**
-

Preliminary steps

1. Remove the pallet from the PHS.
2. Switch **OFF** the system at the control box
3. **Switch OFF onsite power using the onsite ON/OFF switch!**
4. Open the gantry front door and disable rotation with switch S 301.
5. Turn the gantry by hand until the X-ray tube is in the **12 o'clock** position.
6. **Secure the rotating gantry with the safety bolt.**

Removing

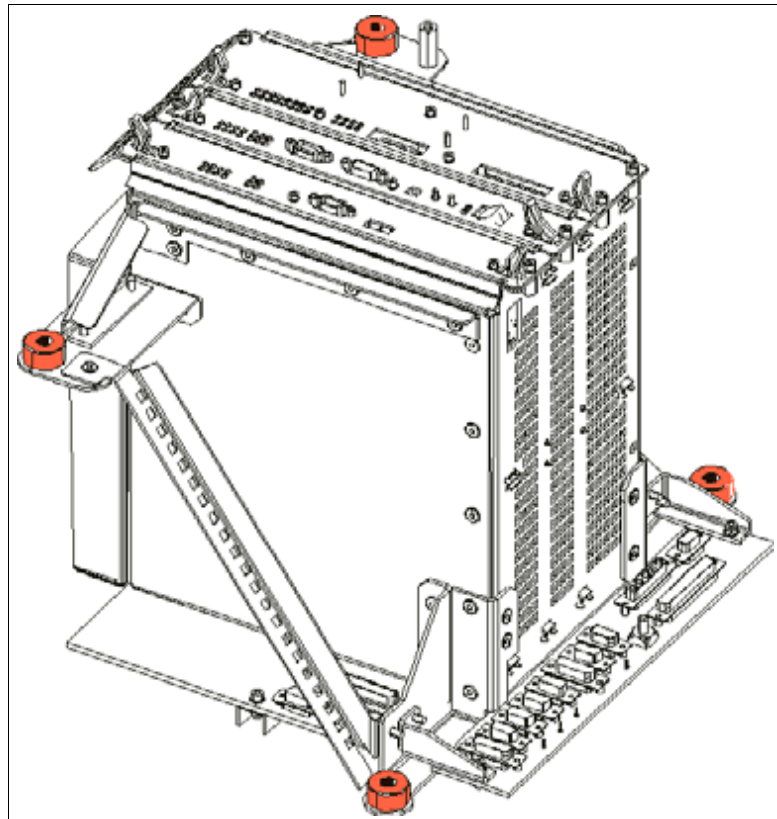


Fig. 143: C-box mounting points

1. Disconnect **all** cables and connectors on the backplane board of the C-box including X100 (ground strap).
2. Disconnect the cables from the boards
 - D540 X3, X4
 - D514 X3, X4
 - D513 Fiber optic cable at bottom
3. Remove all cable ties and cable clamps necessary to facilitate removal of the C-box.
Note the routing of the PHI_Z_MON cable to D514, X3. It is important to run it the same way on reassembly.
4. Remove the 4 M6 mounting bolts (Red) and remove the C-Box.

Installing

1. Remove all necessary parts (fans, brackets, etc.) from the old C-Box and install them on the replacement C-box.
2. Tighten the 4 M6 mounting bolts using a torque of **45Nm** .
⇒ **Replace all self-locking nuts and safety bolts that have been removed 3-4 times.**
3. Reconnect cables/cable brackets and cable clamps according the original configuration.
4. Secure all cables and connectors with cable ties and cable clamps.
5. **Ensure compliance with the C-MOS guidelines** when swapping boards from the old to the new C-Box.

Final steps

Mechanical

1. Unlock safety bolt for gantry rotation.
2. Rotate the gantry by hand and check for potential damage caused during rotation (e.g loose cables).
3. Switch rotation switch S301 to "1 ".
4. Close the Gantry door
5. Reattach pallet to PHS.
6. Switch on the system (**SYSTEM/ON**).

Software

1. The 3 controllers in the C_Box and the 2 parameter files for the collimator in FOC must now be flashed.
2. The floppy included with the collimator contains two files:
0xxxx_1.fsh
0xxxx_2.fsh
xxxx is the serial number of your collimator. The two files must be renamed before they can be used with the flashload tool. (this may have been done already!)
3. Rename file **0xxxx_1.fsh to FOC1xxxx.fsh** (the 1 after FOC indicates parameter set 1).
Rename file **0xxxx_2.fsh to FOC2xxxx.fsh** (the 2 after FOC indicates parameter set 2).
4. Open the NT Explorer at the Navigator console; copy the renamed files from the floppy to the following directory -> **C:/Somaris/Config/ControlSystemFlash** .

5. Close NT explorer.
6. Go to Local Service and select **Control -> Flashload** and select **FOC** , **ACU** and **DOM** by clicking the respective buttons.
7. The fields will show the **Firmware** on disk in the firmware field, **Vxxx** in Param1 field and **Vxxx** in Param2 field. Use the drop-downs to select the latest/valid versions and click "**GO**" to perform the flashload.
8. After you get the message **flashload successfully done** , continue with the next step.

Adjustments

1. No tune-up steps are required

Tests

1. Run quality tests.

FOC D514

Prerequisites

Tools and auxiliary equipment

- Standard service kit.

Time and manpower

- 1 person
- 1 hour

Safety



Avoid accident and injury or damage of parts.

Risk of accident and injury!

- ⇒ **Read and observe the safety information contained in the Biograph Safety Procedure and Biograph Safety Report.**
-

Preliminary steps

1. Remove the pallet from the PHS.
2. Switch off the gantry using the **COMP/ ON** switch at the control box.
3. **Switch off circuit breaker F8 and F9 in the PDC**
4. Open the gantry front door and disable rotation with switch S 301.
5. Turn the gantry by hand until the X-ray tube is in the 12 o'clock position.
6. **Secure the rotating part with the safety bolt.**

Removing

1. Remove plugs X3 X4.
2. Remove all cable ties and cable clamps necessary to facilitate removal of the FOC board.

Installing

1. Plug in the new board. Tighten screws to **0.25 Nm**
2. Reconnect cables/cable brackets and cable clamps according to the original configuration.

Final steps

Mechanical

- Unlock safety bolt for gantry rotation.
- Switch rotation switch S301 to "1".
- Close the gantry door
- Reattach pallet to PHS.
- Switch on circuit breaker **F8** and **F9** inside the PDC.
- Switch on the system (**SYSTEM/ON**) at the control panel.

Software

1. The controller and the 2 parameter files for the collimator in FOC must now be flashed.
2. The floppy included with collimator contains two files:
0xxxx_1.fsh
0xxxx_2.fsh
xxxx is the serial number of your collimator. The two files must be renamed before they can be used with the flashload tool. (this may have been done already!)
3. Rename file **0xxxx_1.fsh to FOC1xxxx.fsh** (the 1 after FOC indicates parameter set 1).
Rename file **0xxxx_2.fsh to FOC2xxxx.fsh** (the 2 after FOC indicates parameter set 2).
4. Open the NT Explorer at the Navigator console; copy the renamed files from the floppy to the following directory -> **C:/Somaris/Config/ControlSystemFlash**.
5. Close NT explorer.
6. Go to Local Service and select **Control -> Flashload** and select **FOC**, by clicking the respective buttons.
7. The fields will show the **Firmware** on disk in the firmware field, **Vxxx** in Param1 field and **Vxxx** in Param2 field. Use drop-down boxes to select the latest/valid versions.
Click "**GO**" to perform the flashload.
8. After you get the message **flashload successfully done**, continue with the next step.

Adjustments

No tune-up steps are required.

Tests

- Run quality tests

ADU D513

Prerequisites

Tools and auxiliary equipment

- Standard service kit.

Time and manpower

- 1 person
- 1 hour

Safety



Avoid accident and injury or damage of parts.

Risk of accident and injury!

- ⇒ **Read and observe the safety information contained in the Biograph Safety Procedure and Biograph Safety Report.**
-

Preliminary steps

1. Remove the pallet from the PHS.
2. Switch off the gantry using the **COMP/ ON** switch at the control box.
3. **Switch off circuit breaker F8 and F9 in the PDC**
4. Open the gantry front door and disable rotation with switch S 301.
5. Turn the gantry by hand until the X-ray tube is in the 12 o'clock position.
6. **Secure the rotating part with the safety bolt.**

Removing

1. Unplug the fiber optic cable from underneath the board.
2. Remove all cable ties and cable clamps necessary to facilitate removal of the ADU board.

Installing

1. Plug in the new board. Tighten screws to **0.25 Nm**
2. Reconnect cables/cable brackets and cable clamps according to the original configuration.

Final steps

Mechanical

1. Unlock safety bolt for gantry rotation.
2. Switch rotation switch S301 to "1 ".
3. Close the gantry door
4. Reattach pallet to PHS.
5. Switch on circuit breaker **F8** and **F9** inside the PDC.
6. Switch on the system (**SYSTEM/ON**) at the control panel.

Software

1. The controllers on ADU must now be flashed.
2. Go to Local Service and select **Control -> Flashload** and select **ACU** and **DOM** by clicking the respective buttons.
3. The fields will show the **firmware** available on disk in the firmware field, click the drop-down menu and select the latest/valid version, then click "**GO** " to perform the flashload.
4. After you get the message **flashload successfully done** , continue with the next step.

Adjustments

1. No tune-up steps are required

Tests

1. Run quality tests

DPU D540

Prerequisites

Tools and auxiliary equipment

- Standard service kit.

Time and manpower

- 1 person
- 1 hour

Safety



Avoid accident and injury or damage of parts.

Risk of accident and injury!

- ⇒ **Read and observe the safety information contained in the Biograph Safety Procedure and Biograph Safety Report.**
-

Preliminary steps

1. Remove the pallet from the PHS.
2. Switch **OFF** the gantry using the **COMP/ ON** switch at the control box.
3. **Switch off circuit breaker F8 and F9 in the PDC**
4. Open the gantry front door and disable rotation with switch S 301.
5. Turn the gantry by hand until the X-ray tube is in the 12 o'clock position.
6. **Secure the rotating part with the safety bolt.**

Removing

1. Remove plugs X3 X4.
2. Remove all cable ties and cable clamps necessary to facilitate removal of the DPU board.

Installing

1. Plug in the new board, tighten screws to **0.25 Nm**
2. Reconnect cables/cable brackets and cable clamps according to the original configuration.

Final steps

Mechanical

1. Unlock safety bolt for gantry rotation.
2. Switch rotation switch S301 to "1".
3. Close the gantry door
4. Reattach pallet to PHS.
5. Switch on circuit breaker **F8** and **F9** inside the PDC.
6. Switch on the system (**SYSTEM/ON**) at the control panel.

Adjustments

1. No tune-up steps are required.

Tests

1. Run quality tests

Overview

- The gantry right stand includes the following components:
 - Dehumidifier
 - Tilt spindle
 - D311 GPC board
 - 3-way valve

Dehumidifier

Prerequisites

Tools and auxiliary equipment

- Standard service kit.
- Insulating tape

Time and manpower

- 1 person
- 1 hour

Safety



Avoid accident and injury or damage of parts.

Risk of accident and injury!

- ⇒ **Read and observe the safety information contained in the Biograph Safety Procedure and Biograph Safety Report.**
-

Preparation

1. Switch the system to **COMP/ON** at the control panel.
2. **Switch off circuit breakers F8 and F9 in the PDC**
3. Remove the upper cover from the right side of the gantry.

Removing

1. Disconnect the power plug E321.X1 and the Canon plug X2 (Circled) and remove all cable ties that interfere with the cables.



Fig. 144: Dehumidifier

2. Remove the 4 screws (arrows).
3. Carefully remove the air flow insulating tape.
4. Remove the defective dehumidifier.

Installing

1. Insert the new dehumidifier and attach it with the 4 screws. Tighten to **5 Nm**
2. Attach the airflow insulating tape.
3. Connect plug E321.X1 and plug X2 and secure the cables with cable ties.

Final steps

Mechanical

1. Switch on circuit breakers **F8** and **F9** inside the PDC.
2. Switch the system back to **SYS/ON**.

Tests

- Verify that no air is escaping from the rotating part of the gantry. If air is escaping, cover the leak with insulating tape.
- Replace the covers.

Tilt Spindle

NOTE

CT Gantry tilt is disabled for a Biograph 40/64

Prerequisites

Tools and auxiliary equipment

- Standard service kit.
- Retaining bolt to prevent the gantry from tilting (M12 x 70)

Time and manpower

- 1 person
- 1 hour

Safety

**WARNING**

Avoid accident and injury or damage of parts.

Risk of accident and injury!

- ⇒ Read and observe the safety information contained in the Biograph Safety Procedure and Biograph Safety Report.

Preparation

Case 1: The gantry can still be tilted.

1. Remove the upper and lower cover from the right side of the gantry.
2. Insert the retaining bolt

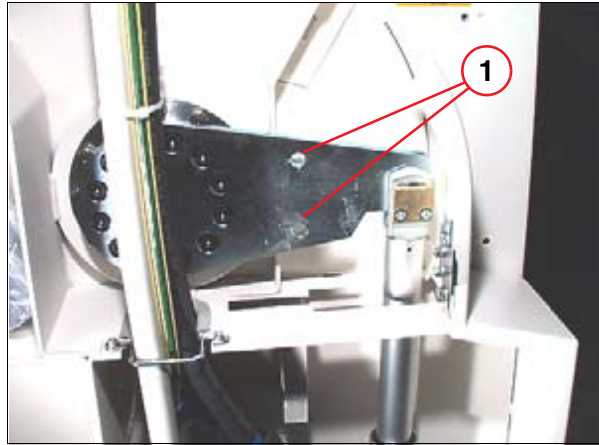


Fig. 145: Tilt spindle, upper

- Tilt the gantry until the retaining bolt can be inserted into either the upper or lower hole, whichever is aligned first. (item 1)
3. Switch **OFF** the system at the control box.
 4. **Switch OFF onsite power using the onsite ON/OFF switch!**
 5. Remove the mechanical lock from the height adjustment mechanism.



Fig. 146: Tilt spindle, lower

- Remove lock (item 2)

Case 2: The gantry cannot be tilted.

1. Switch **OFF** the system at the control box.
2. **Switch OFF onsite power using the onsite ON/OFF switch!**
3. Remove the upper and lower covers from the right side of the gantry.
4. Insert retaining bolt

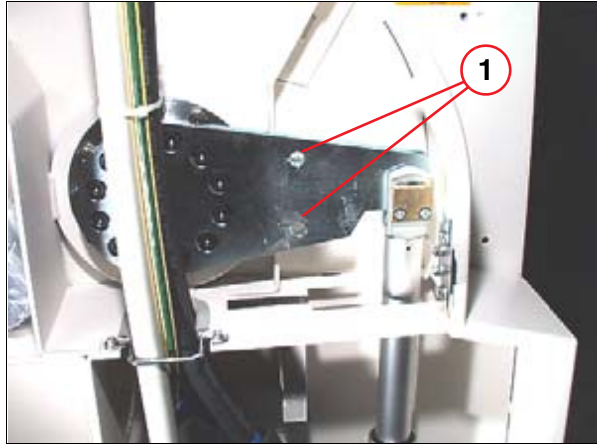


Fig. 147: Tilt spindle, upper



Fig. 148: Tilt spindle, lower

- Remove the mechanical lock from the height adjustment mechanism (item 2).
- Turn the height adjustment nut until the retaining bolt can be inserted into either the upper or lower hole (item 1), whichever is aligned first. **Note the number of turns!**
 - ⇒ Do not force the gantry into the 0° position if it is stuck. The spindle can be replaced even when the gantry is tilted.
 - ⇒ The number of turns are noted so that the adjuster can later be returned close to its original position.

Removing

1. Remove the cover of the tilt motor connection box.
2. Remove the cables from the terminal block. Check the positions against (Tab. 3 / p. 215) as shown in the reassembly instructions.

3. Release the tilt assembly.



Fig. 149: Tilt spindle upper / lower

- Remove the safety bracket for the upper retaining pin (item 1).
 - Remove the safety bracket for the lower retaining pin (item2).
 - Turn the height adjustment nut until you can remove both retaining pins.
4. Remove the defective tilt spindle.

Installing

1. Turn the spindle of the new tilt motor until it corresponds to the length of the defective spindle outside the gantry.

If the length of the new spindle is not identical to that of the old one, the tilt spindle motor may not be able to achieve $\pm 30^\circ$ gantry tilt and will have to be adjusted.

2. Install the new tilt spindle.

3. Insert retaining pins.



Fig. 150: Tilt spindle upper / lower

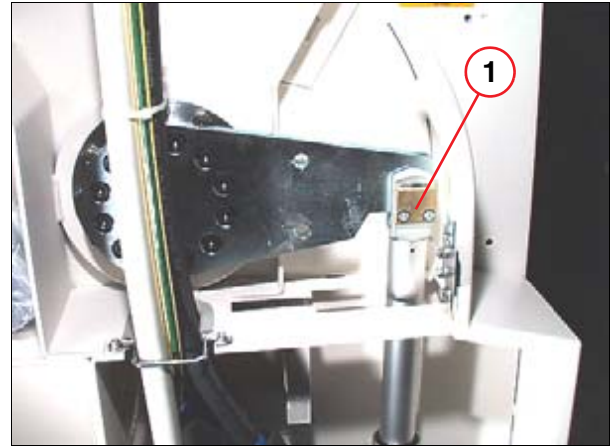


Fig. 151: Tilt spindle upper/lower

- Insert the upper retaining pin using a small amount of grease.
- Attach the safety bracket (item 1) for the upper retaining pin.
- Insert the lower retaining pin using a little grease.

If it is not possible to insert the retaining pin, turn the height adjustment nut slightly until it fits.

- Attach the safety bracket (Item 2) for the lower retaining pin.

4. Connect tilt motor.

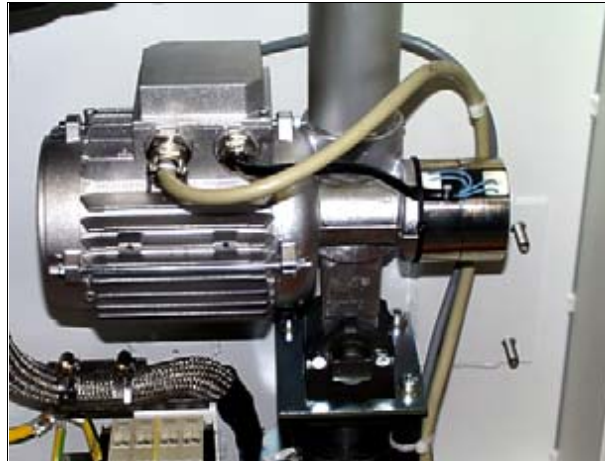


Fig. 152: Tilt motor detail

- Route the cables as shown.
 - Connect the cables to the terminal block (see [Tab. 3 / p. 215](#) below).
5. Check the electrical connections and ensure that they are secure.
 6. Attach the cover of the connections box.

Tab. 3 Connection Table

Terminal block	1	2	3	4	5	6	7	8	9
Where used	U	V	W	Temp	Temp	Brake 1	Brake 1	Brake 2	Brake 2
	Power cable			Control cable and brake cable					

Final steps

Adjustments

1. Remove the retaining bolt.
2. If you have to turn the adjuster to insert the retaining bolt (i.e. you were unable to tilt), make sure you note the number of turns. Turn the adjuster back to its original position.
3. Switch **on** the system.
4. Test the tilting function of the gantry using the operating panels.

If you are unable to tilt the gantry to $\pm 30^\circ$ use the height adjustment mechanism for adjustment.

Mechanical

1. Replace the lock for the mechanical adjuster
2. Replace the covers

Tests

1. Perform a test scan

GPC Board

Prerequisites

Tools and auxiliary equipment

- Standard service kit.

Time and manpower

- 1 person
- 1 hour

Safety



Avoid accident and injury or damage of parts.

Risk of accident and injury!

- ⇒ **Read and observe the safety information contained in the Biograph Safety Procedure and Biograph Safety Report.**
-

Preparation

1. Switch the system to **COMP/ON** .
2. **Switch off circuit breakers F8 and F9 in the PDC**
3. Remove the upper cover for the right side of the gantry.

Removing



Fig. 153: GPC board

1. Remove the metal cover from board D311.
2. Disconnect connectors X1-X10.
3. Remove the eight screws (e.g. circled)
4. Remove the defective board

Installing

1. Install the new board and the old MCB in the reverse order.
2. Ensure that connectors X1-X10 are connected properly.

Final steps

Mechanical

1. Switch on circuit breakers **F8** and **F9** inside the PDC.
2. Install the upper cover for the right side of the gantry.

Tests

1. Switch the system to **SYSTEM/ON** and test all possible tilt/feed functions, including the light markers, on both key panels.

3-way valve

Prerequisites

Tools and auxiliary equipment

- Standard service kit.

Time and manpower

- 1 person
- 1 hour

Safety



Avoid accident and injury or damage of parts.

Risk of accident and injury!

- ⇒ **Read and observe the safety information contained in the Biograph Safety Procedure and Biograph Safety Report.**
-

Preparation

1. Switch the system to **COMP/ON** .
2. **Switch off circuit breakers F8 and F9 in the PDC**
3. Remove the lower cover for the right side of the gantry.

Removing



Fig. 154: 3-way valve

1. Disconnect power cable
2. Open the screw (arrow) and remove the valve

Installing

1. Install the valve in the reverse order.

Final steps

Mechanical

- Switch on circuit breakers **F8** and **F9** inside the PDC.
- Switch the system to **SYSTEM/ON** and verify that water flow is ok.

Tests

- Perform a test scan

Overview

- The additional parts include the following components:
 - light markers
 - LPS board
 - Plexi strip

Rotating light markers

Prerequisites

Tools and auxiliary equipment

- Standard service kit.

Time and manpower

- 1 person
- 1 hour

Safety



Avoid accident and injury or damage of parts.

Risk of accident and injury!

- ⇒ **Read and observe the safety information contained in the Biograph Safety Procedure and Biograph Safety Report.**

Preliminary Steps

- Lower the patient table in order to open the gantry front door.
- Switch off the gantry using the **COMP/ON** switch.
- **Switch off circuit breakers F8 and F9 in the PDC**
- Remove pallet from PHS.
- Open the gantry front door and disable rotation with switch S 301.
- Position the X-ray tube at **3 o'clock**.

Removing

The example shows light marker X582. The other 3 light markers in the rotating gantry are configured in the same way. The table below shows the locations of the light markers.

Tab. 4 Light marker position

Light marker	Location	Connections	Position tube at:
D2 E581	Left of tube collimator	X2	12 o'clock
D2 E582	Right of tube collimator	X3, X4	12 o'clock

Light marker	Location	Connections	Position tube at:
D2 E583	Right of detector	X4	9 o'clock
D2 E584	Left of detector	X2, X3, X4	3 o'clock

1. Removing the light marker.



Fig. 155: Light marker

2. Remove the connectors at the light markers.
3. Remove the two attachment bolts M6 (circled) and remove the complete light marker device with the mounting bracket.

Installing

1. Install the light marker in the reverse order.

Final steps

Adjustments

NOTE

The light marker is preadjusted. Check for correct alignment with the other light markers before you make any adjustments.

The light marker is properly adjusted if the laser beam overlaps with the other laser beams in the slice plane.

Correct adjustment requires that the other light markers are properly adjusted.

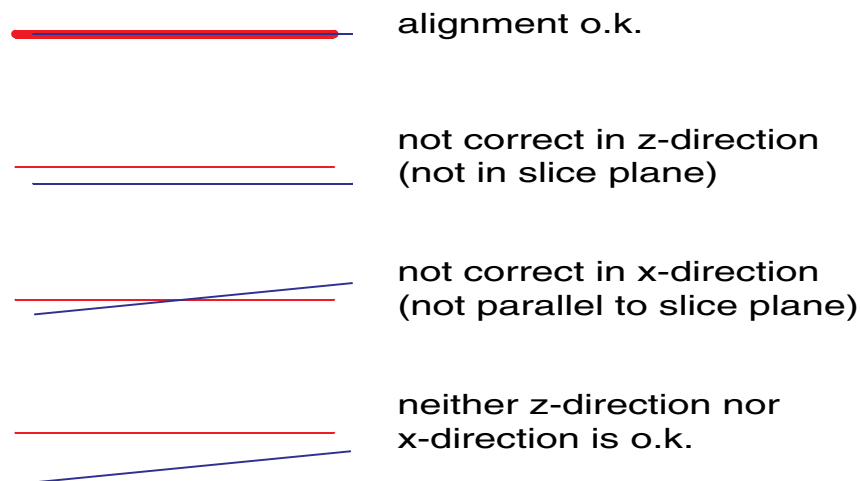


Fig. 156: Example of light marker adjustment

1. Switch on circuit breakers **F8** and **F9** inside the PDC.
2. Switch on the gantry (**SYS/ON**).
3. Position the light marker to approximately 11 o'clock.
4. Switch on the light markers.
5. Feed in the tabletop so that the laser beams are visible.

6. Adjust the X direction



Fig. 157: Adjusting laser

Correction in the X direction

- Loosen 3 of the 4 attachment screws (arrow) slightly.
- With the 13 mm crescent wrench or a small adjustable wrench as shown, you can rotate the laser slightly to adjust the laser beam parallel to the slice plane.
- After adjustment, tighten the attachment screws.

7. Adjust the Z direction

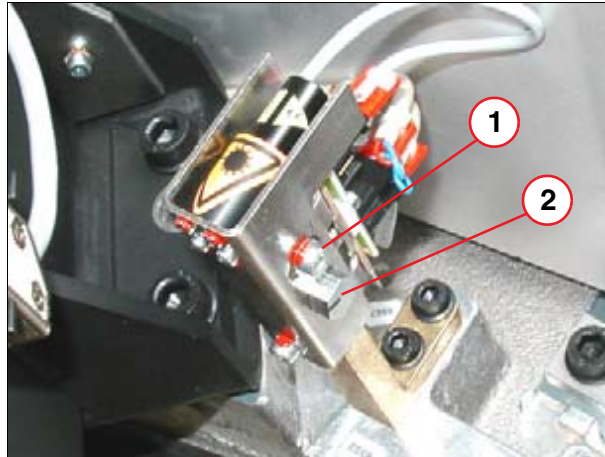


Fig. 158: Adjusting in Z-direction

Correction in the Z direction

- Slightly loosen the attachment screw (item 1).
- Use a 3mm hexagonal wrench to slightly turn the adjustment screw (item 2) for the laser, rotating it in the z-direction until it is on the slice plane.
- After the adjustment, tighten the attachment screw.

Mechanical

1. Close the gantry door.
2. Reattach pallet to PHS.

Tests

1. Perform quality test

LPS Board (D522)

Prerequisites

Tools and auxiliary equipment

- Standard service kit.

Time and manpower

- 1 person
- 1 hour

Safety



Avoid accident and injury or damage of parts.

Risk of accident and injury!

- ⇒ **Read and observe the safety information contained in the Biograph Safety Procedure and Biograph Safety Report.**
-

Preliminary Steps

1. Lower the patient table in order to open the gantry front door.
2. Switch **OFF** the system at the control box
3. **Switch OFF onsite power using the onsite ON/OFF switch!**
4. Remove pallet from PHS.
5. Open the gantry front door and disable rotation with switch S 301.
6. Move the PET gantry into service position to access the rear of the CT gantry.

NOTE

Consult MI12-001.813.01.xx.xx for Biograph Service Positioning Procedure.

7. Remove the CT gantry rear cover

Removing

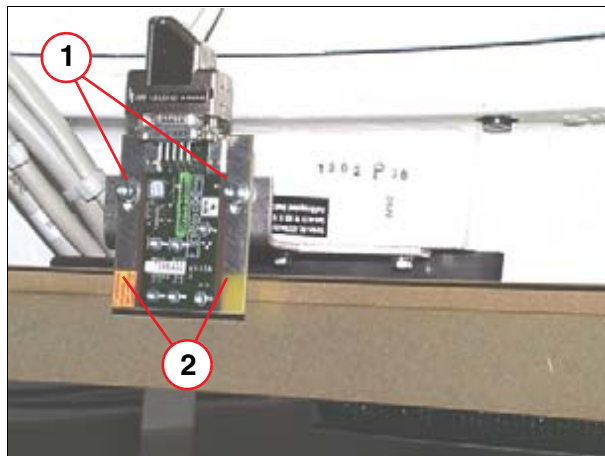


Fig. 159: LPS board

1. Disconnect plug D522, X1 from the LPS board
2. Remove the two screws connecting the board to the holder (item 1).

Installing

1. Install the LPS board in the reverse order; do not tighten the screws yet.

Final steps

Adjustments

1. Adjust the position of the board

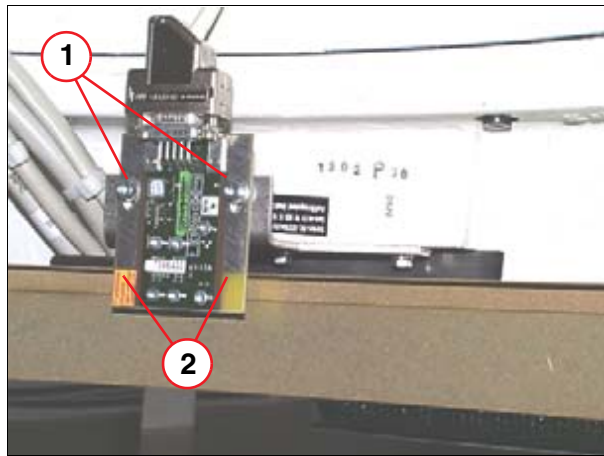


Fig. 160: LPS board

- Adjust the LPS board to align the slot ring with the end of the metal contact foil to the left and right of the LPS board (item 2).

Correct alignment assures that the slot ring is aligned with the center of the 2 light barriers.

- Tighten the two screws (item 1).

2. Check the height of the board.

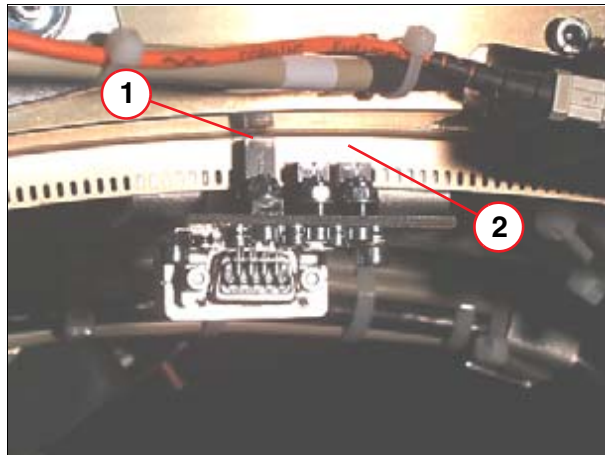


Fig. 161: LPS height adjustment

- Check the height of the board:
 - ⇒ The gap (arrow 1) must be $0.8 \text{ mm} \pm 0.2 \text{ mm}$.
 - ⇒ The gap (arrow 2) must be $5.0 \text{ mm} \pm 0.2 \text{ mm}$.If necessary adjust the height of the LPS board holder.

3. Adjust height of board (only if needed)

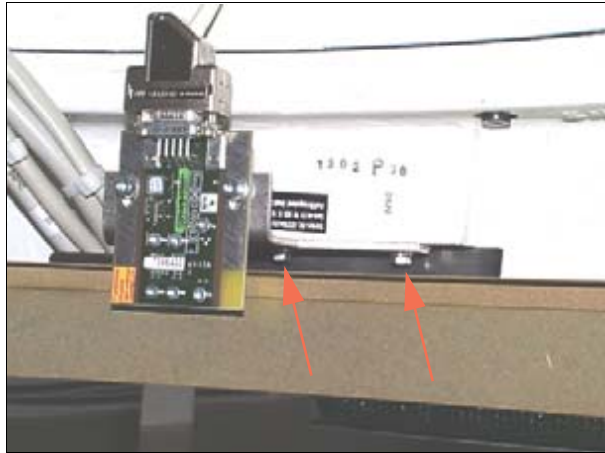


Fig. 162: Height adjustment screws

- **Only if the height needs adjusting:**
 - Loosen the 2 bolts (arrows) with a wrench
 - Adjust the gaps
- Tighten the bolts and re-check the gaps
- **Turn the Gantry by hand and ensure that the light barrier does not touch any part of the slot ring .**

Mechanical

1. Install rear gantry cover.
2. Enable Rotation with switch S301.
3. Close the gantry front door.
4. Reattach pallet to PHS.
5. Move PET gantry back into position.
6. Switch on the system.

Tests

1. Load different rotation modes and verify that they are running without any error messages.

Plexi Strip

Prerequisites

Tools and auxiliary equipment

- Standard service kit.

Time and manpower

- 1 person
- 1 hour

Safety



Avoid accident and injury or damage of parts.

Risk of accident and injury!

- ⇒ **Read and observe the safety information contained in the Biograph Safety Procedure and Biograph Safety Report.**
-

Preliminary Steps

1. Switch off the gantry using the **COMP/ON** switch at the control box

Installing

1. Install the plexi strip and mark the overlapping part of the Plexi strip.
2. Remove the Plexi strip and cut it to the correct length.
3. Install the Plexi strip.
4. **The ends of the Plexi strip must be positioned at the 1 o'clock position. The back cover has a threaded hole at this position.**
5. Insert the locking screw to secure the Plexi strip

Final steps**Mechanical**

1. Switch on the gantry using the **SYSTEM/ON** button at the control box

Tests

1. Perform a test scan

Overview

- The gantry rear includes the following components:
 - Receiver slip ring
 - Power slip rings
 - Intercom boards D346, NOISE_FILT and D344, NOISE_CAN
 - Brake assembly

Receiver ring

Prerequisites

Tools and auxiliary equipment

- Standard service kit
- Four 8mm guide pins for proper alignment, delivered with part.
- Loctite 243

Time and manpower

- 1 person
- 2 hours

Safety



Avoid accident and injury or damage of parts.

Risk of accident and injury!

- ⇒ **Read and observe the safety information contained in the Biograph Safety Procedure and Biograph Safety Report.**
-

Preparation

1. Lower the patient table in order to open the gantry front door.
2. Switch **OFF** the system.
3. **Switch OFF onsite power using the onsite ON/OFF switch!**
4. **Open the fuse holders F100 and F200 in PDC.**
5. Remove pallet from PHS.
6. Open the gantry front door.
7. Move the PET gantry into service position to access the rear of the CT gantry.

NOTE

Consult MI12-001.813.01.xx.xx for Biograph Service Positioning Procedure.

8. Remove the CT gantry rear cover.

Removing

1. Rotate the gantry until tube is 20 cm before 12 o'clock position to align the guide pin holes.
2. Insert the guide pins.

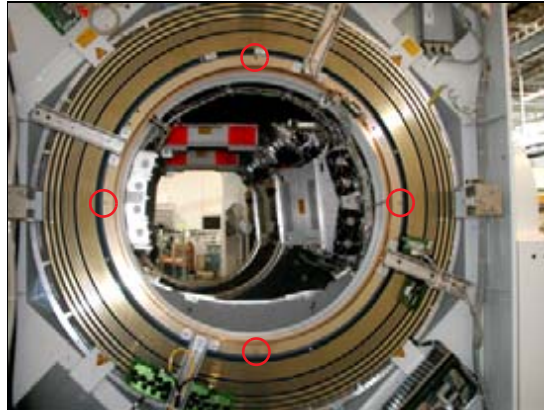


Fig. 163: Location of alignment pins

- Insert the four guide pins in holes (circled), slightly rotating the gantry as necessary.
3. Remove the LPS board ([LPS Board \(D522\)](#) / p. 228).
 4. Remove receiver board ([Transmitter and receiver board](#) / p. 250).

5. Remove upper left bracket

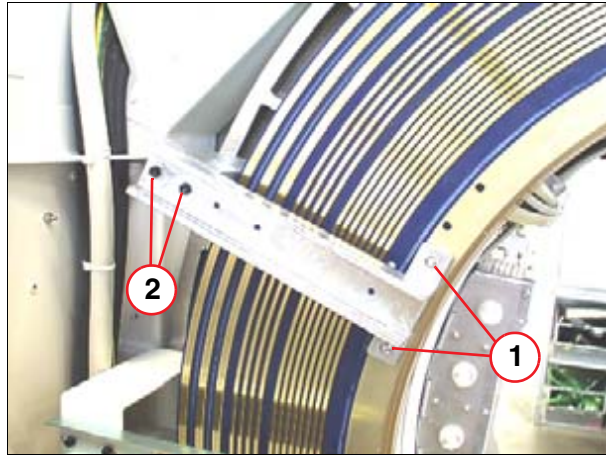


Fig. 164: Upper left bracket

- Loosen 2 screws from receiver ring upper left bracket (item 1).
Keep the plastic washers
- Loosen two screws from receiver ring bracket on frame and remove bracket (item 2).

6. Remove lower left bracket

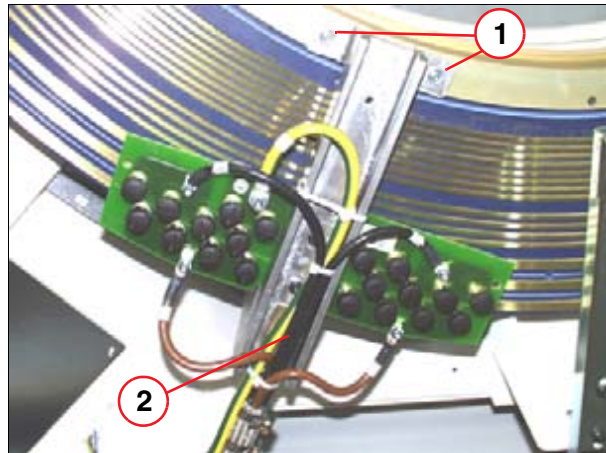


Fig. 165: Lower left bracket

- Remove the 2 screws from receiver ring lower left bracket (item 1)
Keep the plastic washers
- Mark the position of the bracket with a line. This will help to align the brushes on reassembly
- Loosen two screws from receiver ring bracket on frame (item 2) and put the bracket on the ground (be careful with connected cables and brush holder).

7. Remove the lower right bracket

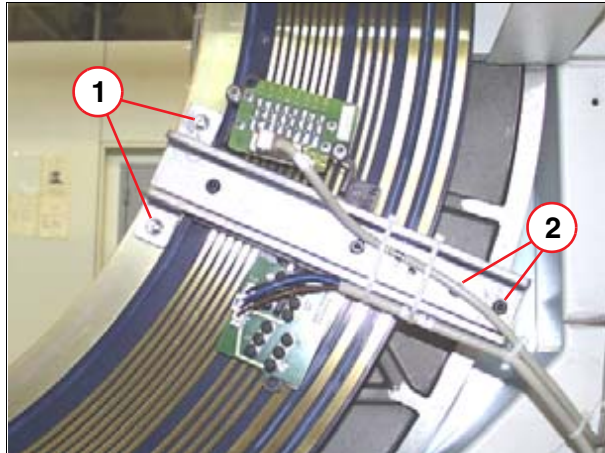


Fig. 166: Lower right bracket

- Remove the 2 screws from receiver ring bracket (item 1)
Keep the plastic washers
- Mark the position of the holder with a line.
- Remove the two screws securing the bracket on the frame (item 2) and tie back the assembly.

8. Remove the upper right bracket

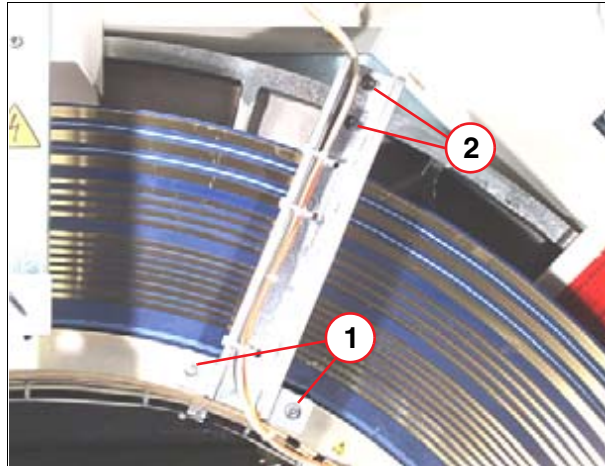


Fig. 167: Upper right bracket

- Remove all cable ties and cable connections E400_X3 ,X4 to the receiver board.
- Remove the screws from receiver ring bracket (item 1).
Keep the plastic washers
- Remove the 2 screws securing bracket to frame (item 2) and remove it.

9. Remove the receiver ring. Make sure not to touch/scratch the surface of the ring. The pins will act as a guide.

Installing

1. Install the new receiver ring into the gantry, making sure not to touch/scratch the surface of the ring.

Make sure that the Index Pulse of the slot ring is at the 10 o'clock position

Use the 4 guide pins to align the receiver ring and slip ring.

2. Install the brackets

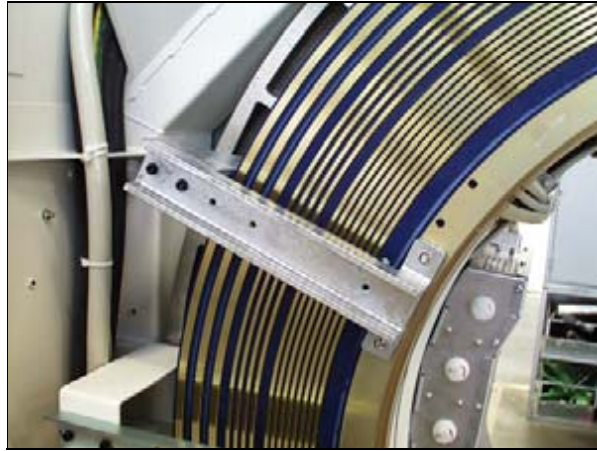


Fig. 168: Example of bracket

- Reinstall all 4 brackets but do not tighten the screws at this point.
Remember to install the plastic washers.
- Tighten the receiver ring holding the screws with a torque of **3 Nm**.
Use the marks to help align the brush holders.
Check that the plastic washers are correctly installed - (isolation -> measure it).
- Tighten the lower left bracket to frame screws with **24 Nm** (item 1).
- Tighten the other three bracket-to-frame screws to **28 Nm**.

3. Remove the guide pins.

4. Adjust the signal brush holder

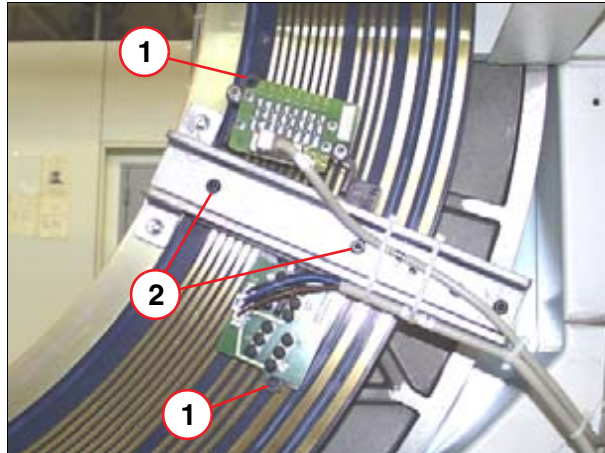


Fig. 169: Signal brush holder

- Check the alignment of the right side brush holder, using the guide pins in the holes (item 1) to properly center the brush holders. If necessary, adjust the holders by opening the holding screws (item 2). Tighten the screws with **6 Nm** (use Loctite 243).

5. Adjust the power brush holder

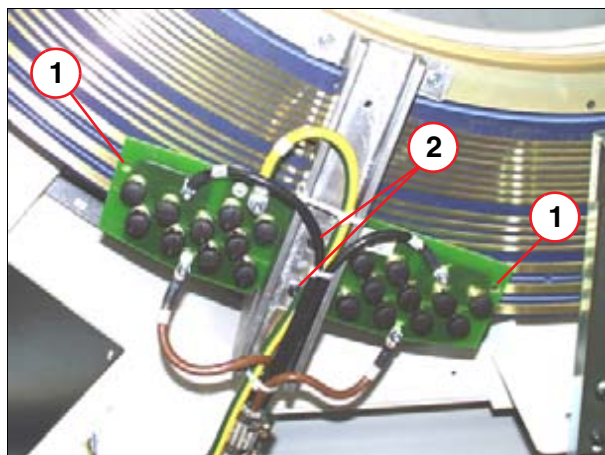


Fig. 170: Power brush holder

- Check the alignment of the left side brush holder, using the guide pins in the holes to properly center the brush holders (item 1). If necessary, adjust the holders by opening the holding screws (item 2). Tighten the screws with **6 Nm** (use Loctite 243).

6. Reinstall the LPS board (light barrier). Refer to [\(LPS Board \(D522\) / p. 228\)](#) for more information.

7. Reinstall the receiver board. Refer to [\(Transmitter and receiver board / p. 250\)](#) for more information.

Final steps

Mechanical

1. Move the gantry by hand and make sure that the receiver ring does not touch the slip ring part.
2. Install the gantry rear cover.
3. Move the PET gantry back to the scan position.
4. Close the gantry front door and install the plexi ring.
5. Reattach pallet to PHS.
6. Close the fuse holders **F100, F200** in the PDC.
7. Switch on the system.

Tests

1. Perform a test scan

Slip ring

Prerequisites

Tools and auxiliary equipment

- Standard service kit
- Four 8 mm guide pins for proper alignment, delivered with part.
- Loctite 243

Time and manpower

- 2 persons
- 3 hours

Safety



Avoid accident and injury or damage of parts.

Risk of accident and injury!

- ⇒ **Read and observe the safety information contained in the Biograph Safety Procedure and Biograph Safety Report.**
-

Preparation

1. Lower the patient table in order to open the gantry front door.
2. Switch **OFF** the system.
3. **Switch OFF onsite power using the onsite ON/OFF switch!**
4. **Open the fuse holders F100 and F200 in PDC.**
5. Remove pallet from PHS.
6. Open the gantry front door.
7. Move the PET gantry into service position to access the rear of the CT gantry.

NOTE

Consult MI12-001.813.01.xx.xx for Biograph Service Positioning Procedure.

8. Remove the rear CT gantry cover.

Removing

1. Mark the position of the 4 funnel holders.
This assures that the holders can be fixed to their previous position during reinstallation.
2. Remove 4 funnel holders.
3. Remove the receiver ring. Instructions are located in [\(Receiver ring / p. 236\)](#).
4. At the front left of the gantry remove the following plugs.
 - E510_X6 (C-box) backplane
 - E510_X4 (C-box) DPU board
 - E 603_X22 (not used, but release cable ties)
 - E535_X11 Oil cooler<may be marked X2 in older systems> and ground.
5. At the front right of the gantry remove the following plugs and cables.
 - E603_X1 E-box
 - D440_X1 (2 DCC cables) E-Box
 - D440_X22 DCC cables) E-box
 - D440_X599 (DCC ground)
 - E607_X1 DMS power supply (only for Sensation 40/64/Cardiac 64, Open)
6. Remove the transmitter board. Instructions can be found in [\(Transmitter and receiver board / p. 250\)](#)

7. Prepare slip ring for removal

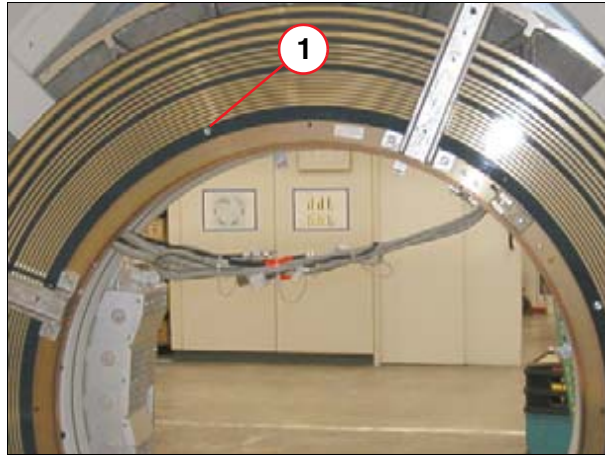


Fig. 171: Slip ring with cables tied together

- Remove cable ties as required to release all of the cables note the routing of the cables
- Secure the cables together in the center with cable ties as shown in picture
- Remove the 8 screws holding the slip ring (e.g. item 1)

8. Hold the slip ring (at least 2 people required) and remove from the gantry.

Installing

1. Check the cable connections on the new slip ring, making sure the cables are connected properly and compare with the old one.

Torques for screws of cable connections:

- **M3x6: max. 0.7 Nm**
- **M4x6: max. 1.4 Nm**
- **M5x8: max. 2.9 Nm**

2. Prepare and install slip ring



Fig. 172: Cables secured to brackets

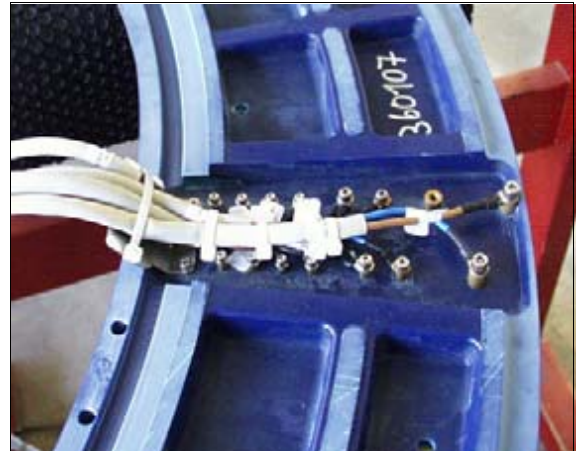


Fig. 173: Cables secured to brackets

- Secure the cables to the metal brackets of the slip ring with cable ties
 - Secure the cables together in the center with cable ties and install the new slip ring.
 - Secure the eight screws of the slip ring with **10 Nm** and **Loctite 243**.
3. Reconnect and secure all cables in the original configuration with cable ties.
 4. Install the receiver slip ring as detailed in [\(Receiver ring / p. 236\)](#).
 - ⇒ **It is important to follow these instructions as they include important information on alignment of brush holders and the isolation of the receiver ring.**
 5. Reinstall the transmitter board [\(Transmitter and receiver board / p. 250\)](#).
 6. Reinstall the 4 funnel holders to their original position.
 7. Reconnect all removed cables. Use cable ties to fix the cables.

Final steps

Mechanical

1. Move the gantry by hand and make sure that the receiver ring does not touch the slip ring part.
2. Install the gantry rear cover.
3. Move the PET gantry back into scan position.
4. Close the gantry front door and install the plexi ring.
5. Reattach the pallet to the PHS.
6. Close the fuse holders **F100**, **F200** in the PDC.

7. Switch on the system.

Tests

1. Perform a test scan

Transmitter and receiver board

Prerequisites

NOTE

The electronics of transmitter and receiver are highly sensitive.

It is mandatory to follow the ESD instructions

Tools and auxiliary equipment

- Standard service kit.
- ESD protection aids.

Time and manpower

- 1 person
- 1 hour

Safety

 WARNING

Avoid accident and injury or damage of parts.

Risk of accident and injury!

- ⇒ Read and observe the safety information contained in the Biograph Safety Procedure and Biograph Safety Report.

Preparation

1. Lower the patient table in order to open the gantry front door.
2. Switch **OFF** the system at the control box.
3. **Switch OFF onsite power using the onsite ON/OFF switch!**
4. Remove pallet from PHS.
5. Loosen the gantry front screws to open the front cover.
6. Move the PET gantry into service position to access the rear of the CT gantry.

NOTE

Consult MI12-001.813.01.xx.xx for Biograph Service Positioning Procedure.

7. Remove the back CT gantry panel

Removing



1. Remove the transmitter or receiver

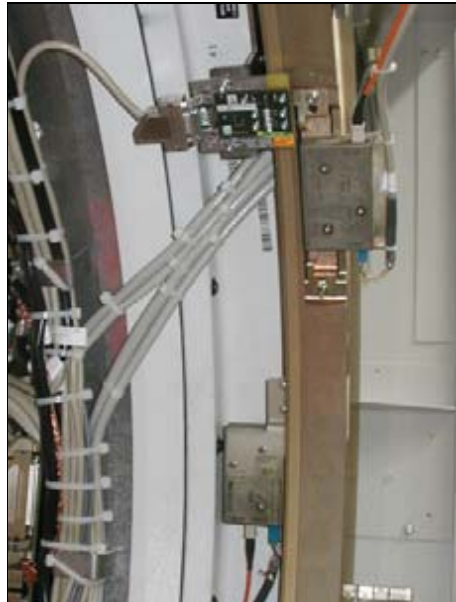


Fig. 174: Transmitter and receiver

- To remove the receiver
 - Disconnect E400_X3 and E400_X4. **Follow C-MOS guidelines**
 - Remove the holding screws
- To remove the transmitter
 - Disconnect E400_X3 and E400_X4. **Follow the C-MOS guidelines**
 - **Sensation 10/16/Cardiac:**
Remove the two screws and the stator ring brush, then remove the third screw.
 - **Sensation 40/64/Cardiac 64/Open:**
Remove the 4 holding screws

Installing

1. Install the two boards in reverse order. Use **Loctite 243** and tighten the screws with a torque of:

- **0.9 Nm** Sensation 10, Sensation 16 and Cardiac

- **0.5 Nm** Sensation 40/64, Cardiac 64 and Open

Make sure the stator ring brush is contacting the stator ring. (**ONLY Sensation 10/16/Cardiac**).

NOTE

The receiver has to be adjusted parallel to the antenna. Follow the instruction as shown in figures [\(Fig. 175 / p. 252\)](#) [\(Fig. 176 / p. 253\)](#) [\(Fig. 177 / p. 253\)](#)

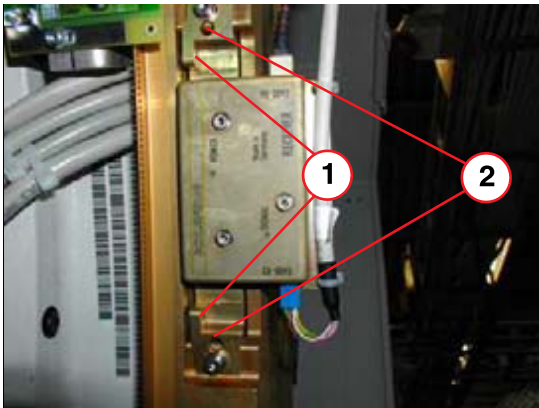


Fig. 175: Receiver

Pos. 1 Use 2 screws M4x12 for the fine adjustment of the receiver.

Pos. 2 Marker for adjustment

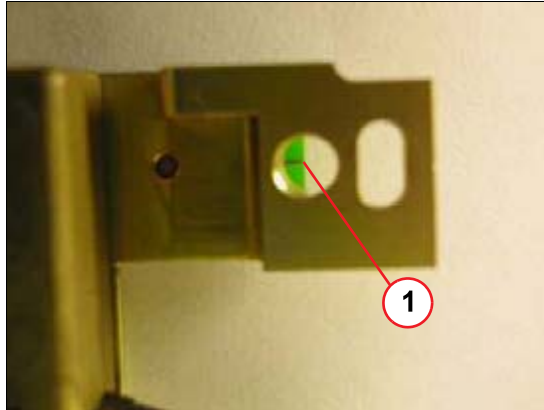


Fig. 176: Marker of Receiver

Pos. 1 Marker for adjustment (one side of receiver)

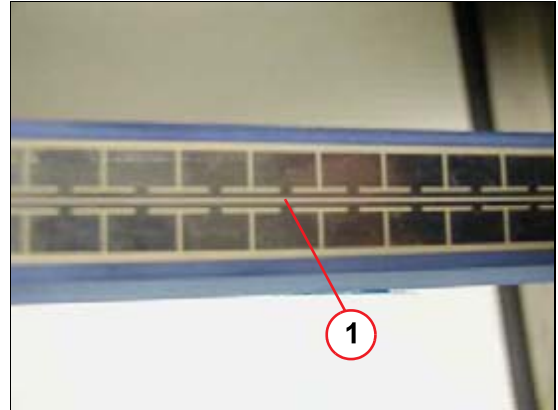


Fig. 177: Antenna

Pos. 1 Center line of antenna

- The marker of the receiver(2/Fig. 175 / p. 252) (1/Fig. 176 / p. 253) must be adjusted congruent with the center line of the antenna (1/Fig. 177 / p. 253). The marker is located on the printed circuit board behind the hole (2/Fig. 175 / p. 252)(1/Fig. 176 / p. 253).
- Use 2 screws M4x12 (1/Fig. 175 / p. 252) for fine adjustment of the receiver. Press the receiver against the inner ring of the gantry and turn the screws, so that both markers (2/Fig. 175 / p. 252) resp. (1/Fig. 176 / p. 253) of the receiver are congruent with the center line of the antenna.
- Remove the 2 screws after completing the adjustment

Difference between Sensation 16 and 40/64

- New receiver and transmitter with additional LED for Sync Lock.
The structure of the transmitting antenna is finer.
- No stator ring brush
- Same procedure for removal and installation
- The transmitter is attached with 4 instead of 3 screws
- The adjustment procedure is the same as for S16
- Tighten all screws for transmitter and receiver with a torque of **0.5 Nm**.
Use **Loctite 243**



Fig. 178: Transmitter for Sensation 40/64/Cardiac64/Open



Fig. 179: Receiver for Sensation 40/64/Cardiac64/Open

Final steps

Mechanical

1. Install the gantry rear cover.
2. Move the PET gantry back into scan position.
3. Close the gantry front door and install the plexi ring.
4. Reattach pallet to PHS.
5. Switch on the system.

Tests

1. Perform a test scan

D344 NOISE_CAN board

Prerequisites

Intercom with D344, NOISE_CAN and D345, MIC_GAN2

- The intercom equipment comprises the two boards D344 NOISE-CAN and D345 MIC_GAN2.

Tools and auxiliary equipment

- Standard service kit.

Time and manpower

- 1 person
- 1 hour

Safety



WARNING

Avoid accident and injury or damage of parts.

Risk of accident and injury!

- ⇒ **Read and observe the safety information contained in the Biograph Safety Procedure and Biograph Safety Report.**

Preparation

1. Switch **off** the system.
2. **Switch OFF onsite power using the onsite ON/OFF switch!**
3. Move the PET gantry into service position to access the rear of the CT gantry.

NOTE

Consult MI12-001.813.01.xx.xx for Biograph Service Positioning Procedure.

4. Remove the rear CT gantry cover.

Removing

1. Remove connector D344_X1, X3 and X4.
2. Remove the 4 mounting screws.



Fig. 180: Noise / CAN board

- Remove the NOISE-CAN board

Installing

1. Install the NOISE_CAN board in the reverse order.

Final steps

Mechanical

1. Install the gantry rear cover.
2. Move the PET gantry back into scan position.
3. Switch on the system.

Tests

1. Test the intercom function.

D346 NOISE_FILT board

Prerequisites

Intercom with D346 NOISE FILTER

- The NOISE FILTER board D346 replaces the earlier intercom equipment with the two boards D344 NOISE_CAN and D345 MIC_GAN2. The NOISE FILTER D346 is mounted at the position of MIC_GAN2 D345 in the upper left corner of the back of the gantry.

Tools and auxiliary equipment

- Standard service kit

Time and manpower

- 1 person
- 1 hour

Safety



Avoid accident and injury or damage of parts.

Risk of accident and injury!

- ⇒ **Read and observe the safety information contained in the Biograph Safety Procedure and Biograph Safety Report.**

Preparation

1. Switch **off** the system.
2. **Switch OFF on-site power using the on-site ON/OFF switch!**
3. Move the PET gantry into service position to access the rear of the CT gantry.

NOTE

Consult MI12-001.813.01.xx.xx for Biograph Service Positioning Procedure.

4. Remove the back cover of the CT gantry.

Removing

1. Remove connector D346_X1, X4, X102, X103 and X104.
2. Remove the four mounting screws.



Fig. 181: D346 NOISE FILTER

Installing

1. Install the NOISE FILTER board in the reverse order.

Final steps

Mechanical

1. Install the back cover of the gantry.
2. Move the PET gantry back into scan position.
3. Switch on the system.

Tests

1. Test the intercom function.

Brake assembly

Prerequisites

Tools and auxiliary equipment

- Standard service kit.

Time and manpower

- 1 person
- 1 hour

Safety



Avoid accident and injury or damage of parts.

Risk of accident and injury!

- ⇒ **Read and observe the safety information contained in the Biograph Safety Procedure and Biograph Safety Report.**
-

Preparation

1. Switch **OFF** the system at the control box.
2. **Switch OFF onsite power using the onsite ON/OFF switch!**
3. Move the PET gantry into service position to access the rear of the CT gantry.

NOTE

Consult MI12-001.813.01.xx.xx for Biograph Service Positioning Procedure.

4. Remove the CT gantry rear cover.

Removing

1. Prepare for removal



Fig. 182: Brake assembly

- Remove the following cables:
 - Motor cable at connector E332_X3, terminals U, V, W
 - Supply cable E322_L1 at X1 terminals 1U1, 1V1, 1W1 and clamp
 - 2 ground connections near clamp
 - Control cable E322_K1 at contactor K1, connectors A1, A2 (top).
2. Cut tie wraps as needed to free all cables.
 3. Remove the 4 screws and lift the assembly out.

Installing

1. Install in the reverse order.

Final steps

Mechanical

1. Install the gantry rear cover.
2. Move the PET gantry back into scan position.
3. Close the gantry front door and install the plexi ring.
4. Reattach pallet to PHS.
5. Switch on the system.

Tests

1. Perform a test scan

H hz_serdoc_F13G02U01M01.....50